

ADVANCED ENERGY STORAGE SYSTEMS MARKET

GLOBAL FORECAST TO 2022

BY TECHNOLOGY (ELECTRO CHEMICAL TECHNOLOGY, MECHANICAL TECHNOLOGY, THERMAL STORAGE TECHNOLOGY, AND OTHERS), BY APPLICATION (TRANSPORTATION AND GRID STORAGE) AND BY REGION

MarketsandMarkets is a global market research and consulting company. It is World's No. 1 in terms of premium market research studies published annually. Serving as a business intelligence partner to Fortune 500 companies across the world, it provides multi-client reports, company profiles, databases, and custom research services.

MarketsandMarkets covers seventeen industry verticals, including advanced materials, aerospace and defense, agriculture, automotive and transportation, biotechnology, building and construction, chemicals, energy and power, food and beverages, industrial automation, medical devices, mining, minerals and metals, packaging, pharmaceuticals, semiconductor and electronics, and telecommunications and IT.

Copyright © 2017 MarketsandMarkets

All Rights Reserved. This document contains highly confidential information and is the sole property of MarketsandMarkets. No part of it shall be circulated, copied, quoted, or otherwise reproduced without the prior written approval of MarketsandMarkets.

TABLE OF CONTENTS

1	INTRODUCTION	14
1.1	OBJECTIVES OF THE STUDY	14
1.2	MARKET DEFINITION.....	14
1.3	MARKET SCOPE.....	15
1.3.1	MARKETS COVERED	15
1.3.2	YEARS CONSIDERED FOR THE STUDY.....	15
1.4	CURRENCY.....	16
1.5	LIMITATIONS	16
1.6	STAKEHOLDERS	16
2	RESEARCH METHODOLOGY.....	17
2.1	RESEARCH DATA	18
2.1.1	SECONDARY DATA.....	18
2.1.2	PRIMARY DATA.....	19
2.1.2.1	Key data from primary sources	19
2.1.2.2	Key industry insights.....	20
2.1.2.3	Breakdown of primary interviews.....	20
2.2	MARKET SIZE ESTIMATION	21
2.3	MARKET BREAKDOWN & DATA TRIANGULATION	23
2.4	RESEARCH ASSUMPTIONS	24
3	EXECUTIVE SUMMARY.....	25
4	PREMIUM INSIGHTS	29
4.1	ATTRACTIVE OPPORTUNITIES IN THE ADVANCED ENERGY STORAGE SYSTEMS MARKET, 2017-2022.....	29
4.2	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION, 2017-2022.....	30
4.3	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2017 & 2022	31
4.4	TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION, 2017 & 2022.....	31
4.5	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2017 & 2022	32
5	MARKET OVERVIEW.....	33
5.1	INTRODUCTION	34
5.2	TECHNOLOGY OVERVIEW	34
5.3	MARKET DYNAMICS.....	35
5.3.1	DRIVERS	35
5.3.1.1	Increasing renewable energy investments	35
5.3.1.2	Energy Arbitrage	36

5.3.1.3	Regulations demanding energy efficiency	36
5.3.1.4	Declining technology cost due to technology advancement and expanded deployment	37
5.3.1.5	Electrification of the transport sector	37
5.3.2	RESTRAINTS.....	38
5.3.2.1	High capital investments	38
5.3.2.2	Hazardous environmental impact.....	38
5.3.3	OPPORTUNITIES.....	38
5.3.3.1	Grid expansion and investment in the modern infrastructure.....	38
5.3.3.2	Growing opportunity for commercial and residential applications	38
5.3.4	CHALLENGES	39
5.3.4.1	Dynamic valuations	39
5.3.4.2	Regulatory treatment.....	39
5.3.4.3	Lack of common codes and standards	39
6	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION.....	40
6.1	INTRODUCTION	41
6.2	TRANSPORTATION	42
6.3	GRID STORAGE.....	43
7	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY	45
7.1	INTRODUCTION	46
7.2	ELECTRO CHEMICAL.....	47
7.2.1	LITHIUM-ION BATTERY.....	49
7.2.2	LEAD ACID BATTERY	50
7.2.3	SODIUM SULFUR (NAS) BATTERY	51
7.2.4	FLOW BATTERY.....	52
7.2.5	NICKEL METAL HYDRIDE (NIMH) & NICKEL CADMIUM BATTERY (NICD)	54
7.3	THERMAL STORAGE.....	55
7.3.1	THERMO CHEMICAL ENERGY STORAGE.....	57
7.3.2	SENSIBLE HEAT THERMAL ENERGY STORAGE.....	58
7.3.3	LATENT HEAT THERMAL ENERGY STORAGE.....	59
7.4	MECHANICAL ENERGY STORAGE.....	61
7.4.1	PUMPED HYDRO ENERGY STORAGE.....	63
7.4.2	FLYWHEEL ENERGY STORAGE	64
7.4.3	COMPRESSED AIR & LIQUID AIR ENERGY STORAGE	65
7.5	OTHER STORAGE TECHNOLOGIES	66

8	ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION	68
8.1	INTRODUCTION	69
8.2	ASIA PACIFIC.....	70
8.2.1	CHINA.....	74
8.2.2	INDIA.....	75
8.2.3	SOUTH KOREA.....	76
8.2.4	JAPAN.....	76
8.2.5	THE REST OF ASIA PACIFIC	77
8.3	NORTH AMERICA	78
8.3.1	US	81
8.3.2	CANADA.....	82
8.4	EUROPE	83
8.4.1	THE UK.....	86
8.4.2	GERMANY	87
8.4.3	FRANCE	87
8.4.4	THE REST OF EUROPE.....	88
8.5	REST OF THE WORLD	88
8.5.1	SOUTH AFRICA	91
8.5.2	CHILE.....	91
8.5.3	REST OF ROW.....	92
9	COMPETITIVE LANDSCAPE	93
9.1	OVERVIEW	93
9.2	KEY PLAYERS OF THE ADVANCED ENERGY STORAGE SYSTEMS MARKET.....	93
9.2.1	GROWTH STRATEGIES IN THE ADVANCED ENERGY STORAGE SYSTEM MARKET	93
9.3	MARKET RANKING, ADVANCED ENERGY STORAGE SYSTEMS, 2016.....	95
9.4	COMPETITIVE SITUATIONS & TRENDS.....	95
9.5	COMPETITIVE LEADERSHIP MAPPING	97
9.5.1	VISIONARY LEADERS.....	98
9.5.2	DYNAMIC DIFFERENTIATORS	98
9.5.3	INNOVATORS	98
9.5.4	EMERGING COMPANIES.....	98
9.6	STRENGTH OF PRODUCT PORTFOLIO (25 COMPANIES).....	99
9.7	BUSINESS STRATEGY EXCELLENCE (25 COMPANIES)	100
10	COMPANY PROFILES.....	101
10.1	LG CHEM, LTD.	101
10.1.1	OVERVIEW.....	101
10.1.2	PRODUCTS OFFERED.....	102
10.1.3	STRENGTH OF PRODUCT PORTFOLIO.....	102
10.1.4	BUSINESS STRATEGY EXCELLENCE.....	102
10.1.5	RECENT DEVELOPMENTS.....	103

10.2	ABB LTD.....	105
10.2.1	OVERVIEW.....	105
10.2.2	PRODUCTS OFFERED.....	106
10.2.3	STRENGTH OF PRODUCT PORTFOLIO.....	106
10.2.4	BUSINESS STRATEGY EXCELLENCE.....	106
10.2.5	RECENT DEVELOPMENTS.....	107
10.3	GS YUASA CORPORATION	108
10.3.1	OVERVIEW.....	108
10.3.2	PRODUCTS OFFERED.....	109
10.3.3	STRENGTH OF PRODUCT PORTFOLIO.....	109
10.3.4	BUSINESS STRATEGY EXCELLENCE.....	109
10.3.5	RECENT DEVELOPMENTS.....	110
10.4	SAMSUNG SDI CO., LTD.....	112
10.4.1	OVERVIEW.....	112
10.4.2	PRODUCTS OFFERED.....	113
10.4.3	STRENGTH OF PRODUCT PORTFOLIO.....	113
10.4.4	BUSINESS STRATEGY EXCELLENCE.....	113
10.4.5	RECENT DEVELOPMENTS.....	114
10.5	GENERAL ELECTRIC COMPANY	116
10.5.1	OVERVIEW.....	116
10.5.2	PRODUCTS OFFERED.....	117
10.5.3	STRENGTH OF PRODUCT PORTFOLIO.....	117
10.5.4	BUSINESS STRATEGY EXCELLENCE.....	118
10.5.5	RECENT DEVELOPMENTS.....	118
10.6	SAFT GROUPE S.A.	120
10.6.1	OVERVIEW.....	120
10.6.2	PRODUCTS OFFERED.....	120
10.6.3	STRENGTH OF PRODUCT PORTFOLIO.....	120
10.6.4	BUSINESS STRATEGY EXCELLENCE.....	121
10.6.5	RECENT DEVELOPMENTS.....	121
10.7	TESLA, INC.	123
10.7.1	OVERVIEW.....	123
10.7.2	PRODUCTS OFFERED.....	124
10.7.3	STRENGTH OF PRODUCT PORTFOLIO.....	124
10.7.4	BUSINESS STRATEGY EXCELLENCE.....	124
10.7.5	RECENT DEVELOPMENTS.....	125
10.8	EVAPCO, INC.	126
10.8.1	OVERVIEW.....	126
10.8.2	PRODUCTS OFFERED.....	126
10.8.3	STRENGTH OF PRODUCT PORTFOLIO.....	126
10.8.4	BUSINESS STRATEGY EXCELLENCE.....	127

10.9	CALMAC.....	128
10.9.1	OVERVIEW.....	128
10.9.2	PRODUCTS OFFERED.....	128
10.9.3	STRENGTH OF PRODUCT PORTFOLIO.....	129
10.9.4	BUSINESS STRATEGY EXCELLENCE.....	129
10.9.5	RECENT DEVELOPMENTS.....	130
10.10	BALTIMORE AIRCOIL COMPANY, INC.....	131
10.10.1	OVERVIEW.....	131
10.10.2	PRODUCTS OFFERED.....	131
10.10.3	STRENGTH OF PRODUCT PORTFOLIO.....	131
10.10.4	BUSINESS STRATEGY EXCELLENCE.....	132
10.10.5	RECENT DEVELOPMENTS.....	132
10.11	BYD COMPANY LIMITED.....	133
10.11.1	OVERVIEW.....	133
10.11.2	PRODUCTS OFFERED.....	134
10.11.3	STRENGTH OF PRODUCT PORTFOLIO.....	134
10.11.4	BUSINESS STRATEGY EXCELLENCE.....	134
10.11.5	RECENT DEVELOPMENTS.....	135
10.12	HITACHI, LTD.	137
10.12.1	OVERVIEW.....	137
10.12.2	PRODUCTS OFFERED.....	138
10.12.3	STRENGTH OF PRODUCT PORTFOLIO.....	138
10.12.4	BUSINESS STRATEGY EXCELLENCE.....	138
10.12.5	RECENT DEVELOPMENTS.....	139
10.13	SIEMENS AG	140
10.13.1	OVERVIEW.....	140
10.13.2	PRODUCTS OFFERED.....	141
10.13.3	STRENGTH OF PRODUCT PORTFOLIO.....	141
10.13.4	BUSINESS STRATEGY EXCELLENCE.....	142
10.13.5	RECENT DEVELOPMENTS.....	142
10.14	PANASONIC CORPORATION.....	144
10.14.1	BUSINESS OVERVIEW.....	144
10.14.2	PRODUCTS OFFERED.....	145
10.14.3	STRENGTH OF PRODUCT PORTFOLIO.....	145
10.14.4	BUSINESS STRATEGY EXCELLENCE.....	146

11	APPENDIX.....	148
11.1	INSIGHTS OF INDUSTRY EXPERTS	148
11.2	DISCUSSION GUIDE	149
11.3	KNOWLEDGE STORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL.....	152
11.4	INTRODUCING RT: REAL-TIME MARKET INTELLIGENCE.....	154
11.5	AVAILABLE CUSTOMIZATIONS.....	155
11.6	RELATED REPORTS	156
11.7	AUTHOR DETAILS.....	157

LIST OF TABLES

TABLE 1	ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT	26
TABLE 2	COMPARISON OF MAJOR ENERGY STORAGE TECHNOLOGIES BY TECHNICAL CHARACTERISTICS	34
TABLE 3	ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY APPLICATION, 2015–2022 (USD MILLION)	41
TABLE 4	TRANSPORTATION: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	43
TABLE 5	GRID STORAGE: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	44
TABLE 6	ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD MILLION)	46
TABLE 7	ELECTRO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	47
TABLE 8	ELECTRO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD MILLION)	48
TABLE 9	LITHIUM-ION BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	49
TABLE 10	LITHIUM-ION BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY APPLICATION, 2015–2022 (USD MILLION)	50
TABLE 11	LEAD ACID BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	51
TABLE 12	SODIUM SULFUR (NAS) BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	52
TABLE 13	FLOW BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	53
TABLE 14	NICKEL METAL HYDRIDE & NICKEL CADMIUM BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	54
TABLE 15	THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	56
TABLE 16	THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD THOUSANDS)	57
TABLE 17	THERMO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)	57
TABLE 18	SENSIBLE HEAT THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)	59
TABLE 19	LATENT HEAT THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)	60
TABLE 20	MECHANICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	61
TABLE 21	MECHANICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD MILLION)	62
TABLE 22	PUMPED HYDRO: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	63
TABLE 23	FLYWHEEL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	64
TABLE 24	CAES & LAES: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	66

TABLE 25	OTHERS: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)	67
TABLE 26	ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015– 2022 (USD MILLION)	70
TABLE 27	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY COUNTRY, 2015–2022 (USD MILLION)	72
TABLE 28	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	72
TABLE 29	ASIA PACIFIC: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	72
TABLE 30	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	73
TABLE 31	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)	73
TABLE 32	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD THOUSAND)	74
TABLE 33	ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)	74
TABLE 34	CHINA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	75
TABLE 35	INDIA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	75
TABLE 36	SOUTH KOREA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	76
TABLE 37	JAPAN: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	76
TABLE 38	THE REST OF ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	77
TABLE 39	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY COUNTRY, 2015–2022 (USD MILLION)	79
TABLE 40	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	79
TABLE 41	NORTH AMERICA: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	79
TABLE 42	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	80
TABLE 43	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)	80
TABLE 44	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD THOUSAND)	81
TABLE 45	NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)	81
TABLE 46	US: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	82
TABLE 47	CANADA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	82
TABLE 48	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY COUNTRY, 2015–2022 (USD MILLION)	83
TABLE 49	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	83
TABLE 50	EUROPE: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	84

TABLE 51	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	84
TABLE 52	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)	85
TABLE 53	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD MILLION)	85
TABLE 54	EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)	86
TABLE 55	THE UK: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	86
TABLE 56	GERMANY: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	87
TABLE 57	FRANCE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	87
TABLE 58	THE REST OF EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	88
TABLE 59	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY COUNTRY, 2015–2022 (USD MILLION)	88
TABLE 60	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	89
TABLE 61	REST OF THE WORLD: TRANSPORTATION APPLICATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	89
TABLE 62	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)	89
TABLE 63	REST OF WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)	90
TABLE 64	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD MILLION)	90
TABLE 65	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)	90
TABLE 66	SOUTH AFRICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	91
TABLE 67	CHILE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	91
TABLE 68	REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)	92
TABLE 69	MARKET RANKING, 2016	95

LIST OF FIGURES

FIGURE 1	MARKETS COVERED: ADVANCED ENERGY STORAGE SYSTEMS MARKET SEGMENTATION	15
FIGURE 2	ADVANCED ENERGY STORAGE SYSTEMS MARKET: RESEARCH DESIGN	17
FIGURE 3	BREAKDOWN OF PRIMARY INTERVIEWS: BY COMPANY TYPE, DESIGNATION, & REGION	20
FIGURE 4	MARKET SIZE ESTIMATION METHODOLOGY: BOTTOM-UP APPROACH	21
FIGURE 5	MARKET SIZE ESTIMATION METHODOLOGY: TOP-DOWN APPROACH	22
FIGURE 6	DATA TRIANGULATION METHODOLOGY	23
FIGURE 7	ASIA PACIFIC DOMINATED THE ADVANCED ENERGY STORAGE SYSTEMS MARKET IN 2016	27
FIGURE 8	ADVANCED ENERGY STORAGE SYSTEMS MARKET: APPLICATION SNAPSHOT	27
FIGURE 9	ADVANCED ENERGY STORAGE SYSTEMS MARKET: TECHNOLOGIES SNAPSHOT	28
FIGURE 10	ATTRACTIVE OPPORTUNITIES IN THE ADVANCED ENERGY STORAGE SYSTEMS MARKET	29
FIGURE 11	ASIA PACIFIC IS EXPECTED TO DOMINATE THE ADVANCED ENERGY STORAGE SYSTEMS MARKET IN 2017	30
FIGURE 12	THE TRANSPORTATION SEGMENT IS EXPECTED TO HOLD THE LARGEST SHARE DURING THE FORECAST PERIOD	31
FIGURE 13	ASIA PACIFIC TO DOMINATE THE MARKET FOR TRANSPORTATION APPLICATION OF ADVANCED ENERGY STORAGE SYSTEMS DURING THE FORECAST PERIOD	31
FIGURE 14	ELECTRO CHEMICAL TO DOMINATE THE MARKET FOR TECHNOLOGY IN THE ADVANCED ENERGY STORAGE SYSTEMS DURING THE FORECAST PERIOD	32
FIGURE 15	MARKET DYNAMICS: ADVANCED ENERGY STORAGE SYSTEMS MARKET	35
FIGURE 16	INVESTMENT IN RENEWABLE ENERGY SOURCES, 2004–2016 (USD BILLION)	36
FIGURE 17	TRANSPORTATION APPLICATION IS PROJECTED TO GROW RAPIDLY DURING 2017–2022	41
FIGURE 18	ASIA PACIFIC IS EXPECTED TO DRIVE THE GROWTH OF ADVANCED ENERGY STORAGE SYSTEMS IN THE TRANSPORTATION APPLICATION DURING 2017–2022	42
FIGURE 19	ASIA PACIFIC IS EXPECTED TO DRIVE THE GROWTH OF ADVANCED ENERGY STORAGE SYSTEMS IN THE GRID STORAGE APPLICATION DURING 2017–2022	44
FIGURE 20	ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET IS PROJECTED TO GROW RAPIDLY DURING 2017–2022	46
FIGURE 21	ASIA PACIFIC IS EXPECTED TO GROW RAPIDLY IN THE ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	47
FIGURE 22	LITHIUM-ION BATTERY IS EXPECTED TO GROW RAPIDLY IN THE ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	48
FIGURE 23	ASIA PACIFIC IS EXPECTED TO GROW RAPIDLY IN THE LITHIUM-ION BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	49
FIGURE 24	NORTH AMERICA IS EXPECTED TO HAVE THE HIGHEST SHARE IN THE LEAD ACID BATTERY ENERGY STORAGE MARKET DURING 2017–2022	50
FIGURE 25	EUROPE IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE NAS BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	51
FIGURE 26	ASIA PACIFIC IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE FLOW BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	53
FIGURE 27	ASIA PACIFIC IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE NIMH & NICD BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	54

FIGURE 28	EUROPE IS EXPECTED TO GROW RAPIDLY IN THE THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	55
FIGURE 29	SENSIBLE THERMAL STORAGE TECHNOLOGY IS EXPECTED TO GROW RAPIDLY DURING 2017–2022	56
FIGURE 30	REST OF THE WORLD IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE SENSIBLE HEAT THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	58
FIGURE 31	REST OF THE WORLD IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE LATENT HEAT THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	60
FIGURE 32	EUROPE IS EXPECTED TO GROW RAPIDLY IN THE MECHANICAL ENERGY ADVANCED STORAGE SYSTEMS MARKET DURING 2017–2022	61
FIGURE 33	PUMPED HYDRO STORAGE IS EXPECTED TO GROW RAPIDLY IN THE MECHANICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	62
FIGURE 34	EUROPE IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE PUMPED HYDRO ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	63
FIGURE 35	NORTH AMERICA IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE FLYWHEEL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	64
FIGURE 36	NORTH AMERICA IS EXPECTED TO HAVE THE LARGEST SHARE IN THE CAES & LAES ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022	65
FIGURE 37	NORTH AMERICA HAS THE LARGEST SHARE IN THE OTHER ADVANCED ENERGY STORAGE SYSTEMS MARKET	67
FIGURE 38	REGIONAL SNAPSHOT: SOUTH KOREA, CHINA AND INDIA ARE THE FASTEST GROWING COUNTRIES, BY VALUE, DURING THE FORECAST PERIOD	69
FIGURE 39	ADVANCED ENERGY STORAGE SYSTEMS MARKET SHARE (VALUE), BY REGION, 2017	69
FIGURE 40	ASIA PACIFIC, ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT	71
FIGURE 41	NORTH AMERICA, ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT	78
FIGURE 42	COMPANIES ADOPTED NEW PRODUCT LAUNCHES AND CONTRACTS AND AGREEMENTS AS KEY STRATEGIES TO CAPTURE THE MARKET, 2013– JULY 2017	94
FIGURE 43	MARKET EVALUATION FRAMEWORK: NEW PRODUCT LAUNCHES, MERGERS & ACQUISITIONS, AND CONTRACTS & AGREEMENTS HAVE FUELED THE GROWTH OF COMPANIES, 2013– JULY 2017	96
FIGURE 44	ADVANCED ENERGY STORAGE SYSTEM MARKET (GLOBAL): COMPETITIVE LEADERSHIP MAPPING, 2017	97
FIGURE 45	LG CHEM, LTD.: COMPANY SNAPSHOT	101
FIGURE 46	ABB LTD.: COMPANY SNAPSHOT	105
FIGURE 47	GS YUASA CORPORATION: COMPANY SNAPSHOT	108
FIGURE 48	SAMSUNG SDI CO., LTD.: COMPANY SNAPSHOT	112
FIGURE 49	GENERAL ELECTRIC COMPANY: COMPANY SNAPSHOT	116
FIGURE 50	TESLA, INC.: COMPANY SNAPSHOT	123
FIGURE 51	BYD COMPANY LIMITED: COMPANY SNAPSHOT	133
FIGURE 52	HITACHI, LTD.: COMPANY SNAPSHOT	137
FIGURE 53	SIEMENS AG: COMPANY SNAPSHOT	140
FIGURE 54	PANASONIC CORPORATION: COMPANY SNAPSHOT	144

LIST OF ABBREVIATIONS

Abbreviation	Full form
AESS	Advanced Energy Storage Systems
APAC	Asia Pacific
BEV	Battery Electric Vehicle
CAES	Compressed Air Energy Storage
CAGR	Compound Annual Growth Rate
CEO	Chief Executive Officer
DESS	Distributed Energy Storage Systems
DG	Distributed Generation
DoE	Department of Energy
EIA	Energy Information Administration
ESS	Energy Storage System
EV	Electric Vehicle
EVI	Electric Vehicle Initiative
GW	Giga Watt
HEV	Hybrid Electric Vehicle
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
KW	Kilo Watt
kWh	Kilowatt Hour
LAES	Liquid Air Energy Storage
MW	Mega Watt
NiCd	Nickel Cadmium
NiMH	Nickel Metal Hydride
PHEV	Plugin Hybrid Electric Vehicle
PHS	Pumped Hydro Storage
R&D	Research & Development
RES	Renewable Energy Sources
RoW	Rest of the World
RT	Revenue Tree
UK	United Kingdom
UPS	Uninterruptible Power Supply
US	United States
USD	United States Dollar
VP	Vice President
Wh/kg	Watt hour/ Kilogram

1 INTRODUCTION

1.1 OBJECTIVES OF THE STUDY

- To define and segment the global advanced energy storage systems market with respect to technology (including electro chemical, thermal storage, mechanical, and others), application, and region
- To estimate the market size, in terms of value, of the global advanced energy storage systems market
- To provide a detailed information regarding major factors influencing the market growth (drivers, restraints, opportunities, and industry-specific challenges)
- To strategically analyze micro markets with respect to individual growth trends, future prospects, and contribution to the total market covered by types of technology, application, and region
- To estimate the market size of the global advanced energy storage systems market in terms of the market value
- To study individual growth trends of the global advanced energy storage systems market, their future expansions, and analysis of their contribution to the market
- To strategically profile key market players and comprehensively analyze their market shares and core competencies
- To forecast the revenue of market segments with respect to four main regions, namely, North America, Europe, Asia Pacific, and the rest of the world.
- To track and analyze competitive developments, such as joint ventures, mergers & acquisitions, new product developments, awards and recognitions, and research and developments in the global advanced energy storage systems market
- To benchmark players within the market using proprietary Competitive Leadership Mapping which analyzes market players on various parameters within the broad categories of the business and product strategy

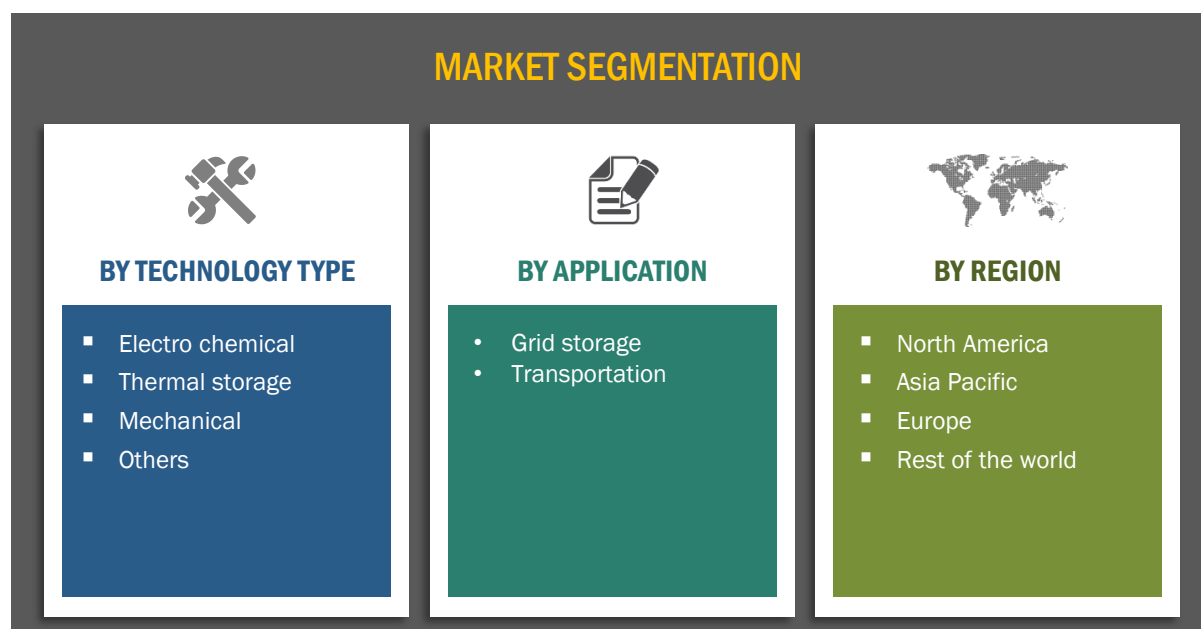
1.2 MARKET DEFINITION

Advanced Energy Storage Systems (AESS) involve energy storage systems having applications in the grid storage and transportation industry. Single Energy Storage System (ESS) units at the grid side, can be remotely controlled to manage its individual charge and discharge activities in response to the demand at the circuit, substation, or system level. These types of units are envisioned to support grid peak loads during summers and provide the backup support as needed. In the transportation industry, batteries are used to store energy in vehicles that can be used as standalone power supply to the vehicle or with other energy sources, such as in the case of Hybrid Electric Vehicles (HEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).

1.3 MARKET SCOPE

1.3.1 MARKETS COVERED

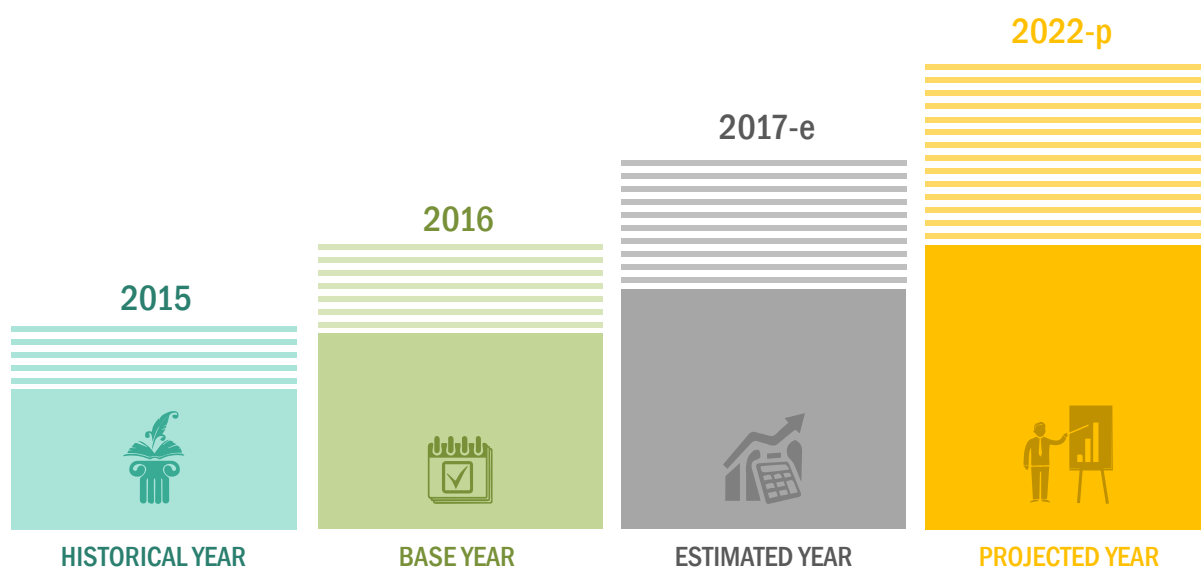
FIGURE 1 MARKETS COVERED: ADVANCED ENERGY STORAGE SYSTEMS MARKET SEGMENTATION



Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

Note: The Rest of the World includes Chile, South Africa, and other Middle Eastern, African, and South American countries

1.3.2 YEARS CONSIDERED FOR THE STUDY



1.4 CURRENCY

- For companies reporting revenues in USD, the data has been obtained from their annual report.
- For companies reporting revenue in other currencies, the average annual currency conversion rate has been used for a particular year to convert the value to USD.
- For the conversion of various currencies to USD, average historical exchange rates have been used according to the year specified.

1.5 LIMITATIONS

Some of the systems considered in advanced energy storage system technologies are in a very initial stage and the commercial data availability for such technologies is very limited. The transportation segment considered only includes BEV and PHEV. The quantitative information for some of the market segments is kept confidential by the players. Hence, the qualitative insights gathered during the course of the study have been used to arrive at the market size of such segments.

1.6 STAKEHOLDERS

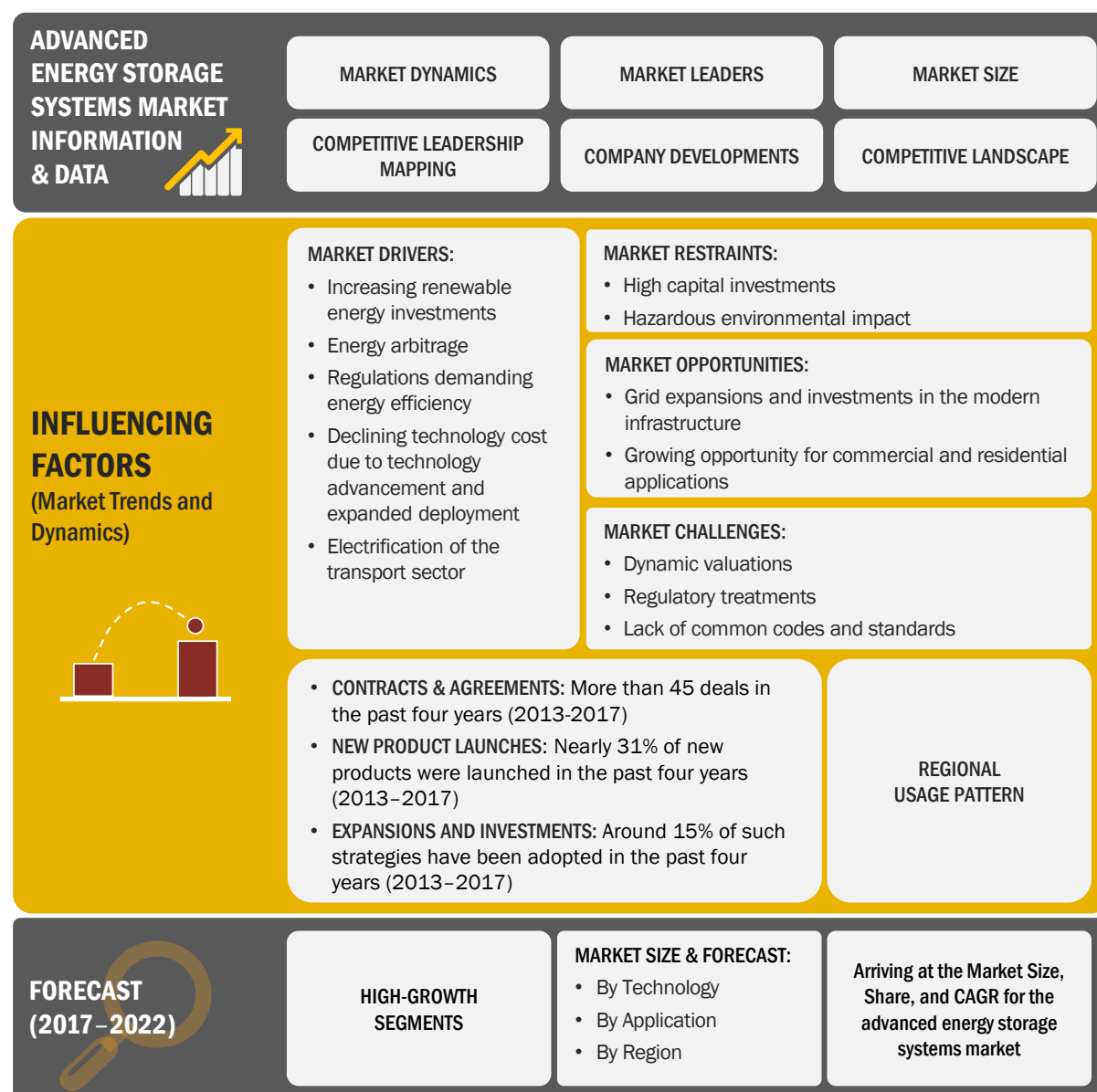
- Associations and industrial bodies
- End customers
- Energy storage research institutions
- Government and research organizations
- Grid operators
- Retail service providers
- Solid state battery, electrochemical capacitor, and other energy storage technology manufacturers
- State regulators
- Utility providers

2 RESEARCH METHODOLOGY

Extensive secondary sources, press releases, investment reports, industry white papers, company annual reports and presentations, publications, and other publicly available sources have been used to identify and collect information useful for this extensive, technical, market oriented, and commercial study of the global AESS market. Primary sources include several industry experts from the core and related industries and preferred service providers, administrators, solution providers, technology developers, alliances, standards, and regulatory organizations. All primary sources have been interviewed to obtain and verify critical qualitative and quantitative information as well as to assess future prospects.

The following illustrative figure shows the market research methodology applied in this report on the global advanced energy storage systems market.

FIGURE 2 ADVANCED ENERGY STORAGE SYSTEMS MARKET: RESEARCH DESIGN



2.1 RESEARCH DATA

2.1.1 SECONDARY DATA

Secondary sources include annual reports, press releases, and investor presentations of companies, white papers, certified publications, and articles by recognized authors, and various company and related association databases. Secondary research has mainly been used to obtain key information about the supply chain of the industry, the monetary chain of the market, the total pool of key players, the market classification and segmentation according to industry trends to the bottom-most level, regional markets, and key developments from both the market and technology-oriented perspectives.

PARAMETER	SOURCE
MARKET SIZE	• Company Financials • Magazines • Journals • Press Releases • Paid Databases • MarketsandMarkets Data Repository
REVENUE OF COMPANIES	• Annual Reports • Company Websites • Public Databases • MarketsandMarkets Data Repository
QUALITATIVE INFORMATION (Market Dynamics, Market Trends)	• Company Websites • Annual Reports • Press Releases • MarketsandMarkets Data Repository

2.1.2 PRIMARY DATA

Extensive primary research has been conducted after gaining knowledge about the global advanced energy storage systems market through secondary research. Various primary interviews (via questionnaires, e-mails, and telephonic interviews) were conducted with global advanced energy storage systems market experts, from both the supply- and demand- sides, to obtain the qualitative and quantitative information for this report. Primary sources from the supply-side include industry experts, such as CEOs, vice presidents, marketing directors, technology and innovation directors, and related key executives of various companies and organizations operating in the global advanced energy storage systems market.

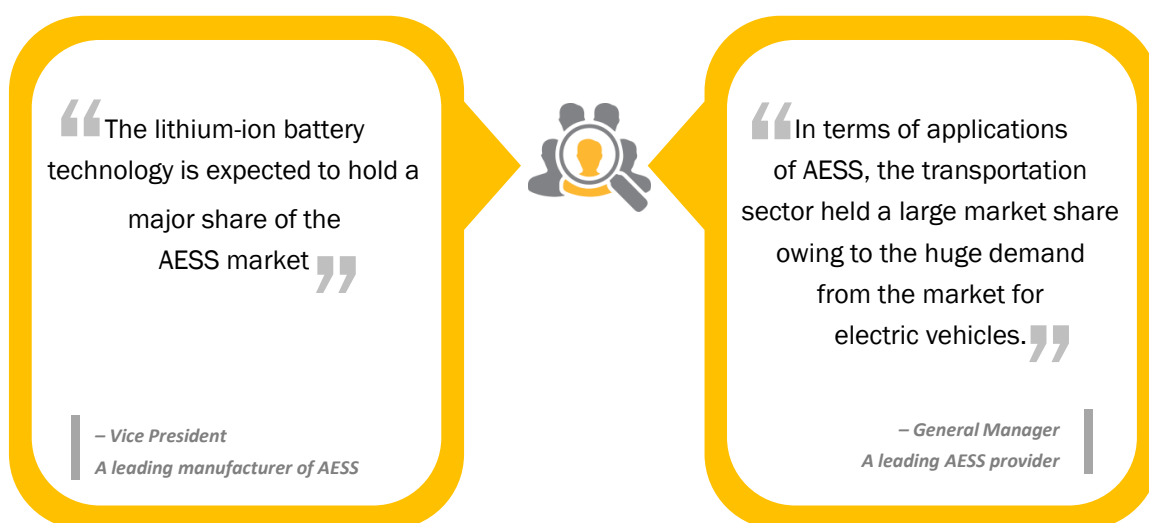
2.1.2.1 Key data from primary sources

TYPE	PARAMETER	KEY DATA
REGIONAL SPLIT	- Overall market and sub-segments in 2016 - CAGR of each region in the forecast period (2017–2022)	The global advanced energy storage systems market, by region: North America, Europe, Asia Pacific, and the rest of the world
GLOBAL MARKET SIZE	- The global advanced energy storage systems market size for the year 2016 - CAGR for the forecast period (2017– 2022)	- The global advanced energy storage systems market - The global advanced energy storage systems market by sub-segments
MARKET SPLIT	Technology: Electro chemical, thermo chemical, mechanical, and others Application: Grid storage and transportation Region: North America, Asia Pacific, Europe, and the rest of the world	- The global advanced energy storage systems market, by technology - The global advanced energy storage systems market, by application - The global advanced energy storage systems market, by region

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

Note: The rest of the world includes Chile, South Africa, and other Middle Eastern, African, and South American countries

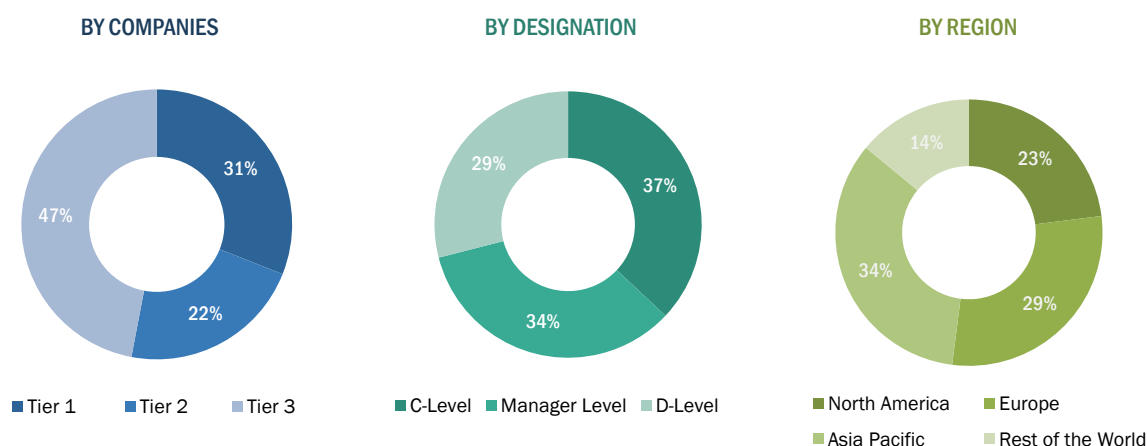
2.1.2.2 Key industry insights



2.1.2.3 Breakdown of primary interviews

After arriving at the overall market size from the estimation process explained above, the numbers were verified by primary respondents that include industry experts, such as CEOs, vice presidents, marketing directors, and related key executives. The figure given below shows the breakdown of the primary interviews by company type, designation, and region.

FIGURE 3 BREAKDOWN OF PRIMARY INTERVIEWS: BY COMPANY TYPE, DESIGNATION, & REGION



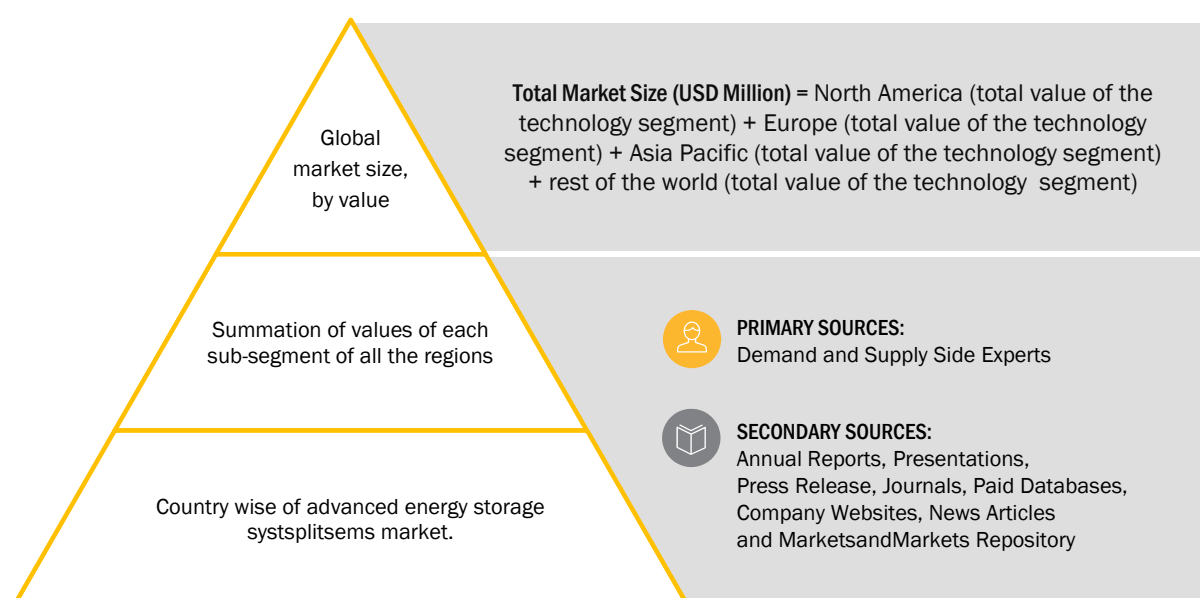
Note: RoW is the rest of the world that includes Middle Eastern, African, and South American countries.

The tier for companies has been defined on the basis of their total revenue, as of 2016: Tier 1=>USD 5 billion, Tier 2=from USD 1 billion to USD 5 billion, and Tier 3=<USD 1 billion

2.2 MARKET SIZE ESTIMATION

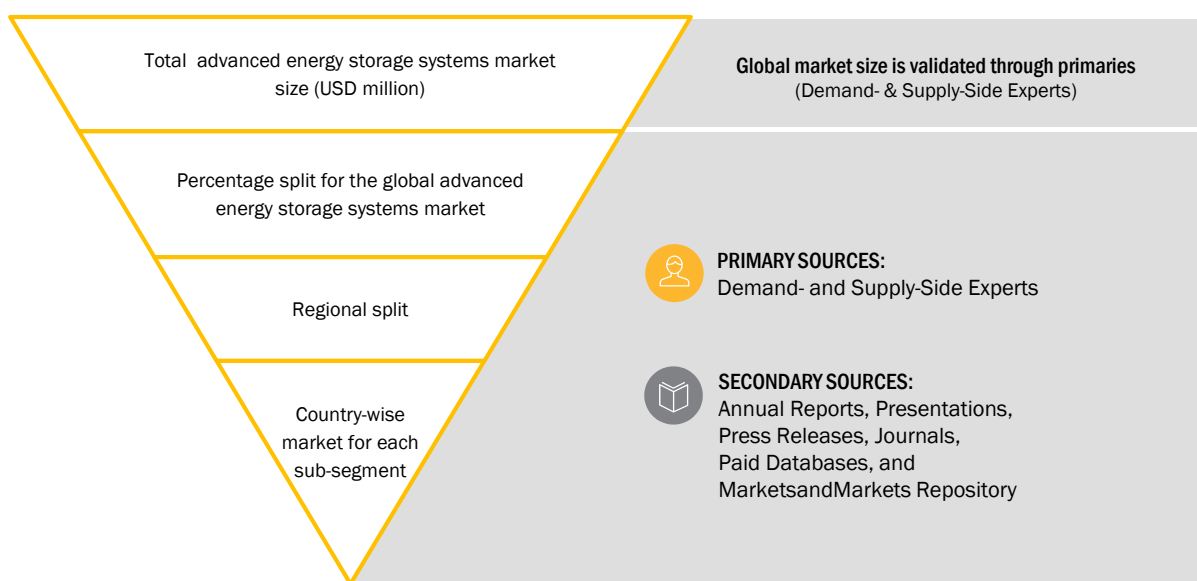
The top-down and bottom-up approaches have been used to estimate and validate the size of the global advanced energy storage systems market. Extensive qualitative and quantitative analyses have been performed during the complete market engineering process to list key information and insights throughout the report. The percentage shares, splits, current market size, and sales per company have been estimated using annual sales data taken from press releases of the global advanced energy storage systems supplying companies and associations, and are subsequently confirmed with primary respondents. Using secondary sources, such as directories, industry publications, and databases, the key players in the global advanced energy storage systems market are identified. The historic data has been collected from different energy associations and company websites. In this report, data triangulation has been conducted by using secondary and primary researches, as well as MarketsandMarkets analysis. The company-wise revenue of a particular region has been identified and summed-up to derive the total market size, which has been further split into sub-segments for the market size estimation.

FIGURE 4 MARKET SIZE ESTIMATION METHODOLOGY: BOTTOM-UP APPROACH



The bottom-up approach has been applied to arrive at the size of the global advanced energy storage systems market. This approach is employed on an authentic data to validate the obtained market segment, in terms of value. The advanced energy storage systems market at the global level is calculated by estimating the total deployment of the advanced energy storage systems, based on the adoption by the grid and the transportation segment at the regional and country level. Thus, by combining the global market size (in terms of value) of each specific category, the global advanced energy storage systems market has been derived. .

Subsequently, market shares have been estimated for each company to verify the revenue shares used earlier in the bottom-up process. With the data triangulation process and the validation of data through primaries, the exact values of the overall parent and individual market sizes have been determined and confirmed in this study.

FIGURE 5 MARKET SIZE ESTIMATION METHODOLOGY: TOP-DOWN APPROACH

In the top-down approach, the overall market size (in terms of value) has been used to estimate the size of individual markets (mentioned in the market segmentation) through percentage splits from secondary and primary research. For the calculation of specific market segments and the immediate parent market, the global market size has been used to implement the top-down approach.

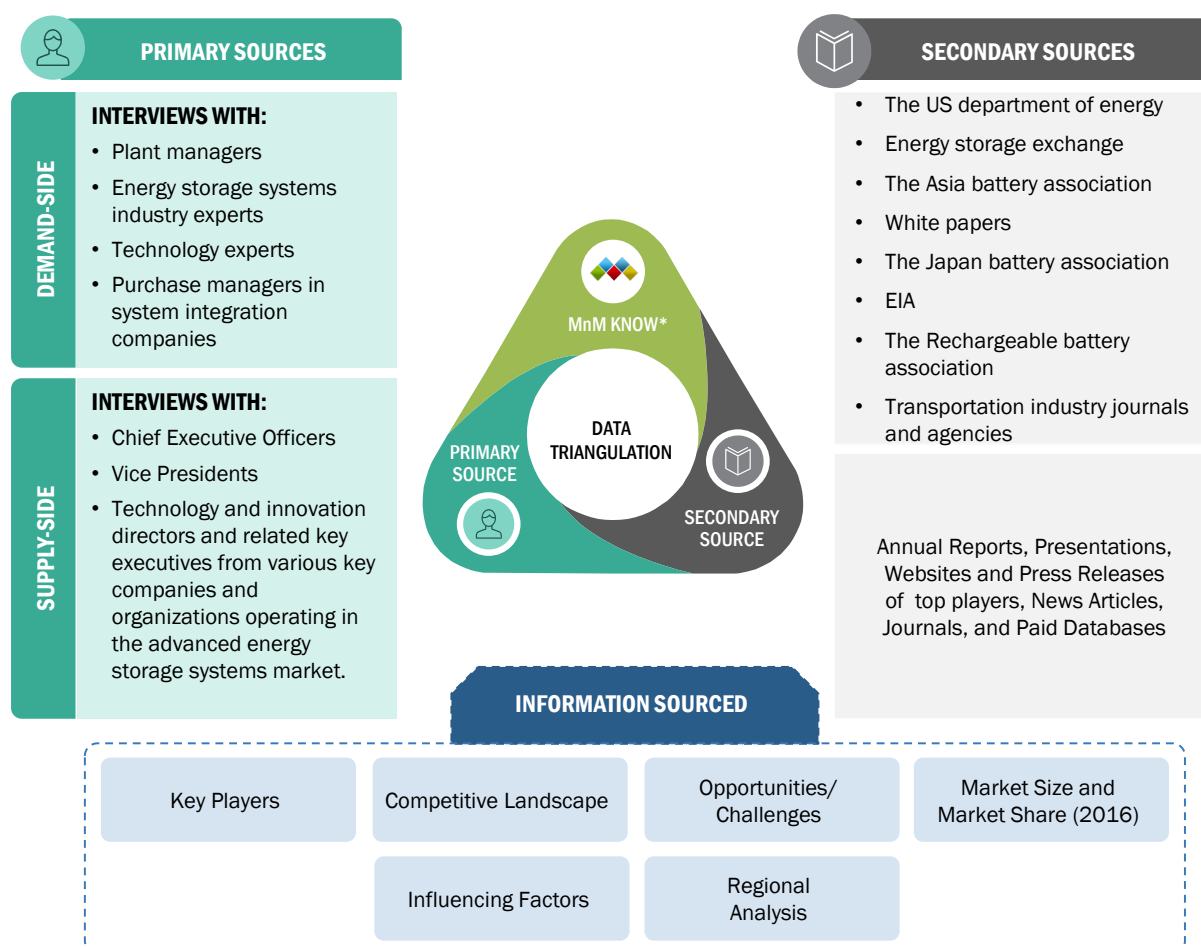
Market shares have then been estimated for each company to verify the revenue shares used earlier in the bottom-up approach. With the data triangulation process and the validation of data through primaries, the overall parent and each individual market sizes have been determined and confirmed in this study. The data triangulation process used for this study has been explained in the next section.

2.3 MARKET BREAKDOWN & DATA TRIANGULATION

The total market has been split into different segments and sub-segments, after arriving at the overall market size from the market size estimation process explained above. The data triangulation and market breakdown processes have been employed, wherever applicable, to complete the overall market engineering process and to arrive at the exact statistics for all segments and sub-segments. The data has been triangulated by studying various factors and trends from both the demand- and supply-sides. Along with this, the market size has been validated using both the top-down and bottom-up approaches.

The figure given below shows the market breakdown structure and the data triangulation methodology implemented in making this report.

FIGURE 6 DATA TRIANGULATION METHODOLOGY



MnM KNOW* stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "RT", subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

2.4 RESEARCH ASSUMPTIONS

PARAMETER	ASSUMPTION
PRICE TRENDS	The global average of different advanced energy storage system prices, has been assumed at a prevalent market cost. The price of lithium-ion batteries are considered to decline at a rapid rate over the forecast period.
FUTURE INVESTMENT TRENDS	The market growth over the forecast period is assumed, considering the capital expenditures in the renewable and non- renewable energy generation and transportation projects, involving energy storage systems. Many ongoing, planned and announced new projects, and capacity expansions, have also been considered for the regional market growth forecast.
FUTURE INFLATION TRENDS	Inflation in the price of advanced energy storage systems has not been considered during the forecast period. The price fluctuation based on advancement in technology, especially in the case of battery storage has been considered in the future forecast.
EXCHANGE RATES & CURRENCY VALUE	• For conversion of various currencies to the U.S. Dollar, the average historical exchange rates for that particular year has been utilized. • All the forecasts for market size have been undertaken under the standard assumption, that the value of the U.S. Dollar remains constant over the next five years.
INDUSTRY	The industry segments considered in this report are aimed to give a holistic view of the entire advanced energy storage systems market. Primaries with various industry experts confirmed their assumption.

3 EXECUTIVE SUMMARY

Advanced energy storage technologies capture the energy as it is produced and provide it when it is needed. These systems offer greater reliability and control over the usage of energy during peak demands. It is gaining importance day-by-day, due to increasing and fluctuating energy consumption trends and the emphasis on the use of clean energy. This has played a critical role in increasing the number of Distributed Generation (DG) technologies and Renewable Energy Sources (RES), which primarily uses various advanced energy storage systems in order to store the excess generated power. The use of such advanced energy storage systems makes DG or RES as reliable as the traditional source of energy.

Advanced energy storage is a growing technology for portable electronics explosion. In less than 40 years, technologies, such as lithium and NiMH battery have taken over the electronic industry and also growing in the transportation industry and the utility grid. Advanced energy storage systems offer greater reliability and control over the usage of energy during peak demands. Advanced energy storage conjunction capabilities with the smart grid are expected to increase in the next 25 years. Energy storage is used to store electricity which can be utilized during power shortages, blackouts, and peak hours of demand. Advanced energy storage is gaining importance due to increasing and fluctuating energy consumption trends and the emphasis on the use of clean energy.

There are various types of AESS, such as Electro Chemical which includes lithium-ion battery, sodium sulfur (NaS) battery, lead acid battery, flow battery, and nickel metal hydride and nickel cadmium batteries; Thermal Energy Storage Technology that includes thermo chemical, latent thermal, and sensible thermal; Mechanical Energy Storage Technology comprising of pumped hydro, flywheels, and Compressed Air Energy Storage as well as Liquid Air Energy Storage (CAES and LAES); and others which includes hydrogen storage, synthetic natural gas, and supercapacitors.

The electro chemical technology segment is estimated to account for the largest market share in 2017 and is also estimated to grow at the highest rate. The global AESS market is estimated to be USD 12.73 billion in 2017, and is projected to reach USD 19.04 billion by 2022, at a CAGR of 8.38% from 2017 to 2022. The transportation segment is expected to hold the largest market share of 65.8% in the global AESS market in 2017. Its market size is projected to reach USD 13.26 billion by 2022, at a CAGR of 9.63% during the forecast period. The electro chemical technology segment is expected to account for the largest share of 76.5% in the global AESS market in 2017. It is projected to reach a market size of USD 15.32 billion by 2022, at a CAGR of 9.49% during the forecast period.

The market in Asia Pacific is projected to account for the maximum share in the global AESS market with a size of USD 7.52 billion by 2022, at a CAGR of 9.79% during the forecast period. Among the countries in Asia Pacific, the market in South Korea is expected to grow at the highest CAGR of 14.44% from 2017 to 2022, due to the growing transportation and grid storage sector in the country.

The major players in the AESS market are LG Chem, Ltd. (South Korea), Samsung SDI Co., Ltd. (South Korea), GS Yuasa Corporation (Japan), ABB Limited (Switzerland), BYD Company Limited (China), and Tesla, Inc. (US) among others.

TABLE 1 ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT

Particulars	2017 (E)	2022 (P)
Global Value (USD Billion)	12.73	19.04
Regional Share (%)	- Asia Pacific: 37.0 - North America: 27.3 - Europe: 24.2 - Rest of the World: 11.4	- Asia Pacific: 39.5 - North America: 26.4 - Europe: 24.4 - Rest of the World: 9.7
Application Share (%)	- Transportation: 65.8 - Grid storage: 34.2	- Transportation: 69.7 - Grid storage: 30.3
Technology Share (%)	- Electro chemical: 76.5 - Mechanical: 18.4 - Thermal: 4.9 - Others: 0.2	- Electro chemical: 80.4 - Mechanical: 15.3 - Thermal: 4.1 - Others: 0.2
Global Market CAGR (2017–2022)	8.38%	

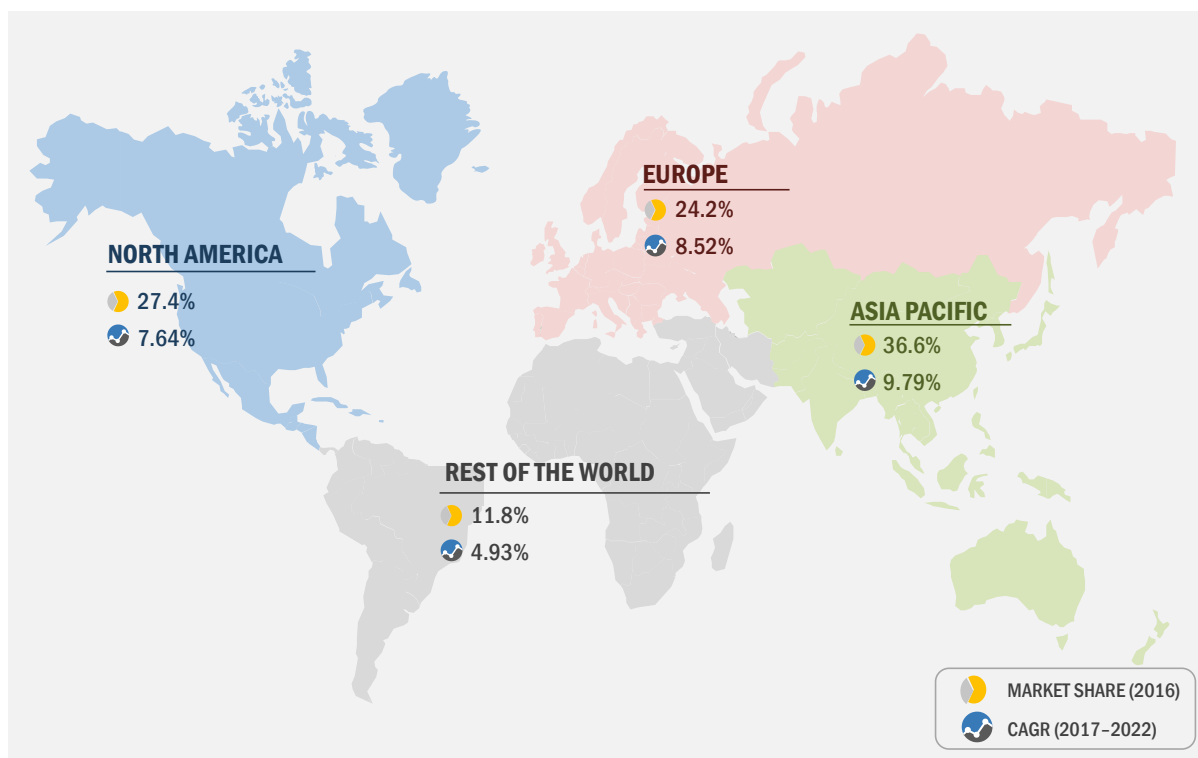
E- Estimated, P- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

In 2016, the market in Asia Pacific held the largest share in the AESS market, followed by the markets in North America and Europe. In terms of countries, the AESS market is dominated by the market in the US, because of increasing adoption on lithium-ion batteries in the country. The market in South Korea is projected to have the highest CAGR during the next five years, given the increase in number of EVs in the country.

FIGURE 7 ASIA PACIFIC DOMINATED THE ADVANCED ENERGY STORAGE SYSTEMS MARKET IN 2016

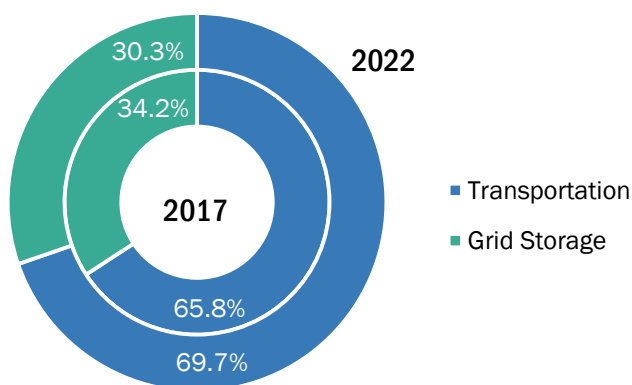


Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

Renewable energy installations and growing EV market in developed and developing countries, are primary reasons for the growth of the AESS market.

FIGURE 8 ADVANCED ENERGY STORAGE SYSTEMS MARKET: APPLICATION SNAPSHOT

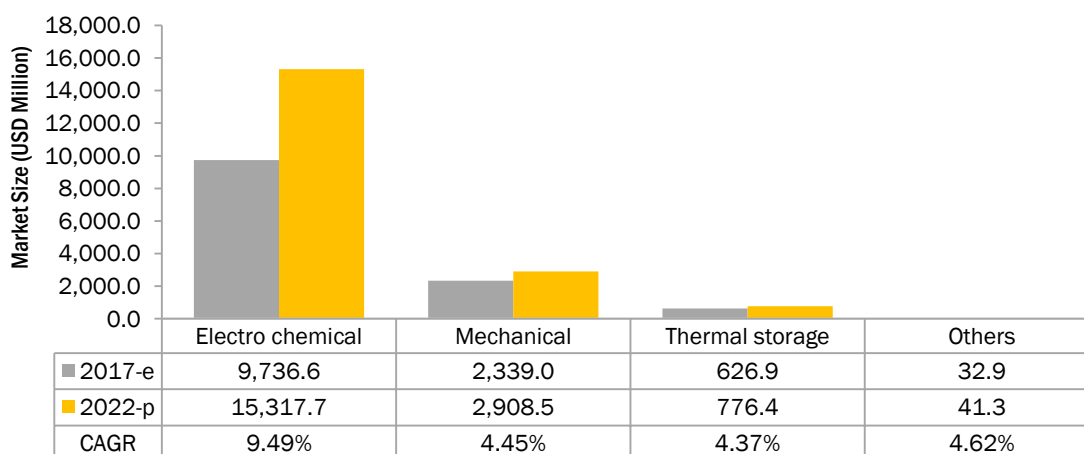


Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The inner circle in the above figure indicates the market share by application in 2017, while the outer circle depicts the share in 2022.

The transportation segment is expected to lead the market with a share of 65.8% in 2017. This segment considers batteries used in BEVs and PHEVs. The grid storage application considers technologies used to store energy at utility scales, such as lithium-ion batteries, pumped hydro, battery, CAES, flywheels, thermal storage, and others.

FIGURE 9 ADVANCED ENERGY STORAGE SYSTEMS MARKET: TECHNOLOGIES SNAPSHOT



Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

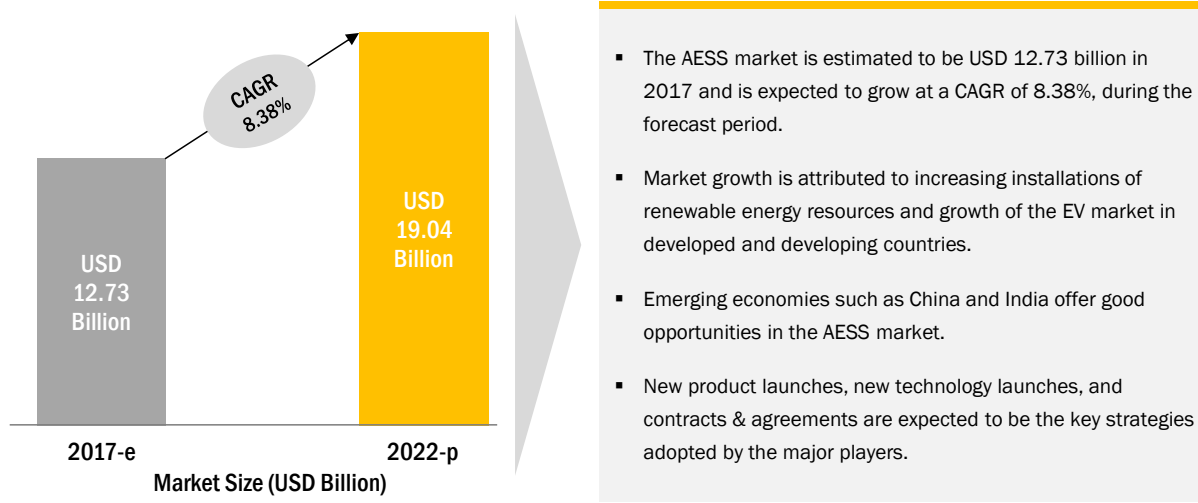
Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The electro chemical segment is expected to lead the market with a share of 76.5% in 2017. This segment was worth USD 8.92 billion in 2016 and is expected to reach USD 15.32 billion by 2022, at a CAGR of 9.49% during 2017 to 2022. The growth in this segment was majorly driven by the increase in the lithium-ion battery sub-segment, in the market in Asia Pacific. Various features offered by lithium-ion, such as higher cell voltage, requirement of less cells for the same power output, short response time, and high life expectancy have made this technology to sustain holding a large share in the electro chemical storage technology segment. The increasing applications in residential, utility, and EV sectors have supported the growth of this technology.

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE OPPORTUNITIES IN THE ADVANCED ENERGY STORAGE SYSTEMS MARKET, 2017–2022

FIGURE 10 ATTRACTIVE OPPORTUNITIES IN THE ADVANCED ENERGY STORAGE SYSTEMS MARKET



e- estimated, p- projected

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.2 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION, 2017-2022

FIGURE 11 ASIA PACIFIC IS EXPECTED TO DOMINATE THE ADVANCED ENERGY STORAGE SYSTEMS MARKET IN 2017

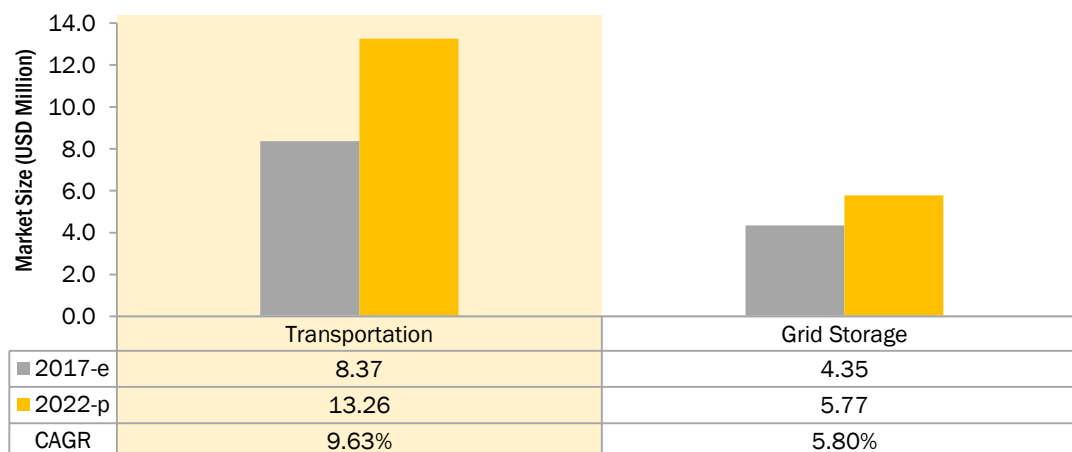


Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.3 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2017 & 2022

FIGURE 12 THE TRANSPORTATION SEGMENT IS EXPECTED TO HOLD THE LARGEST SHARE DURING THE FORECAST PERIOD

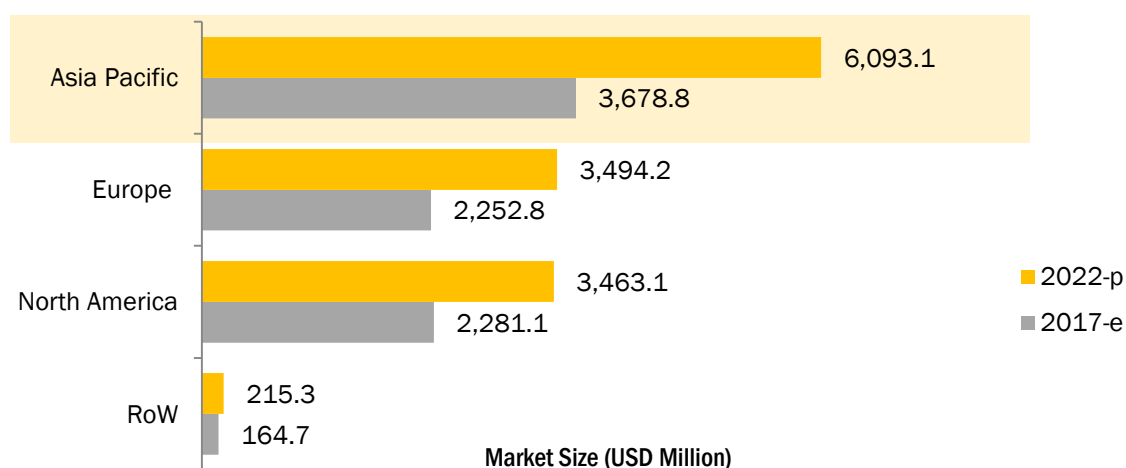


Note: e- Estimated, p- Projected

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.4 TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION, 2017 & 2022

FIGURE 13 ASIA PACIFIC TO DOMINATE THE MARKET FOR TRANSPORTATION APPLICATION OF ADVANCED ENERGY STORAGE SYSTEMS DURING THE FORECAST PERIOD



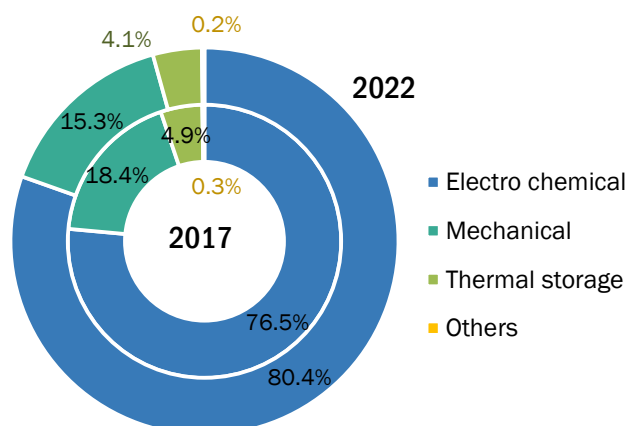
e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.5 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2017 & 2022

FIGURE 14 ELECTRO CHEMICAL TO DOMINATE THE MARKET FOR TECHNOLOGY IN THE ADVANCED ENERGY STORAGE SYSTEMS DURING THE FORECAST PERIOD



e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

KEY FINDINGS

- Increasing investments in renewables and regulations demanding energy efficiency are the key drivers for the AESS market.
- The application in EVs is expected to drive the innovation and growth in this market.
- High capital investments and hazardous environmental impact of certain technologies, acts as key restraints for the growth of the AESS market.
- The requirement of energy storage in commercial and residential applications is expected to create huge opportunities in this market.
- Regulatory treatment, lack of common codes and standards, and dynamic valuation are the major challenges in the AESS market.

5.1 INTRODUCTION

Advanced Energy Storage Systems (AESS) are used to satisfy the intermittent, as well as unpredictable increase in energy demand. They also help in maintaining the power quality, distribution reliability, energy management, and improvement of grid efficiency. The key roles played by AESS in power grids include time shifting to manage peak loads, providing power quality by aiding in frequency regulations, mitigating power congestion on grids, and supplying power uniformly in the distributed generation. They are also used as a primary power source in EVs.

In the field of different existing energy storage technologies, AESS is becoming increasingly important to ensure uninterrupted power supply during power shortages. The future growth opportunity, is likely to arise from its applications, such as the grid storage and transport sector. Currently, countries that generate large amount of renewable energy are required to effectively store this energy for future usage. As a result, large renewable energy generating countries, such as the US, Germany, Japan, India, and China are expected to be the major markets for AESS.

Furthermore, growing investments in renewable sources and increasing demand for energy efficiency along with electrification of the transportation sector are the key drivers for the growth in the AESS market. However, lack of common codes and procedures and not regulatory framework are expected to be the major hindrance for growth in AESS market.

5.2 TECHNOLOGY OVERVIEW

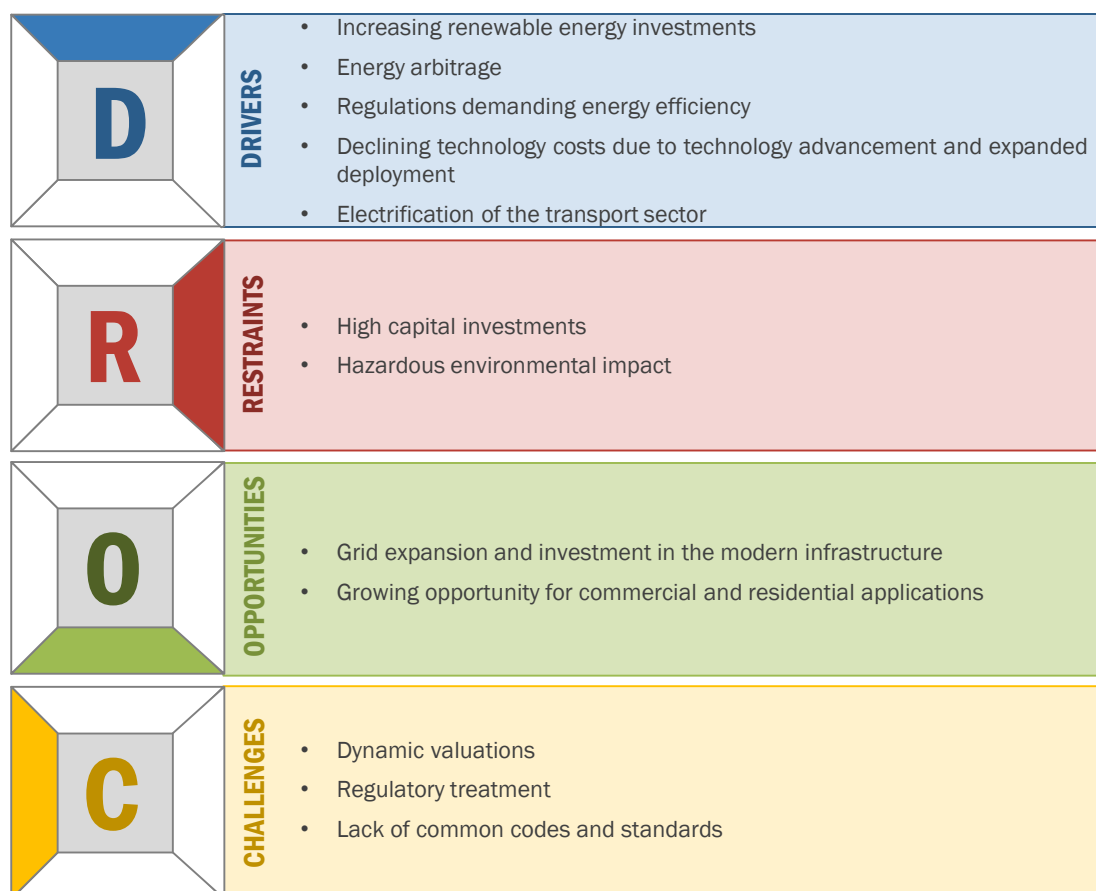
TABLE 2 COMPARISON OF MAJOR ENERGY STORAGE TECHNOLOGIES BY TECHNICAL CHARACTERISTICS

Technology	Power Rating (MW)	Discharge time	Energy density (Wh/l)	Power density (W/l)	Efficiency	Response time
Pumped Hydro	100–2500	4–16 h	0.2–2	0.1–0.2	70–85%	10 sec–min
Compressed Air	10–1000	2–30 h	2–6	0.2–0.6	40–70%	min
Flywheels	0.001–20	sec–min	20–80	5000	70–95%	<sec
Lithium-ion battery	0.05–100	1 min–8 h	200–400	1300–10000	85–95%	<sec
Lead acid battery	0.001–100	1 min–8 h	50–80	90–700	80–90%	<sec
NaS battery	10–100	1 min–8 h	150–300	120–160	70–90%	<sec
Flow battery	0.1–100	hours	20–70	0.5–2	60–85%	<sec
Supercapacitor	0.01–1	ms–min	10–20	40000–120000	80–95%	<sec
Hydrogen storage	0.01–100	min–week	600 (200 bar)	0.2–20	25–45%	sec–min
Synthetic natural gas	1–100	hour–week	1800 (200 bar)	0.2–2	25–50%	sec–min
Molten salt (Latent thermal)	1–150	hours	70–210	n/a	80–90%	min

Source: World Energy Council, Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

5.3 MARKET DYNAMICS

FIGURE 15 MARKET DYNAMICS: ADVANCED ENERGY STORAGE SYSTEMS MARKET



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

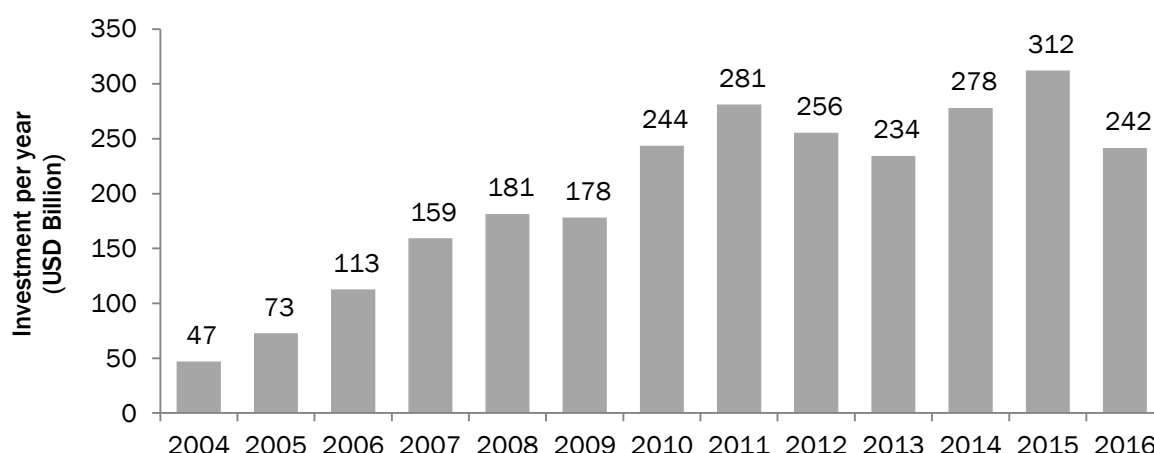
5.3.1 DRIVERS

5.3.1.1 Increasing renewable energy investments

The developed as well as developing countries are revising energy policies to include a considerable portion of generation, to be sourced from renewable energy resources that includes wind and solar. Financial incentives by government bodies and associated returns are increasing investments in renewable energy, which are expected to boost the AEES market.

By the end of 2015, almost 173 countries established renewable energy targets at the provincial, state, or national level. In addition, about 146 countries had renewable energy support policies by the end of 2015. Several countries held auctions to deploy renewables in a structured but in a cost efficient manner. As per International Renewable Energy Agency (IRENA), at least 67 countries held such auctions in 2016 as compared to just six in 2005.

Renewables have now become the first choice for expanding, upgrading, and modernizing power systems around the world. According to Bloomberg New energy finance wind and solar energy accounted for 90% of the 2015 investments in renewable power, as the cost of wind turbines has fallen by nearly a third, since 2009 and that of solar photovoltaic modules fell by 80%. It is reported that, currently the cost of onshore wind, biomass, hydro power, and geothermal are competitive with coal, oil, and gas powered power plants.

FIGURE 16 INVESTMENT IN RENEWABLE ENERGY SOURCES, 2004–2016 (USD BILLION)

Source: Bloomberg New Energy Finance, UNEP, and Other Secondary Literature

Renewable investments have been increasing, with the highest investment in 2015 during the period between 2004 and 2016. Several upcoming renewable energy markets in the developing countries reordered huge investments in 2015, however there was a sharp fall in 2016 in response to delays in the auction schedules and also because of lower dollar-denominated costs.

According to the Bloomberg study, the market in China was the leading investor in renewable energy sources in 2015. The country invested USD 108 billion, followed by the US with USD 51 billion invested towards clean energy. However, there was a fall in investment in renewables in 2016. Decline in average dollar capital expenditure per MW by more than 10% for solar photovoltaics, onshore wind and offshore wind, along with slowdown in finances in China, Japan, and some emerging markets during the course of the year resulted in low investments for renewable resources.

5.3.1.2 Energy Arbitrage

Conventional energy sources, such as coal and natural gas generate power at a definite capacity factor and have a constant supply. Non-conventional sources, such as wind and solar power, on the other hand, have low capacity factor and an intermittent supply. Energy storage technologies help to store energy during low demand so that it can be supplied during peak demand, thus benefitting the power producing companies from electricity price difference. This benefit of energy arbitrage is driving the power producers to invest in AESS.

5.3.1.3 Regulations demanding energy efficiency

Energy storage offer multiple benefits to the power grid, as well as compliments variable renewable energy resources, such as solar and wind. The government support in various countries is likely to act as a key driver for the growth of AESS all over the world. Major countries implementing battery energy storage are the US, Japan, China, and Germany, whereas other countries including Italy and South Korea are also following increasing installations of AESS. The increased variable renewable energy sources, improved system flexibility, and technology advancement factors are pushing governments to install AESS in their respective countries. In the US, AESS installations are driven by the country's 2009 federal stimulus package, the American Recovery and Reinvestment Act (ARRA).

The regulatory changes are helping to integrate and add value-based services provided by the energy storage. The US government focuses on grid reliability issues in parts of the country, state-level storage mandates, and renewable support programs. Moreover, few of the states in the US are focusing on battery storage and also have been introducing incentives to boost this market. Furthermore, in Japan, the

government offers subsidies, thereby increasing the solar self-consumption. Japan's Ministry of Economy, Trade, and Industry, in March 2014, announced a lithium-ion battery subsidy program worth USD 100 million. Additional subsidies also include support for stand-alone renewable energy generation with batteries supporting the growth of the AESS market.

5.3.1.4 Declining technology cost due to technology advancement and expanded deployment

The ongoing R&D work in the field of energy storage focusses on realizing technology cost reductions along with improving the performance of existing and emerging technologies.

It is expected that rapid technical improvements, such as increased discharge cycle for batteries along with other technologies, will be principally aimed at cost reduction. Furthermore, the expectation for falling battery costs is based on the construction of new large scale lithium-ion battery manufacturing plants, mainly in China. It is expected that these new manufacturing facilities would lead to economies of scale which would lower the costs of the battery.

According to the IRENA, the price for the lithium-ion battery reached USD 350/kWh in 2015 as compared to USD 1000/kWh in 2010, with a decline of 65%. Further, it is expected that the price of the lithium-ion battery will fall further over the next decade.

5.3.1.5 Electrification of the transport sector

The growing implementation of batteries in EV, Hybrid Vehicles (HV), and PHEVs is expected to support the growth of the AESS market. EVs are vehicles running with the support of batteries, whereas hybrid vehicles combine an electric engine and battery which can be charged by an internal combustion engine. Plug-in EV run entirely on electricity, powered by rechargeable batteries. They have many advantages over conventional cars, such as they are more efficient, emit less air pollutants, are noiseless with stronger acceleration, and require less maintenance. Moreover, according to the IEA Global EV Outlook (2017), by 2025, there would be around 40 million to 70 million Electric cars on road, compared to more than 2 million in 2016.

To achieve sustainable development with low carbon footprint and low emission of greenhouse gases, it is expected that EVs would aid in energy security and lesser greenhouse gas emissions. Furthermore, with increasing research and development in hybrid and electric cars, coupled with reducing battery costs and increasing efficiencies, the future demand for these vehicles is expected to increase, driving the demand for storage battery devices. Thus, the growth in the EV market is expected to boost this market.

5.3.2 RESTRAINTS

5.3.2.1 High capital investments

Energy storage technologies such as lithium-ion batteries, flow batteries, and sodium sulfur (NaS) batteries require high capital for production as compared to the other technologies due to their benefits of high energy density and improved performance. Lithium, being an expensive metal, increases the cost of battery production; therefore, with its emerging and evolving applications in EV & grid storage for the initial two or three years, the cost of technology is expected to be high; with the increasing adoption, the cost will decrease. The total cost of technology also increases with rated power. High capital investments associated with AESS installations restrain the usage of these technologies by customers such as captive power customers and residential customers among others. Hence, the impact of this factor is expected to be there for next 3-4 years.

5.3.2.2 Hazardous environmental impact

Batteries, superconducting magnets, and flywheels operate with hazardous materials, such as lead, fluoride, rare-earth magnets, and other materials, such as copper and aluminum that require safe disposal for recycling. These materials can harm the ecosystem if not disposed safely. Further, an electric field from the batteries is of low magnitude and causes stimulation of nerves or muscles in humans. Substantial exposure to such materials or fields can cause biological disorders, affecting the human health. Hence, it is considered as a restraining factor to the use of AESS technology.

5.3.3 OPPORTUNITIES

5.3.3.1 Grid expansion and investment in the modern infrastructure

The developing countries, such as India and China require modernization of the aging infrastructure and grid expansion. Currently, there is a need to bring power to an estimated 30% of the rapidly growing global population without access to electricity.

Further, according to the United Nations Sustainable Energy for All initiative (SE4All), around USD 45 billion investment is required by 2030, to provide universal access to the modern electric power. Advanced energy storage is expected to play a prominent role in these investments by strengthening the grid against diverse threats, such as natural disasters and facilitating improved deployment of both new and existing resources.

5.3.3.2 Growing opportunity for commercial and residential applications

As the private customers are looking to mitigate energy costs, residential storage solutions are being introduced to them. Further, more commercial properties are incorporating energy storage into their building plans and are also being encouraged to do so, which can be reflected in current trends, such as zero energy buildings. As there is a growing concern for pollution, consumers today are attracted to invest in companies that causes the minimum pollution.

Additionally, energy storage systems can become a medium for Energy Management Systems (EMS) in homes and buildings. With a Home Energy Management System in place, residential customers are expected to actively being involved, by monitoring their actual consumption and accordingly modifying their energy spending patterns in real time. The success of these types of EMS would require energy storage systems to store electricity from local generation when it is not needed and discharge it when necessary, therefore allowing the EMS to function optimally with less power needed from the grid.

The acceptance and interest in aggregation of behind-the-meter assets is also on rise. Furthermore, several US utilities have already partnered with energy storage system vendors to leverage a fleet of residential energy storage systems to provide grid services. For example, in 2016, Green Mountain Power began shipping Tesla Powerwalls, offering customers an option to allow utility access to the systems in exchange for the electricity bill credit.

5.3.4 CHALLENGES

5.3.4.1 Dynamic valuations

The value of energy storage can be ascertained by the difference between light load hour and heavy load hour energy prices in the wholesale energy market. Decreased natural gas prices and the introduction of low variable cost resources, such as solar and wind, depressed the market prices generally. It may be noted that market prices will not reflect the value of resource additions until major shortages were to occur, such as in case of unforeseen events of market disrupt ability. There is a need to devise a policy or regulation to analyze energy storage valuation to propel growth in the energy storage market.

Further, the current market structure does not allow for a buyer of energy storage equipment to easily capture the value streams, as there are number of services and benefits that the energy storage systems provide cannot be easily monetized. Some of these benefits are reduced incidence of service interruption, increased system reliability, and increased efficiency.

5.3.4.2 Regulatory treatment

The regulatory mechanism for energy storage faces different challenges in traditional vertically integrated utility structures and in the new restructured markets. In market storage, investments are recovered as part of the least cost plan. This methodology is difficult to carry out using standard capacity expansion tools due to the limitation of capturing full value of storage system. Energy storage can also be considered as an alternative to transmission line expansion to avoid congestion. There exists a need for communication between generation expansion and transmission planning functions, which may be further mandated by regulatory bodies.

Further, development risk increases with uncertainties and lack of ability to rely on regulatory structures and markets to reflect the value of energy storage systems. Uncertainties, such as reliance on market prices in nascent markets and volatile arbitrage values, discourage utility planners, management, and third-party developers to pursue the energy storage system project from regulatory commissions.

In addition, utilities in some areas, actively worked to prevent customers from using behind the meter storage to consume more of the electricity they generate on-site. Some of these efforts include enactment of regulations prohibiting third-party system ownerships.

5.3.4.3 Lack of common codes and standards

Currently, there are no industry-wide common codes and standards developed for energy storage devices. This can lead to compatibility issues, driven by each manufacturer developing its own internal standards. Furthermore, unique or custom-made spare parts deployed in different technologies, manufactured by different companies, and deployed in technologies can bring down the effects of economies of scale for the industry.

In addition, the growth in energy storage technologies is uneven and irregular across different states in a same country due to lack of legal uniformity. For example, in the US, state governments are inconsistent in their renewable energy laws, and the federal government has been inconsistent to implement nationwide uniform regulations. As a result, energy storage is booming in certain states, while crawling in other states.

6 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION

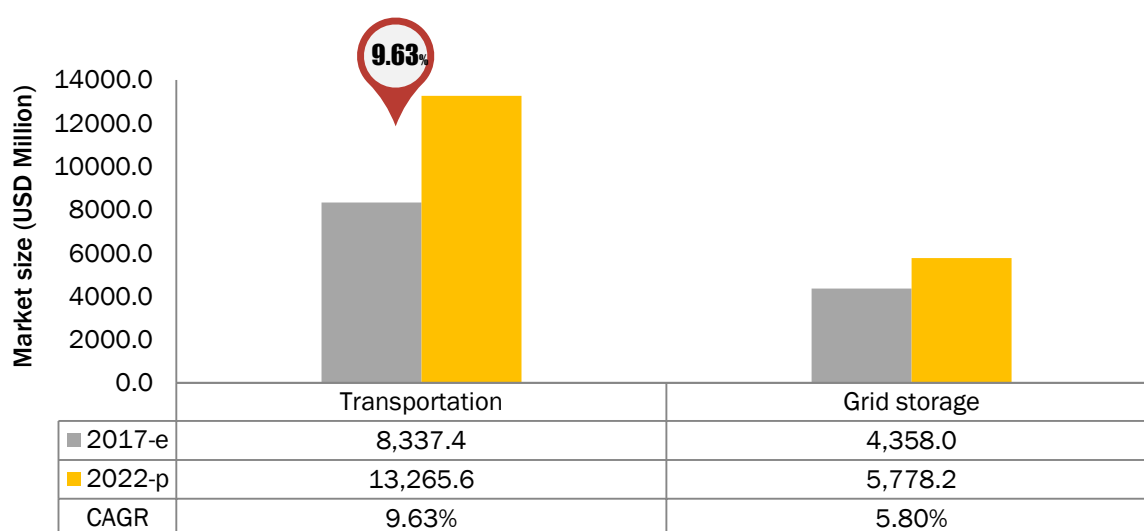
KEY FINDINGS

- The global AESS market is segmented into transportation and grid storage, based on application.
- The market size of the transportation segment is expected to grow at a CAGR of 9.63%, which is majorly driven by the growth in the market in the Asia Pacific region.
- The market in Asia Pacific is expected to dominate the transportation application with estimated 43.9% share in 2017.
- Renewable policies and energy arbitrage are driving the growth of grid storage in the AESS market.
- The grid storage application segment is expected to grow at a CAGR of 5.80% during the forecast period.

6.1 INTRODUCTION

The application of AESS technology considered in this report can be categorized into transportation and grid storage. Among the two mentioned applications, the transportation segment is estimated to have the largest market share of 65.78% in 2017 and is also the fastest growing application at a CAGR of 9.63% during the forecast period.

FIGURE 17 TRANSPORTATION APPLICATION IS PROJECTED TO GROW RAPIDLY DURING 2017–2022



e- estimated, p- projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 3 ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	7,040.8	7,668.5	8,377.4	13,265.6	9.63%
Grid storage	3,877.2	4,123.5	4,358.0	5,778.2	5.80%
Total	10,918.0	11,792.0	12,735.4	19,043.8	8.38%

e- Estimated; p- Projected

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The transportation application is expected to dominate the market with 65.7% share in 2017 and is projected to increase its share to 69.6% by 2022. Growth in the EV market is driving the growth of the transportation application in the AESS market.

In 2016, lithium-ion and nickel metal hydride (NiMH) were the most commonly used batteries in transportation. Battery-powered vehicles available today, includes hybrid vehicles, plug-in hybrid vehicles, and EV. The cost of lithium-ion batteries are expected to decline due to the expansion of battery manufacturing capacity by some of the big manufacturers, such as LG Chem, Ltd. (South Korea), BYD Company Limited (China), Samsung SDI Co., Ltd. (South Korea), Tesla, Inc. (US), and other companies.

Furthermore, with the ongoing R&D activities carried out by companies, such as Honda Motor Company, Ltd. (Japan), Chrysler Group, LLC (U.S), and Toyota Motor Corporation (Japan), lithium-ion batteries are expected to dominate the EV market by 2022.

6.2 TRANSPORTATION

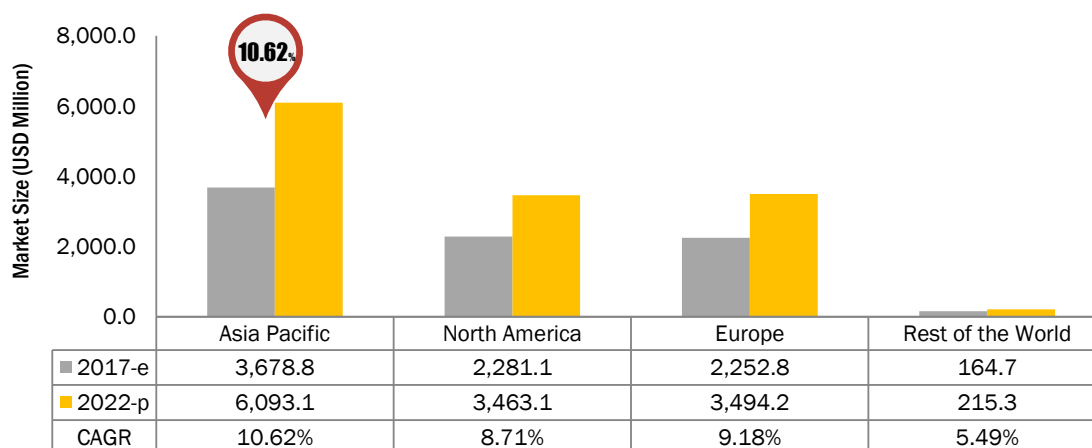
The transport application includes Battery Electric Vehicle (BEV) and Plug-In Hybrid Electric Vehicle (PHEV). A BEV is a type of EV which uses chemical energy stored in rechargeable battery packs. The BEV is also known as battery-only electric vehicle or all-electric vehicle, which uses battery to store the electric energy that powers the motor. A PHEV can be recharged from any external source of electricity, such as wall socket. The electricity is stored in rechargeable battery packs. PHEV is a subcategory of EV. BEVs generally have more battery capacity than PHEVs. Therefore, charging a fully depleted BEV takes longer time than charging a fully depleted PHEV.

Lithium-ion battery is the leading technology platform both for BEV and PHEV, which uses larger-format cells and packs with capacities of 15 kWh to 20 kWh for PHEVs and up to 50 kWh for all-electric vehicles. The lithium-ion battery was first commercially released in 1991, by Sony Corporation (Japan) and Asahi Kasei Corporation (Japan). Lithium-ion batteries gain huge traction as they provide high energy and power density in terms of volume and high charge or discharge efficiency. They are also lightweight than other batteries, such as lead-acid and Ni-MH batteries. Several manufacturers, such as Tesla, Inc. (US), are heavily investing in R&D to reduce their cost, making lithium-ion batteries more stable. They are also developing safety mechanisms to prevent harm, in case the battery catches fire.

Further, NiMH batteries are the most readily available rechargeable batteries that are used in EV. NiMH electric vehicle batteries are used in all-electric plug-in vehicles, such as General Motors EV1, Honda EV Plus, Ford Ranger EV, and Vectrix scooter. They are also used in hybrid vehicles, such as the Toyota Prius, Honda Insight, Ford Escape Hybrid, Chevrolet Malibu Hybrid, and Honda Civic Hybrid.

Nickel cadmium (NiCd) was invented by Waldemar Jungner in 1899. Although the nickel cadmium battery offers several advantages over lead acid, the materials for NiCd are expensive. In the year 1947, an advancement was made which absorbed the gases generated during the charge, which developed the modern sealed NiCd battery.

FIGURE 18 ASIA PACIFIC IS EXPECTED TO DRIVE THE GROWTH OF ADVANCED ENERGY STORAGE SYSTEMS IN THE TRANSPORTATION APPLICATION DURING 2017–2022



e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 4 TRANSPORTATION: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Asia Pacific	3,029.7	3,327.2	3,678.8	6,093.1	10.62%
North America	1,956.8	2,112.8	2,281.1	3,463.1	8.71%
Europe	1,905.2	2,071.8	2,252.8	3,494.2	9.18%
Rest of the World	149.0	156.7	164.7	215.3	5.49%
Total	7,040.8	7,668.5	8,377.4	13,265.6	9.63%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The AESS transportation segment is expected to be USD 8,377.4 million in 2017 and is projected to reach USD 13,265.6 million by 2022 at a CAGR of 9.63%. The market in Asia Pacific accounted for the largest share of the global AESS market in the transportation application in 2016 and it is estimated to continue its dominance during the forecast period. This growth in the market in Asia Pacific is mainly driven by government policies related to EV adoption in these countries. Further, in 2016, China had the maximum number of EVs on road. Growth in the EV market is expected to boost the transportation application. According to the Global EV outlook 2017, the electric car stock surpassed 2 million vehicles in 2016.

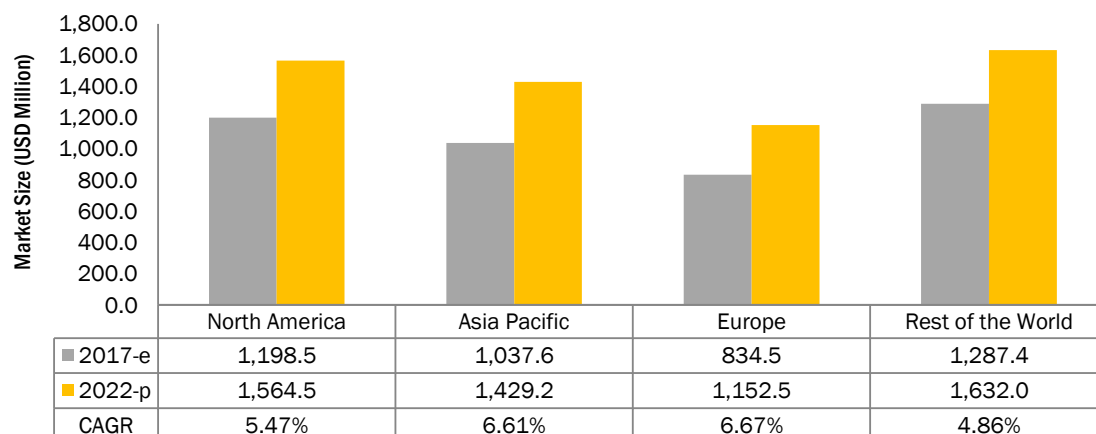
6.3 GRID STORAGE

The grid storage application includes energy storage installations of the utility scale AESS. The electricity storage can provide multiple benefits to the grid, including the ability to levelize load, provide ancillary services, and provide firm capacity etc. Further, the energy storage also provides cost savings to the electric grid. Large scale energy storage allows the electrical system, to run more efficiently, indicating lower prices, less emissions, and more reliable power.

The International Energy Agency (IEA) projected that by 2050, the necessary amount of energy storage system in the world would be between 189 GW to 305 GW, corresponding to an output variation rate of renewable energies of 15% or 30%, respectively.

The advanced energy storage system in grid storage application accounted for 34.2% share of the total market in 2016. Renewable policies and energy arbitrage opportunities are triggering the growth of grid storage in the AESS market.

FIGURE 19 ASIA PACIFIC IS EXPECTED TO DRIVE THE GROWTH OF ADVANCED ENERGY STORAGE SYSTEMS IN THE GRID STORAGE APPLICATION DURING 2017–2022



Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 5 GRID STORAGE: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	1,044.9	1,119.1	1,198.5	1,564.5	5.47%
Asia Pacific	927.3	993.9	1,037.6	1,429.2	6.61%
Europe	726.0	778.4	834.5	1,152.5	6.67%
Rest of the World	1,179.1	1,232.1	1,287.4	1,632.0	4.86%
Total	3,877.2	4,123.4	4,357.9	5,778.2	5.80%

e- Estimated; p- Projected

Note: The rest of the world includes Chile, South Africa, and other Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The AESS market in the grid storage application is expected to be USD 4,357.9 million in 2017 and is projected to reach USD 5,778.2 million by 2022 at a CAGR of 5.80%. The market in North America is expected to account for 27.5% of the market share in 2017, mainly driven by growth of renewable installations and grid optimization efforts in the region. Furthermore, in 2016, there were about 1,060 energy storage projects that were operational, with 196 projects under construction. Out of these, around 39.0% of operational projects and 33.0% of projects under construction were located in the US.

7 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY

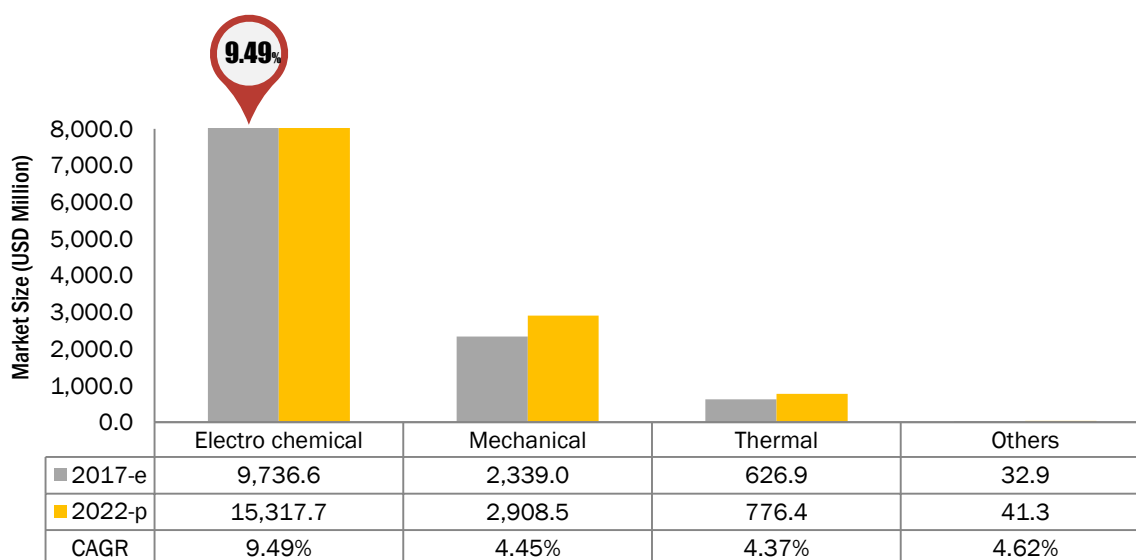
KEY FINDINGS

- The global AEES market based on technology is segmented into electro chemical, mechanical, thermal, and others.
- The electro chemical segment had the maximum share of 75.7% in 2016.
- The market size of the lithium-ion battery, a sub-segment of the electro chemical energy storage market is the largest among other technologies and constituted 80.4% share in the electro chemical segment in 2016
- The growth in the number of EV, energy arbitrage, and renewable policies are the key drivers for the AEES market.
- The lithium-ion battery energy storage market is expected to grow at the highest CAGR of 10.51% during the forecast period.
- Asia Pacific is the fastest growing region in the global AEES market, by technology, from 2017 to 2022.

7.1 INTRODUCTION

There are various technologies available in the AESS market, which perform different functions for the purpose of energy storage. Advanced energy storage is gradually becoming the ideal companion for the storage of renewable energy. Unlike the conventional energy sources, the renewable energy sources paired with the AESS are offering flexibility, energy efficiency, stability, and resilience of supply. This report primarily focuses on major technology types of AESS, such as electro chemical, which includes lithium-ion battery, sodium sulfur (NaS) battery, lead acid battery, flow battery, and NiMH and NiCd; thermal energy storage technology that includes thermo chemical, latent thermal, and sensible thermal; mechanical energy storage technology that includes pumped hydro, flywheels, and Compressed as well as Liquid Air Energy Storage (CAES and LAES); and others which include hydrogen storage, synthetic natural gas, and supercapacitors. Among all these types, electro chemical is expected to command the largest market share of 76.5% in 2017 and is expected to increase its market share to 80.4% by 2022.

FIGURE 20 ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET IS PROJECTED TO GROW RAPIDLY DURING 2017–2022



e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 6 ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Electro chemical	8,192.6	8,924.9	9,736.6	15,317.7	9.49%
Mechanical	2,126.0	2,238.1	2,339.0	2,908.5	4.45%
Thermal	569.8	597.7	626.9	776.4	4.37%
Others	29.6	31.3	32.9	41.3	4.62%
Total	10,918.0	11,792.0	12,735.4	19,043.8	8.38%

e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

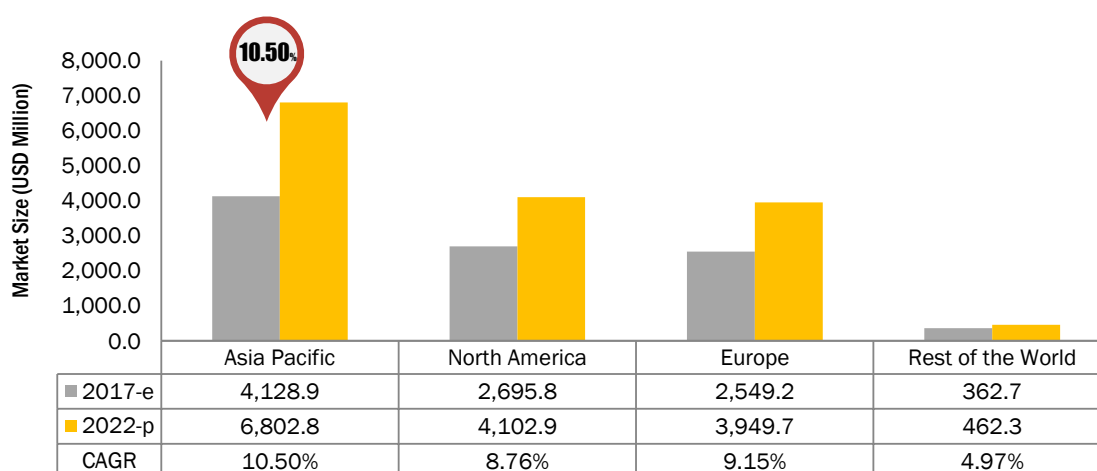
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The electro chemical market has the largest share in the AESS market with an estimated market size of USD 9,736.6 million in 2017 and is expected to reach a market size of USD 15,317.7 million by 2022, growing at a CAGR of 9.49% during the forecast period. Varied applications of this technology type including the large transportation application, is contributing to its large market share.

7.2 ELECTRO CHEMICAL

The electro chemical technology can be classified into secondary batteries and flow batteries. The secondary batteries consist of lead acid, lithium-ion, sodium sulfur, and NiMH & NiCd. The redox flow and hybrid flow are together termed as flow batteries.

FIGURE 21 ASIA PACIFIC IS EXPECTED TO GROW RAPIDLY IN THE ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 7 ELECTRO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Asia Pacific	3,412.7	3,752.0	4,128.9	6,802.8	10.50%
North America	2,295.0	2,486.9	2,695.8	4,102.9	8.76%
Europe	2,153.5	2,339.2	2,549.2	3,949.7	9.15%
Rest of the World	331.5	346.7	362.7	462.3	4.97%
Total	8,192.6	8,924.9	9,736.6	15,317.7	9.49%

e- Estimated; p- Projected

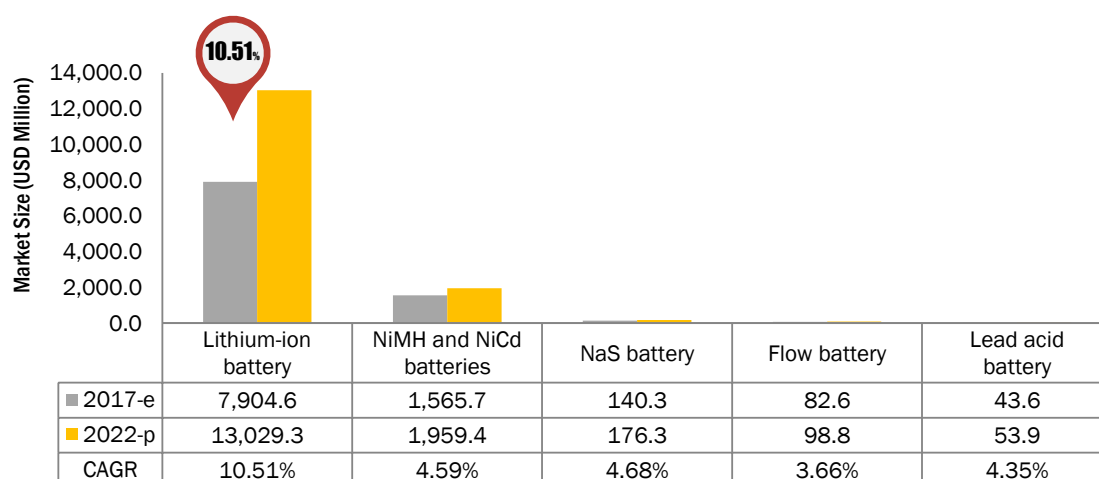
Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

Asia Pacific is estimated to hold the largest market share of 42.4% in the global electro chemical technology market in 2017. The region's market is expected to be USD 4,128.9 million in 2017 and is projected to reach a market size of USD 6,802.8 million by 2022, growing at a CAGR of 10.50% from 2017

to 2022 at the highest CAGR among all the regions. The growth in the electro chemical technology segment was majorly driven by the growth of the lithium-ion batteries in the region, particularly in the EV market.

FIGURE 22 LITHIUM-ION BATTERY IS EXPECTED TO GROW RAPIDLY IN THE ELECTRO CHEMICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017-2022



e- Estimated; p- Projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 8 ELECTRO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015-2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017-2022)
Lithium-ion battery	6,530.8	7,179.9	7,904.6	13,029.3	10.51%
NiMH & NiCd batteries	1,422.5	1,491.6	1,565.7	1,959.4	4.59%
NaS battery	125.7	133.5	140.3	176.3	4.68%
Flow battery	74.5	78.7	82.6	98.8	3.66%
Lead Acid battery	39.1	41.3	43.6	53.9	4.35%
Total	8,192.6	8,924.9	9,736.6	15,317.7	9.49%

e- Estimated; p- Projected

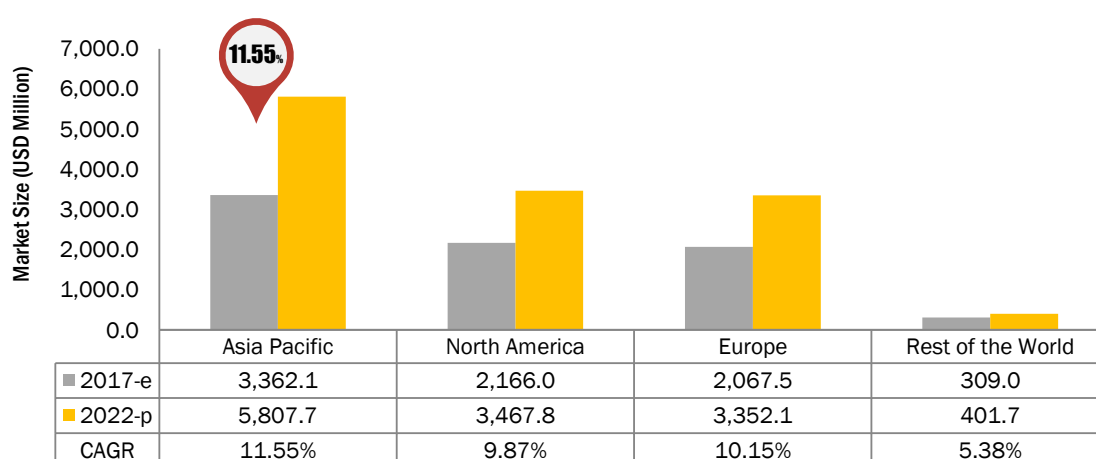
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The lithium-ion battery energy storage market is estimated to hold the largest market share of 81.2% in the global electro chemical technology market in 2017. The market for the lithium-ion battery is expected to be USD 7,904.6 million in 2017 and is projected to reach a market size of USD 13,029.3 million by 2022, growing at a CAGR of 10.51% from 2017 to 2022 at the highest CAGR among all the technologies. The growth in the lithium-ion battery energy storage market is due to the rapid growth in the EV market coupled with the falling cost of these batteries.

7.2.1 LITHIUM-ION BATTERY

The lithium-ion battery was first commercially released in 1991 by Sony Corporation (Japan) and Asahi Kasei Corporation (Japan). The lithium-ion batteries gain huge traction as they provide high energy and power density in terms of volume and high charge and discharge efficiency. These are more lightweight than other batteries, such as lead acid and NiMH batteries. The lithium-ion battery is grouped into two categories of cathode materials and anode materials to complement lithium with iron phosphate and mixed metal. The positively charged lithium-ions travel between the anode and the cathode in an electrolyte. However, these batteries are very expensive and are dangerous under some conditions due to the presence of flammable electrolytes. Several manufacturers are heavily investing in the R&D to reduce their cost, making the lithium-ion batteries more stable and to develop safety mechanisms to prevent harm in case the battery catches fire.

FIGURE 23 ASIA PACIFIC IS EXPECTED TO GROW RAPIDLY IN THE LITHIUM-ION BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 9 LITHIUM-ION BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Asia Pacific	2,727.9	3,028.0	3,362.1	5,807.7	11.55%
North America	1,807.5	1,978.4	2,166.0	3,467.8	9.87%
Europe	1,715.1	1,879.3	2,067.5	3,352.1	10.15%
Rest of the World	280.3	294.3	309.0	401.7	5.38%
Total	6,530.8	7,179.9	7,904.6	13,029.3	10.51%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

Asia Pacific is estimated to hold the largest market share of 42.5% in the global lithium-ion batteries market in 2017. The region's market is expected to be USD 3,362.1 million in 2017 and is projected to reach a market size of USD 5,807.7 million by 2022, growing at a CAGR of 11.55% from 2017 to 2022. Technological advancements and the rapid involvement of domestic manufacturers to cater to the varied application demand for lithium-ion batteries are some of the reasons behind the region's strong hold in the market.

TABLE 10 LITHIUM-ION BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	5,618.2	6,177.0	6,811.8	11,306.2	10.67%
Grid storage	912.6	1,002.9	1,092.8	1,723.1	9.53%
Total	6,530.8	7,179.9	7,904.6	13,029.3	10.51%

e- Estimated; p- Projected

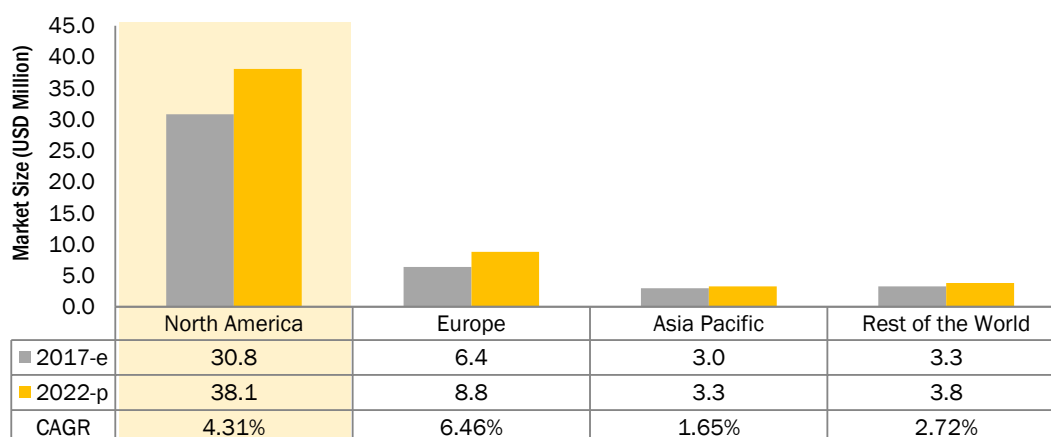
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The transportation application in the lithium-ion battery market is estimated to hold the largest market size of USD 6,811.8 million in 2017 and is expected to reach USD 11,306.2 million by 2022, at a CAGR of 10.67% between 2017 and 2022. The rapid pace of R&D activities by EV manufacturers is expected to fuel the growth in the transportation segment.

7.2.2 LEAD ACID BATTERY

Lead acid batteries were invented over 150 years ago and are the oldest and most commonly used type of rechargeable battery. These batteries consist of lead and lead oxide electrodes in an acid electrolyte. 'Flooded or vented' and 'sealed or valve regulated' are two types of lead acid batteries. A typical lead acid battery has an efficiency of around 70%–90% with an expected lifetime of 5–15 years. The common applications of lead acid batteries are emergency power supply systems, stand-alone systems with Photovoltaic battery systems for mitigation of output fluctuations from wind power, and starter batteries in vehicles.

FIGURE 24 NORTH AMERICA IS EXPECTED TO HAVE THE HIGHEST SHARE IN THE LEAD ACID BATTERY ENERGY STORAGE MARKET DURING 2017–2022



Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 11 LEAD ACID BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	27.5	29.1	30.8	38.1	4.31%
Europe	5.6	6.0	6.4	8.8	6.46%
Asia Pacific	2.8	3.0	3.0	3.3	1.65%
Rest of the World	3.1	3.2	3.3	3.8	2.72%
Total	39.1	41.3	43.6	53.9	4.35%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

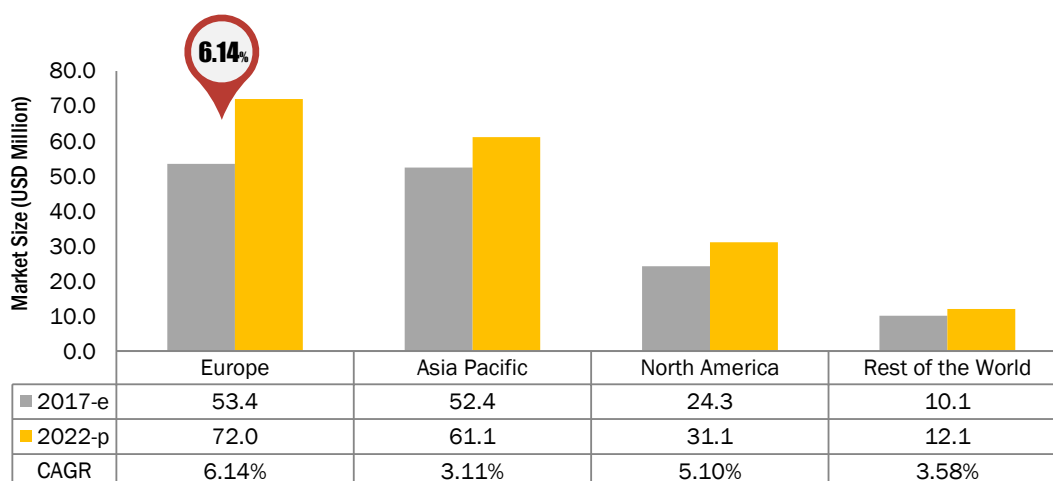
Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The North American advanced energy storage systems market for the lead acid battery held the largest market size of USD 30.8 million in 2017 and is expected to reach USD 38.1 million by 2022, at a CAGR of 4.31% between 2017 and 2022. The lead acid batteries have historically been a cheap, reliable source of storage, however the lithium-ion batteries have started replacing many traditional lead acid applications.

7.2.3 SODIUM SULFUR (NAS) BATTERY

The sodium sulfur (NaS) battery was first released in the 1960s by the Ford Motor Company. The battery was sold to the Japanese battery system manufacturer NGK for stationary application. Sodium sulfur is a type of molten salt constructed from sodium (Na) and sulfur (S). These batteries offer high energy density, long cycle, and high charge and discharge of about (89–92%). NaS batteries are also fabricated from inexpensive materials and this makes them suitable for large-scale grid energy storage applications. The NaS battery storage systems deliver power for several hours at a time, for long-term applications.

FIGURE 25 EUROPE IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE NAS BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 12 SODIUM SULFUR (NAS) BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Europe	47.0	50.1	53.4	72.0	6.14%
Asia Pacific	47.9	50.8	52.4	61.1	3.11%
North America	21.3	22.7	24.3	31.1	5.10%
Rest of the World	9.5	9.8	10.1	12.1	3.58%
Total	125.7	133.5	140.3	176.3	4.68%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

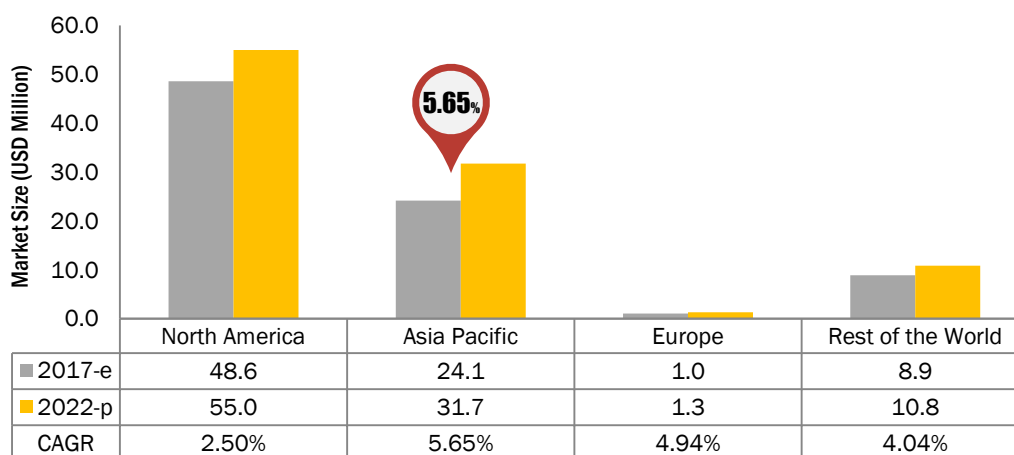
Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

Europe is expected to hold the largest market share of 38.1% in the global NaS battery storage systems market in 2017. The region's market is estimated to be USD 53.4 million in 2017 and is expected to reach a market size of USD 72.0 million by 2022, growing at a CAGR of 6.14% during the forecast period. In 2014, NGK Insulators, Ltd. (Japan) and Terna S.p.A (Italy), Italy's largest Transmission System Operator (TSO) came to an agreement for supply of the NaS battery system. This agreement was stated to be the first large scale NaS battery energy storage system installation in the European grid system.

7.2.4 FLOW BATTERY

Flow batteries comprise two chemical components dissolved in a liquid within the system, which are separated by a membrane. Flow battery systems rely on electrode and electrolyte which store active chemicals separately until the power is needed. They exchange ions across the membrane, thereby generating electricity. Flow batteries are considered safer than most commonly used lithium-ion batteries as they can resist greater temperatures and periods of idleness, making them suitable for storing and releasing renewable energy, such as that from solar panels and wind turbines. The most popular flow batteries include vanadium redox flow battery and hybrid flow batteries which include zinc-bromine and zinc-cerium. The Vanadium-redox flow batteries are cheaper and more reliable for stationary energy storage.

FIGURE 26 ASIA PACIFIC IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE FLOW BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e- estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 13 FLOW BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	43.4	45.9	48.6	55.0	2.50%
Asia Pacific	21.9	23.3	24.1	31.7	5.65%
Europe	0.9	1.0	1.0	1.3	4.94%
Rest of the World	8.2	8.6	8.9	10.8	4.04%
Total	74.5	78.7	82.6	98.8	3.66%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

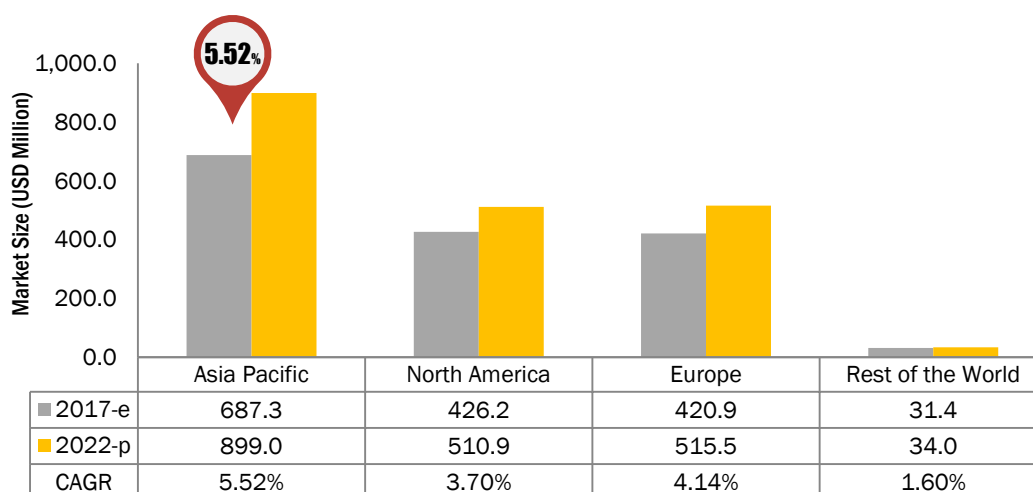
North America is expected to hold the largest market share of 58.8% in the global flow battery storage systems market in 2017. The region's market is estimated to be USD 48.6 million in 2017 and is projected to reach a market size of USD 55.0 million by 2022, growing at a CAGR of 2.50%. However, the market in Asia Pacific is expected to grow at the highest CAGR of 5.65% from 2017 to 2022 due to major technological breakthroughs by manufacturers of flow batteries in the Asia Pacific region.

7.2.5 NICKEL METAL HYDRIDE (NIMH) & NICKEL CADMIUM BATTERY (NICD)

NiMH batteries are the most readily available rechargeable batteries for use in consumer and EV. NiMH electric vehicle batteries are used in all electric plug-in vehicles, such as General Motors EV1, Honda EV Plus, Ford Ranger EV, and Vectrix scooter. They are also used in hybrid vehicles, such as the Toyota Prius, Honda Insight, Ford Escape Hybrid, Chevrolet Malibu Hybrid, and Honda Civic Hybrid.

NiCd was invented by Waldemar Jungner in 1899. Although NiCd battery offers several advantages over lead acid, the materials for NiCd are expensive. In the year 1947, there were advancements that resulted in absorption of the gases generated during charge, which led to the development of the modern sealed NiCd battery.

FIGURE 27 ASIA PACIFIC IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE NIMH & NICD BATTERY ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017-2022



e-estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 14 NICKEL METAL HYDRIDE & NICKEL CADMIUM BATTERY: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015-2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017-2022)
Asia Pacific	612.1	647.0	687.3	899.0	5.52%
North America	395.3	410.8	426.2	510.9	3.70%
Europe	384.9	402.9	420.9	515.5	4.14%
Rest of the World	30.3	30.8	31.4	34.0	1.60%
Total	1,422.5	1,491.6	1,565.7	1,959.4	4.59%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

Asia Pacific is expected to hold the largest market share of 43.9% in the global NiMH and NiCd battery storage systems market in 2017. The region's market is estimated to be USD 687.3 million in 2017 and is projected to reach a market size of USD 899.0 million by 2022, growing at a CAGR of 5.52% during the forecast period. The growth in this market is driven by the growing number of EVs in the region.

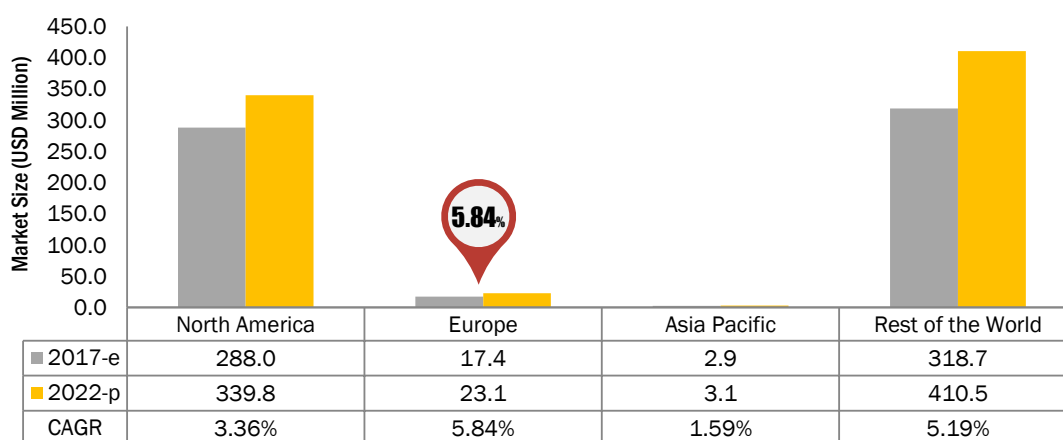
7.3 THERMAL STORAGE

The thermal energy storage includes a number of different technologies that accommodate a wide range of needs. This technology allows for excess electricity to be converted into thermal energy and stored for short term or seasonal use or for later use. Generally the energy is not converted back to electricity except in the case of concentrating solar power technologies.

Thermal storage can be categorized into different technologies, namely, storage of sensible heat, storage of latent heat, and thermo chemical storage. The storage medium may be a liquid, such as water or thermo-oil or a solid, such as concrete or ground. Thermal energy is stored solely through a change of temperature of the storage medium. The capacity of a storage system is defined by the specific heat capacity and the mass of the medium used.

The possibilities of thermal storage applications are diverse, however, in practice, thermal storage applications are relatively limited. There is a developing need for thermal storage to defer air-conditioning loads from peak periods as heating systems can use storage of hot water to match electricity and heat production.

FIGURE 28 EUROPE IS EXPECTED TO GROW RAPIDLY IN THE THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017-2022



e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Rest of the World	290.2	304.1	318.7	410.5	5.19%
North America	261.5	274.4	288.0	339.8	3.36%
Europe	15.3	16.3	17.4	23.1	5.84%
Asia Pacific	2.8	2.9	2.9	3.1	1.59%
Total	569.8	597.7	626.9	776.4	4.37%

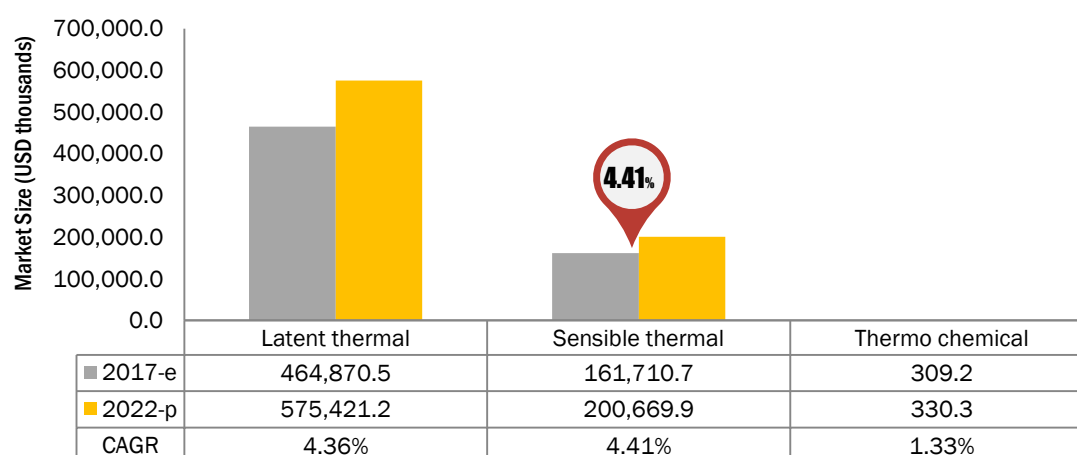
e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The rest of the world is estimated to hold the largest market share of 50.8% in the global thermal energy storage market in 2017. The region's market is expected to be USD 318.7 million in 2017 and is projected to reach a market size of USD 410.5 million by 2022, growing at a CAGR of 5.19% from 2017 to 2022. The market in the rest of the world which includes Middle East and African countries is driven by increasing demand for renewable power, such as CSP plants in countries, such as, South Africa, UAE, Israel, and Morocco. Europe is expected to witness the highest CAGR of 5.84% among all the regions. For instance a district in Denmark, installed the world's largest underground thermal pit storage (200,000 m³) in the city of Vojens. The plant was put into operation in May 2015.

FIGURE 29 SENSIBLE THERMAL STORAGE TECHNOLOGY IS EXPECTED TO GROW RAPIDLY DURING 2017–2022



e- Estimated; p- Projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 16 THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD THOUSANDS)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Latent thermal	4,21,589.8	4,42,740.0	4,64,870.5	5,75,421.2	4.36%
Sensible thermal	1,47,945.2	1,54,676.2	1,61,710.7	2,00,669.9	4.41%
Thermo chemical	260.2	279.8	309.2	330.3	1.33%
Total	5,69,795.2	5,97,696.0	6,26,890.4	7,76,421.4	4.37%

e- Estimated; p- Projected

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The latent heat thermal energy storage market is estimated to hold the largest market share of 74.2% in the global thermal energy storage technology market in 2017. The market for latent heat thermal energy storage is expected to grow at a CAGR of 4.36% from 2017 to 2022, highest CAGR among all other technologies under thermal storage. The growth in the thermal energy storage market is due to the rapid growth in the adoption of molten salt as a phase change material.

7.3.1 THERMO CHEMICAL ENERGY STORAGE

The thermo chemical storage technology is the least explored storage technology, though it can potentially store more energy than sensible or latent heat systems due to the heat of chemical reactions. These systems depend on heat from solar fields to drive reversible chemical reactions. Thus, the storage medium must have the ability to completely dissociate in the temperature range of the solar field. Thermo chemical storage is deemed advantageous over the sensible heat storage because the chemical products used for the dissociation reactions can be stored indefinitely at an ambient temperature, thus, reducing thermal losses.

In addition, the exothermic process occurs at a constant temperature if the heat is dissipated at a rate that prevents combustion. However, due to the higher cost of thermo chemical storage, as compared to other storage technologies, the commercial acceptance of the technology is limited in the present market. Most thermo chemical projects are in the R&D stage with studies being conducted on market feasibility and cost effectiveness. There are ongoing government-funded projects in the European Union, Japan, and the US to commercialize the thermochemical technology.

TABLE 17 THERMO CHEMICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	260.2	279.8	309.2	330.3	1.33%
Total	260.2	279.8	309.2	330.3	1.33%

e- Estimated; p- Projected

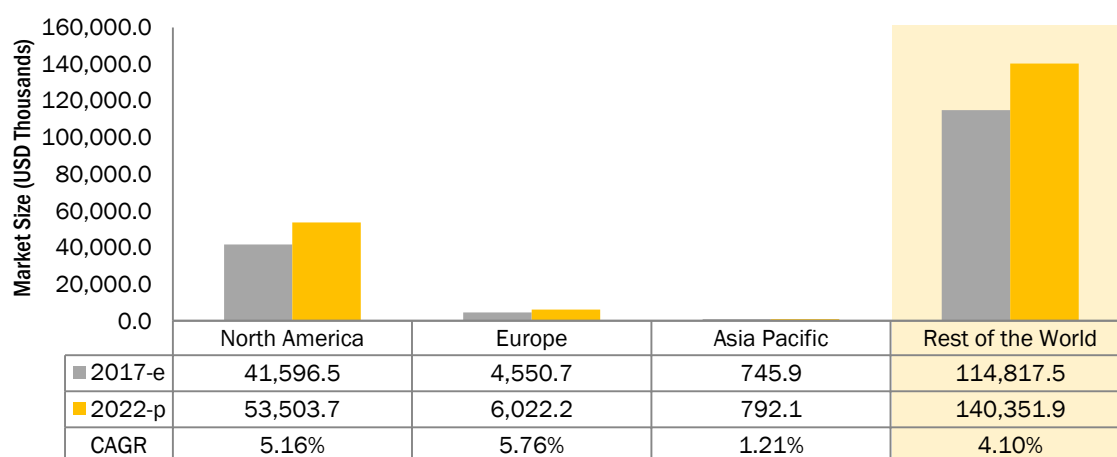
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The thermo chemical storage market in North America is expected to be at 330.3 thousand by 2022. Globally, the thermo chemical storage of thermal energy is still in the demonstration phase and is not commercially available as yet. The governments of the US, the European Union, and Japan are funding the thermo chemical storage projects, applicable to domestic systems, district heating, and industrial needs. Researchers and developers are working on new materials and techniques for thermo chemical-based energy storage systems and their cost, performance, and reliability needs.

7.3.2 SENSIBLE HEAT THERMAL ENERGY STORAGE

Thermal energy storage in the form of sensible heat is based on the specific heat of a storage medium, which is kept in storage tanks with high thermal insulation. Underground storage of sensible heat in both liquid and solid storage medium is also used for typically large-scale applications, such as district heating and cooling systems. However, thermal energy storage systems based on sensible heat storage offer a storage capacity that is limited by the specific heat of the storage medium which keeps the energy density lower than other prevailing storage technologies. Concrete thermal storage and heat thermal storage projects have been considered under the sensible thermal storage.

FIGURE 30 REST OF THE WORLD IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE SENSIBLE HEAT THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e-estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 18 SENSIBLE HEAT THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Rest of the World	1,06,697.2	1,10,685.5	1,14,817.5	1,40,351.9	4.10%
North America	36,480.7	38,956.4	41,596.5	53,503.7	5.16%
Europe	4,027.5	4,281.6	4,550.7	6,022.2	5.76%
Asia Pacific	739.7	752.7	745.9	792.1	1.21%
Total	1,47,945.2	1,54,676.2	1,61,710.7	2,00,669.9	4.41%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The sensible heat thermal energy storage market for the rest of the world, which includes regions, such as Middle East, Africa, and South America is expected to have the largest market, with a share of 71.0% in 2017. Europe is expected to witness the highest CAGR of 5.76% among all the regions.

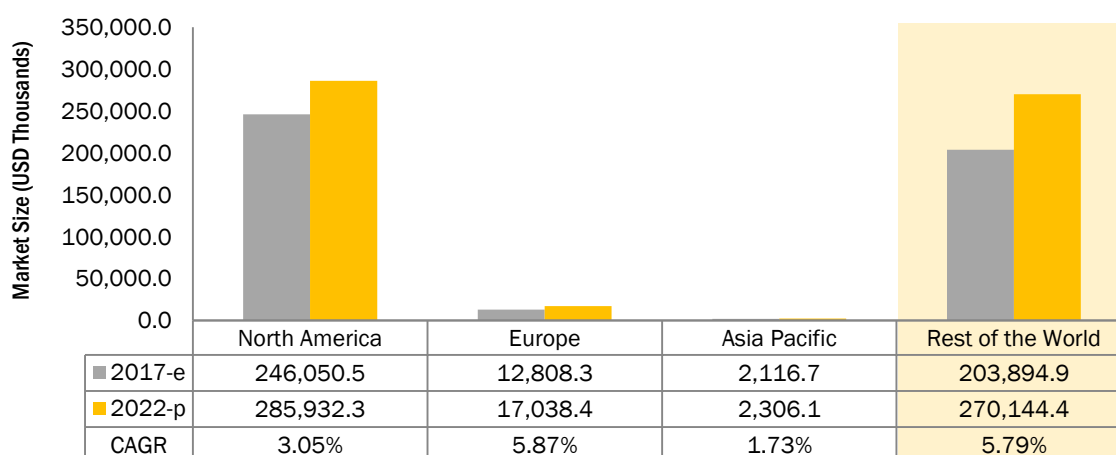
7.3.3 LATENT HEAT THERMAL ENERGY STORAGE

Phase Change Materials (PCM) offer higher storage capacity which is associated with the latent heat of the phase change. There are organic (paraffins) and inorganic PCMs (salt hydrates and molten salt) available for such storage systems. Phase change is a heat-seeking (endothermic) process and, therefore, PCMs absorb heat. Upon storing heat in the storage material, the material begins to melt when the phase change temperature is reached. The temperature then stays constant until the melting process is over. The heat stored during the phase change process (melting process) of the material is called the latent heat.

The latent heat storage has an advantage of storing large amounts of heat with only small temperature changes and, therefore, has a high storage density. Phase change materials also enable a target-oriented discharging temperature that is set by the constant temperature of the phase change. The use of phase change materials can be found in solar energy storage systems for water heating, greenhouses, space heating and cooling, cooking, and waste heat recovery systems.

Phase change materials can offer higher storage efficiencies from 75% to 90%. In most of the cases, storage is based on a solid or liquid phase change, such as in ice.

FIGURE 31 REST OF THE WORLD IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE LATENT HEAT THERMAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e- estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 19 LATENT HEAT THERMAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD THOUSANDS)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	2,24,732.2	2,35,160.5	2,46,050.5	2,85,932.3	3.05%
Rest of the World	1,83,467.6	1,93,416.4	2,03,894.9	2,70,144.4	5.79%
Europe	11,312.2	12,038.2	12,808.3	17,038.4	5.87%
Asia Pacific	2,077.7	2,124.9	2,116.7	2,306.1	1.73%
Total	4,21,589.8	4,42,740.0	4,64,870.5	5,75,421.2	4.36%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

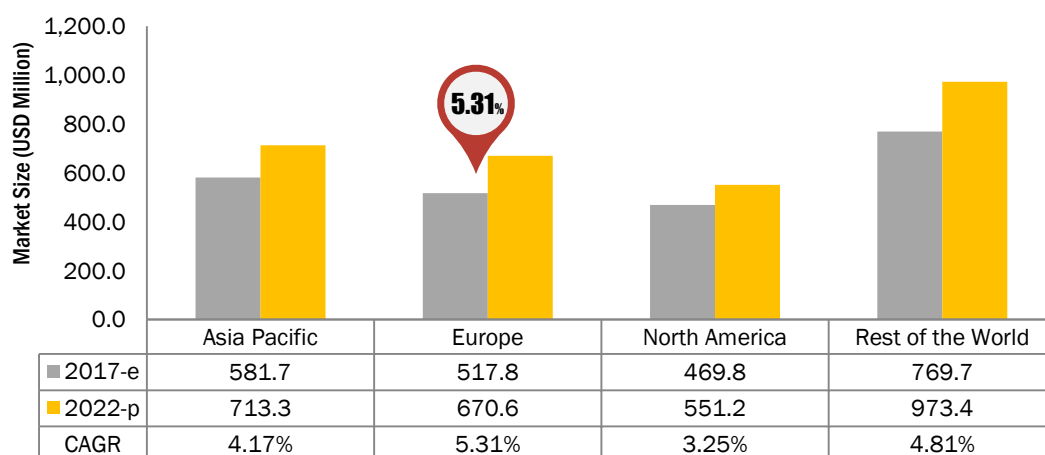
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The latent heat thermal energy storage market for North America is expected to have the largest market share of 52.9% in 2017 and is expected to grow at a CAGR of 3.05%. The market is driven by the increasing demand for renewable power, such as CSP plants, in countries, such as, South Africa, UAE, Israel, and Morocco.

7.4 MECHANICAL ENERGY STORAGE

The mechanical energy storage system is classified into pumped hydro energy storage, flywheel, and compressed air and liquid air energy storage systems. The historic trend was dominated by pumped hydro storage, which has now become a mature technology. Furthermore, pumped hydro and compressed air energy storage are the most advanced electricity storage technologies among all the other storage technologies.

FIGURE 32 EUROPE IS EXPECTED TO GROW RAPIDLY IN THE MECHANICAL ENERGY ADVANCED STORAGE SYSTEMS MARKET DURING 2017–2022



e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 20 MECHANICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Asia Pacific	538.6	563.3	581.7	713.3	4.17%
Europe	459.7	491.9	517.8	670.6	5.31%
North America	422.2	446.0	469.8	551.2	3.25%
Rest of the World	705.5	737.0	769.7	973.4	4.81%
Total	2,126.0	2,238.1	2,339.0	2,908.5	4.45%

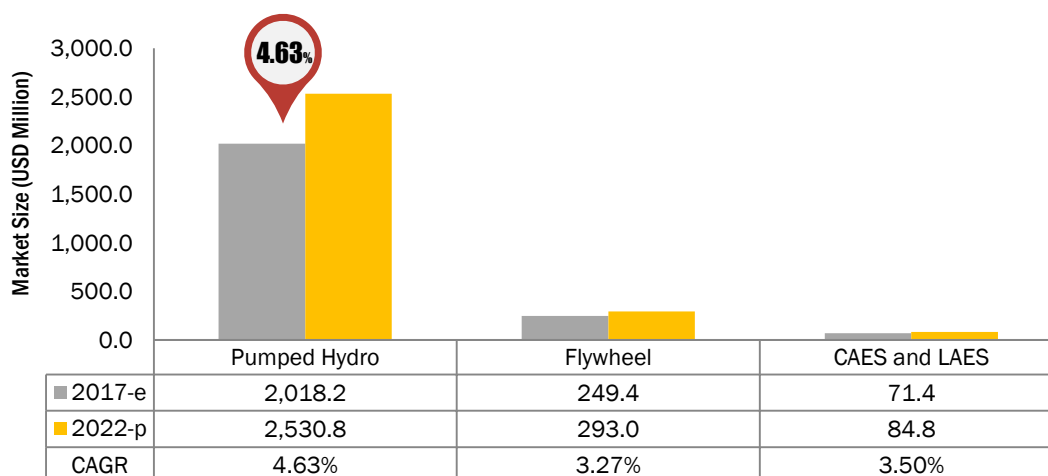
e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

Asia Pacific is estimated to hold the market share of 24.9% in the global mechanical energy storage market in 2017. The region's market is expected to be USD 581.7 million in 2017 and is projected to reach a market size of USD 713.3 million by 2022, growing at a CAGR of 4.17% from 2017 to 2022. Europe is expected to grow at the maximum CAGR of 5.31% during the forecast period, driven by growth of the pumped hydro energy storage in countries, such as, Portugal, Switzerland, and Spain.

FIGURE 33 PUMPED HYDRO STORAGE IS EXPECTED TO GROW RAPIDLY IN THE MECHANICAL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e- Estimated; p- Projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 21 MECHANICAL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Pumped Hydro	1,837.5	1,933.2	2,018.2	2,530.8	4.63%
Flywheel	224.3	237.2	249.4	293.0	3.27%
CAES and LAES	64.2	67.7	71.4	84.8	3.50%
Total	2,126.0	2,238.1	2,339.0	2,908.5	4.45%

e- Estimated; p- Projected

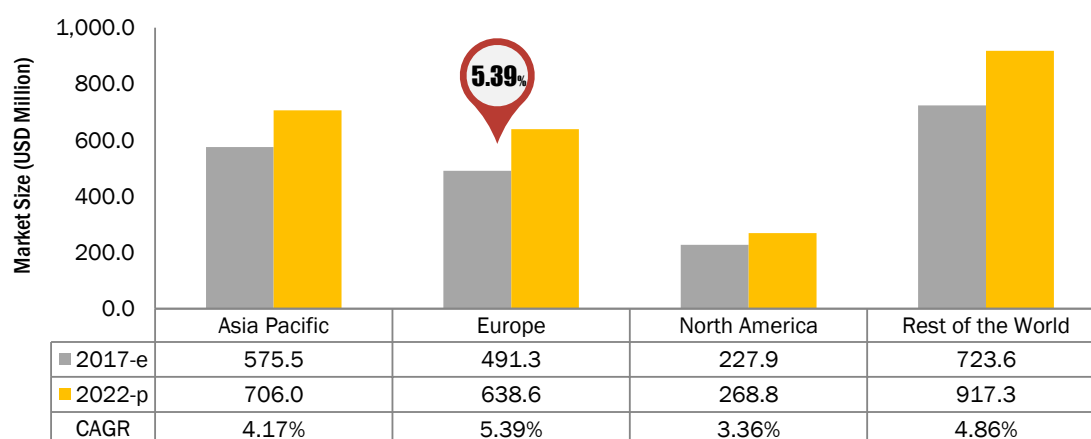
Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The pumped hydro energy storage market is estimated to hold the largest market share of 86.3% in the global mechanical energy storage technology market in 2017. The market for pumped hydro energy storage is expected to be USD 2,018.2 million in 2017 and is projected to reach a market size of USD 2,530.8 million by 2022, growing at a CAGR of 4.63% from 2017 to 2022.

7.4.1 PUMPED HYDRO ENERGY STORAGE

In pumped hydro energy storage, water is pumped to reservoirs at higher altitude into the existing dam by using excess electricity generated during light load durations. During the peak demand, this stored water is released from the upper reservoir to the lower, powering a turbine in the process, generating electrical energy. Pumped hydro storage is one of the oldest technologies, with the first plant been setup in the 1890s. The efficiency of the pumped hydro storage system ranges from 70% to 85%, with the discharge time ranging from several hours to days.

FIGURE 34 EUROPE IS EXPECTED TO BE THE HIGHEST GROWING REGION FOR THE PUMPED HYDRO ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e-estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 PUMPED HYDRO: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Asia Pacific	532.7	557.1	575.5	706.0	4.17%
Europe	435.4	466.4	491.3	638.6	5.39%
North America	206.9	217.2	227.9	268.8	3.36%
Rest of the World	662.6	692.4	723.6	917.3	4.86%
Total	1,837.5	1,933.2	2,018.2	2,530.8	4.63%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

Asia Pacific is estimated to hold the largest market share of 28.5% in the global pumped hydro storage market in 2017. The region's market is expected to be USD 575.5 million in 2017 and is projected to reach a market size of USD 706.0 million by 2022, growing at a CAGR of 4.17%. The reason behind its growth is the commencement of the new utility scale, pumped hydro energy storage projects in the region, primarily in India, China, and Japan. However, Europe is expected to grow at the highest CAGR of 5.39%

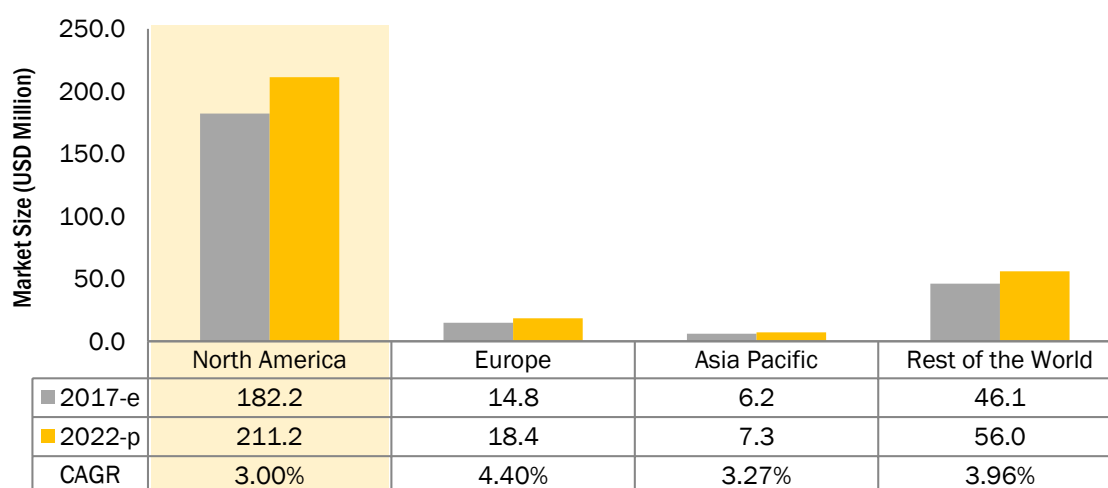
during the forecast period due to expected increase in the capacity of energy storage projects announced or yet to be commissioned, which are based on the pumped hydro storage technology.

7.4.2 FLYWHEEL ENERGY STORAGE

Flywheels are the type of mechanical energy storage, wherein the kinetic energy of fast spinning cylinders contain stored energy. Technological advances regarding the flywheel design and materials have improved the efficiency of flywheels. A flywheel consists of rotating cylinder supported on stator and bearings to eliminate wear and tear. A flywheel acts as a generator during massive demand and as a motor in regular demand patterns. Flywheels have energy densities ranging between 50 Wh/Kg and 100 Wh/Kg and an efficiency of over 90%.

Even though flywheels have been in existence for decades, they recently gained attention for large scale stationary energy storage. This technology is best suited for applications requiring high power for short periods and require little maintenance as compared to other technologies.

FIGURE 35 NORTH AMERICA IS ESTIMATED TO HAVE THE MAXIMUM SHARE IN THE FLYWHEEL ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017–2022



e-estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 FLYWHEEL: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	162.0	172.3	182.2	211.2	3.00%
Europe	13.5	14.1	14.8	18.4	4.40%
Asia Pacific	5.9	6.1	6.2	7.3	3.27%
Rest of the World	43.0	44.5	46.1	56.0	3.96%
Total	224.3	237.2	249.4	293.0	3.27%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The flywheel energy storage market for North America is expected to reach USD 211.2 million by 2022, from an estimated USD 182.2 million, in 2017. Europe is expected to witness the highest CAGR of 4.40% during the forecast period among all the regions.

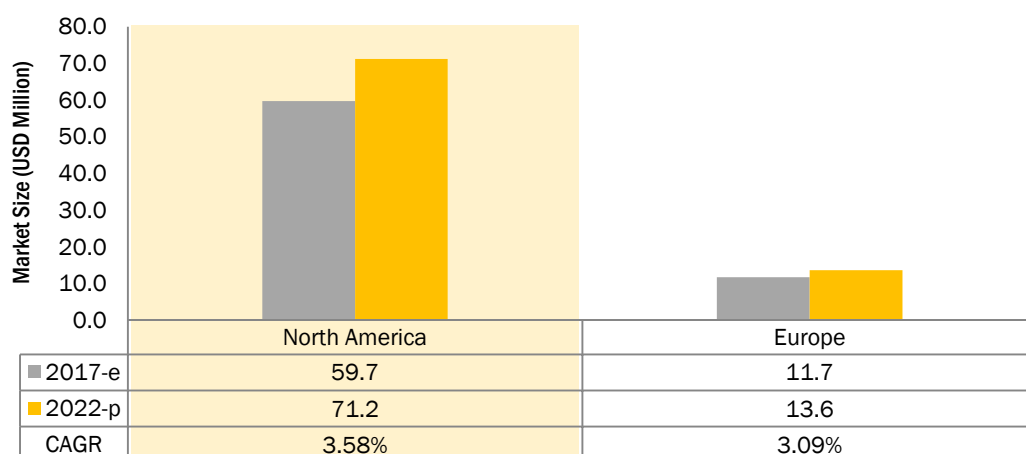
Currently, flywheels are commercially deployed for power quality in industrial and UPS applications, mainly in a hybrid configuration. However, efforts are being made to optimize this technology, to be used as a power storage device in vehicles and power plants for long-duration operations.

7.4.3 COMPRESSED AIR & LIQUID AIR ENERGY STORAGE

Compressed Air Energy Storage (CAES) shares some characteristics with pumped hydro, as it is also mature, commercially viable, can provide significant energy storage at a relatively low cost, and also has topographical constraints. However, as compared to the pumped hydro energy storage, CAES has very limited contribution to the energy storage market. CAES utilizes off-peak electricity to compress air, usually at high pressures, for storage in geological structures, such as, mines or aquifers, salt caverns, and aboveground pressure vessels. The compressed air is then released, preheated, and used to drive a turbine-generator system, to produce electricity when required.

Liquid Air Energy Storage (LAES), is in the demonstration stage. It is not prone to geological constraints or public resistance. Air is compressed and cooled in a refrigeration plant, using cheap, off-peak energy and stored in a relatively large insulated tank or vessel. This liquid air is then converted back to gas, expanded in volume, heated, and used to drive a turbine to generate electricity on demand. It can be particularly suited to locations where there is a source of low-grade heat or cooling, such as on the industrial process. The University of Birmingham's Cryogenic Energy Storage (CES) pilot facility or LAES pilot facility became operational in 2015.

FIGURE 36 NORTH AMERICA IS EXPECTED TO HAVE THE LARGEST SHARE IN THE CAES & LAES ADVANCED ENERGY STORAGE SYSTEMS MARKET DURING 2017-2022



e-estimated, p-projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 24 CAES & LAES: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	53.3	56.5	59.7	71.2	3.58%
Europe	10.9	11.3	11.7	13.6	3.09%
Total	64.2	67.7	71.4	84.8	3.50%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The CAES and LAES energy storage market for North America is expected to reach USD 71.2 million by 2022, from an estimated USD 59.7 million, in 2017, growing at a CAGR of 3.58%. There are very few commercial plants deploying the CAES and LAES technology. Some of them are a 321 MW plant in Huntorf, Germany and a 110 MW plant located in McIntosh, US. A new 330 MW project was developed in Northern Ireland in 2015.

7.5 OTHER STORAGE TECHNOLOGIES

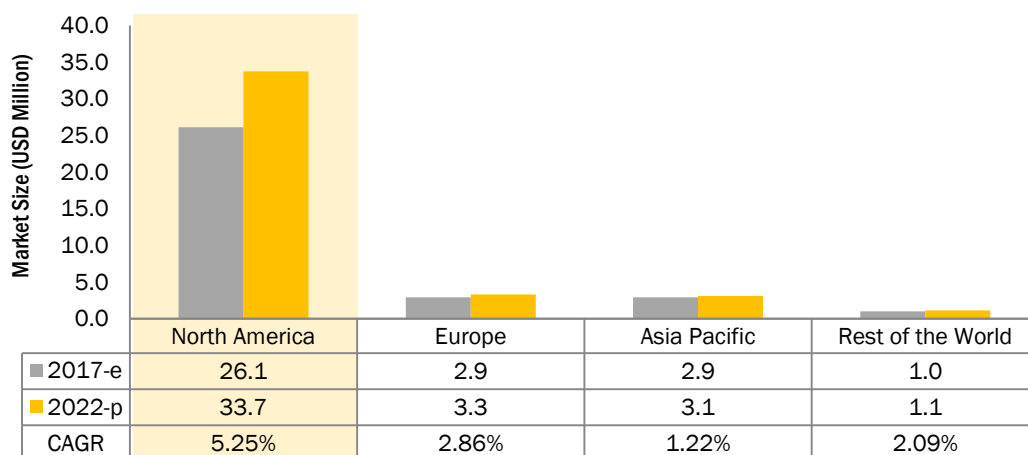
Other energy storage technologies include hydrogen storage, synthetic natural gas, and supercapacitors.

Hydrogen storage technologies are suitable for both large and small applications. In addition, hydrogen storage tanks have the capacity to supply kW of demand over discharge timeframes ranging from hours to months. Typically, the hydrogen fuel cells have a response time of seconds to minutes which makes it suitable for all storage applications. However, the main disadvantages of this technology are safety requirements and low charging cycle efficiency.

Hydrogen can be converted into methane or Synthetic Natural Gas (SNG). The main advantage of this as an energy storage technology is that the amount of hydrogen that can be tolerated in the existing natural gas infrastructure is limited. Furthermore, there are no restrictions on the amount of SNG that can be injected into the existing gas infrastructure.

Supercapacitors are also known as ultracapacitors. It is an established technology which stores much more energy per unit volume or mass than traditional capacitors. The response time is extremely fast. Supercapacitors store energy in the electric field between a pair of charged plates. They contain enlarged electrodes, liquid electrolyte, and a polymer membrane. Supercapacitors are capable of charging and discharging in a relatively less time and, hence can be used for instantaneous power supply. The efficiency levels achieved with a supercapacitor are close to 95.0%. However, the costs per unit of the energy storage capacity are higher than the cost of batteries.

FIGURE 37 NORTH AMERICA HAS THE LARGEST SHARE IN THE OTHER ADVANCED ENERGY STORAGE SYSTEMS MARKET



e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 25 OTHERS: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017-e	2022-p	CAGR (2017–2022)
North America	23.1	24.6	26.1	33.7	5.25%
Europe	2.7	2.8	2.9	3.3	2.86%
Asia Pacific	2.9	2.9	2.9	3.1	1.22%
Rest of the World	0.9	1.0	1.0	1.1	2.09%
Total	29.6	31.3	32.9	41.3	4.62%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

Source: SEC Filings, Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

North America is estimated to hold the largest market share of 79.2% in the other energy storage market in 2017. The region's market is expected to be USD 26.1 million in 2017 and is projected to reach a market size of USD 33.7 million by 2022, growing at a CAGR of 5.25% from 2017 to 2022.

8 ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY REGION

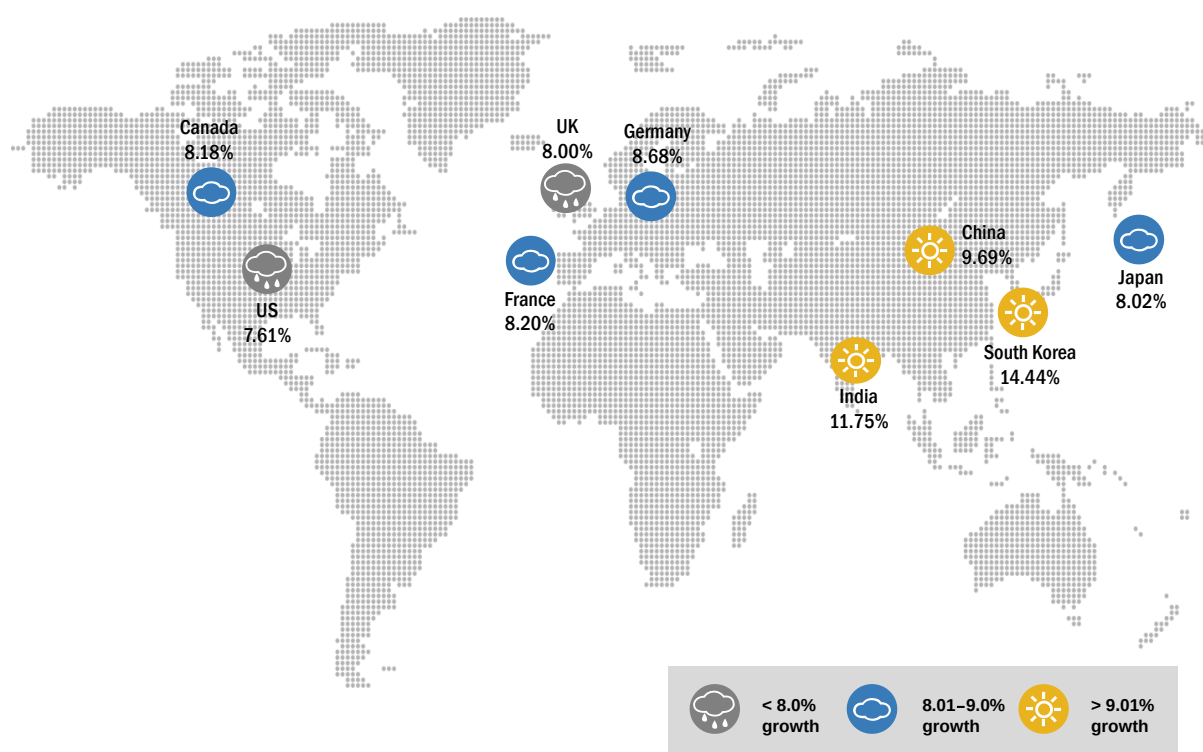
KEY FINDINGS

- In 2017, the market in Asia Pacific is estimated to account for the largest share of 37.0%, of the global AESS market at USD 4,716.3 million. It is projected to grow at the highest CAGR of 9.79% during the forecast period.
- The market in South Korea is expected to witness the fastest growth in in the AESS market in Asia Pacific and is projected to grow at a CAGR of 14.44%, during the forecast period.
- The European AESS market is projected to grow at the second highest CAGR of 8.52%, during the forecast period.
- The need for commercial and residential applications is expected to create huge opportunities in the AESS market.
- Increasing investments in renewables and regulations demanding energy efficiency are the key drivers for the AESS market.

8.1 INTRODUCTION

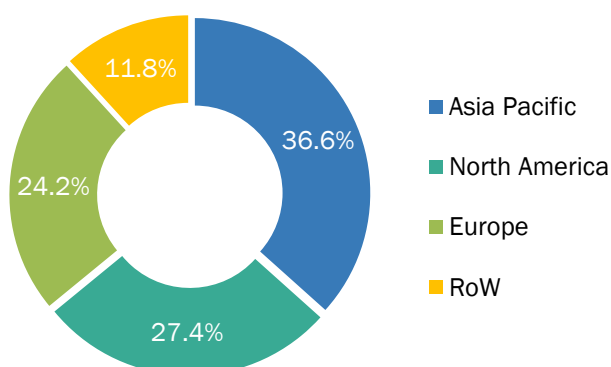
The global AEES market has been categorized into regions, namely, Asia Pacific, Europe, North America, and the rest of the world. In 2017, the market in Asia Pacific held the largest share of the global advanced energy storage systems market due to increasing renewable energy investments, regulations demanding energy efficiency, and electrification of the transport sector.

FIGURE 38 REGIONAL SNAPSHOT: SOUTH KOREA, CHINA AND INDIA ARE THE FASTEST GROWING COUNTRIES, BY VALUE, DURING THE FORECAST PERIOD



Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 39 ADVANCED ENERGY STORAGE SYSTEMS MARKET SHARE (VALUE), BY REGION, 2017



Note: The rest of the world (RoW) includes Middle Eastern, African, and South American countries.

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

In 2017, the market in Asia Pacific is estimated to account for the largest share of 37.0% of the global AEES market, while the North American market is expected to hold the second largest share of around 27.3%.

TABLE 26 ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY REGION, 2015– 2022 (USD MILLION)

Region	2015	2016	2017-e	2022	CAGR (2017-22)
Asia Pacific	3,956.9	4,321.1	4,716.3	7,522.3	9.79%
Europe	2,631.2	2,850.2	3,087.3	4,646.7	8.52%
North America	3,001.7	3,231.9	3,479.6	5,027.6	7.64%
Rest of the world	1,328.1	1,388.8	1,452.1	1,847.3	4.93%
Total	10,918.0	11,792.0	12,735.4	19,043.8	8.38%

e- Estimated; p- Projected

Note: The rest of the world includes Middle Eastern, African, and South American countries

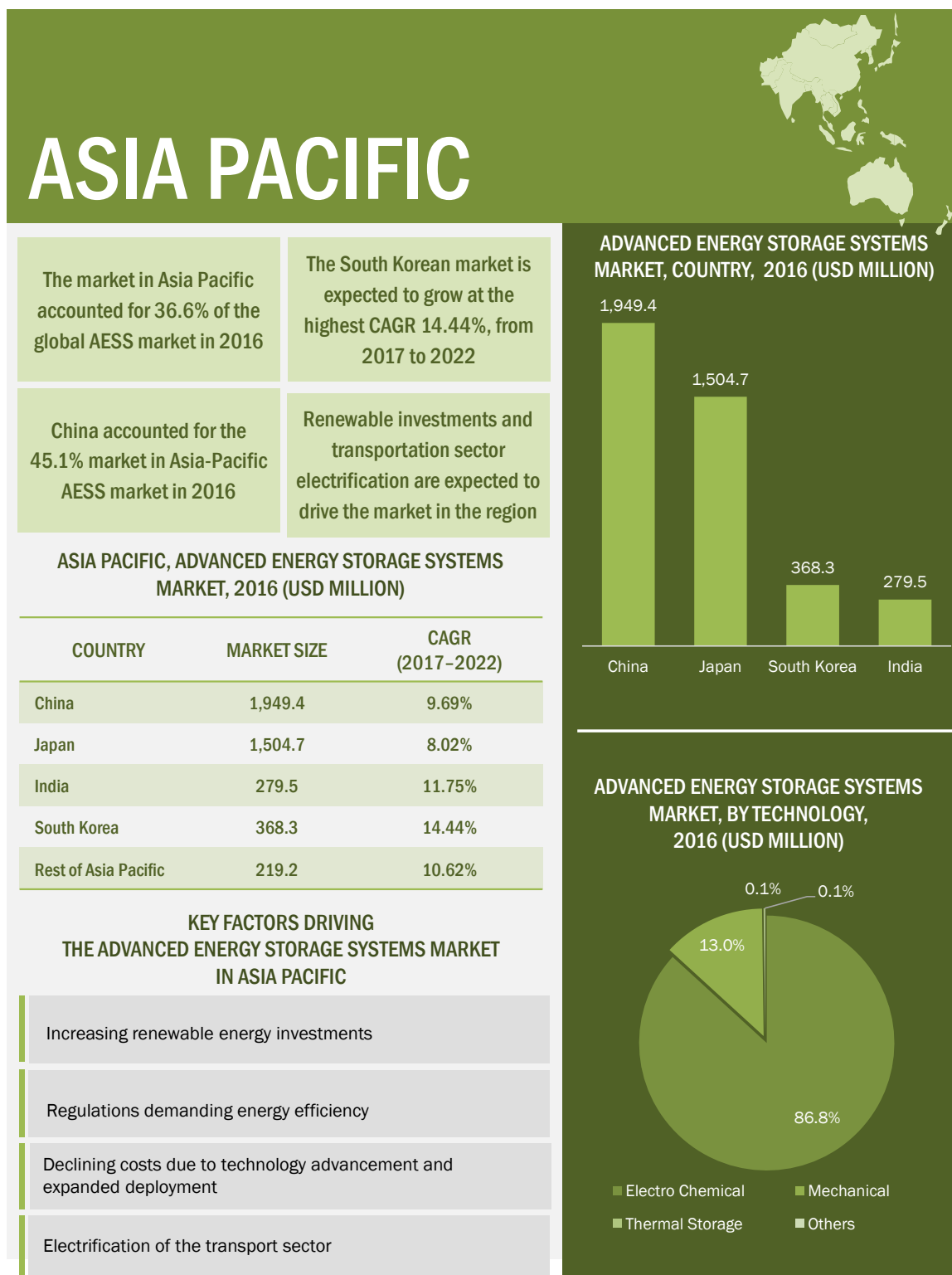
Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market in Asia Pacific for AEES was the largest at USD 4,321.1 million in 2016 and is projected to grow at the highest CAGR of 9.79%, from 2017 to 2022. The North American market is expected to reach USD 5,027.6 million by 2022, at a CAGR of 7.64%, from 2017 to 2022.

8.2 ASIA PACIFIC

The AEES market in Asia Pacific is primarily driven by the markets in countries, such as China, Japan, South Korea, and India. Increasing renewable energy investments, regulations demanding energy efficiency, electrification of the transport sector, and declining costs because of technology advancement and expanded deployment in developing countries of the Asia Pacific region, have driven the AEES market.

FIGURE 40 ASIA PACIFIC, ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT



Note: The rest of Asia Pacific includes Malaysia, Singapore, the Philippines, Australia, and Vietnam

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 27 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY COUNTRY, 2015–2022 (USD MILLION)

Country	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Japan	1,400.5	1,504.7	1,615.9	2,376.4	8.02%
China	1,786.7	1,949.4	2,125.7	3,375.0	9.69%
South Korea	319.1	368.3	423.1	830.5	14.44%
India	251.4	279.5	310.5	541.3	11.75%
Rest of Asia Pacific	199.2	219.2	241.0	399.2	10.62%
Total	3,956.9	4,321.1	4,716.3	7,522.3	9.79%

Note: The Rest of Asia Pacific includes Malaysia, Singapore, the Philippines, Australia and Vietnam

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market in Asia Pacific is estimated to be the largest market at USD 4,716.3 million in 2017 and is projected to reach USD 7,522.3 million by 2022, at a CAGR of 9.79%, from 2017 to 2022. The market in China is expected to reach USD 3,375.0 million by 2022, growing at the highest CAGR of 9.69%, during the forecast period.

TABLE 28 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	3,029.7	3,327.2	3,678.8	6,093.1	10.62%
Grid Storage	927.3	993.9	1,037.6	1,429.2	6.61%
Total	3,956.9	4,321.1	4,716.3	7,522.3	9.79%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for AESS, by application, is estimated to be USD 4,716.3 million in 2017 and is projected to reach USD 7,522.3 million by 2022, at a CAGR 9.79%, during the forecast period. The market for transportation is projected to reach USD 6,093.1 million by 2022, at a CAGR of 10.62%, during the forecast period.

TABLE 29 ASIA PACIFIC: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion	2,417.6	2,680.2	2,991.5	5,194.1	11.67%
NiMH and NiCd	612.1	647.0	687.3	899.0	5.52%
Total	3,029.7	3,327.2	3,678.8	6,093.1	10.62%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for transportation application for the AESS market, by technology, is estimated to be USD 3,678.8 million in 2017 and is projected to reach USD 6,093.1 million by 2022, at a CAGR of 10.62%, during the forecast period. The market for lithium-ion batteries is projected to reach USD 5,194.1 million by 2022, at a CAGR of 11.67%, during the forecast period. The market for the NiMH and NiCd in technology application is projected to reach USD 899 million by 2022, at a CAGR of 5.52%, during the forecast period.

TABLE 30 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015-2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017-2022)
Electro chemical	3,412.7	3,752.0	4,128.9	6,802.8	10.50%
Thermal storage	2.8	2.9	2.9	3.1	1.59%
Mechanical	538.6	563.3	581.7	713.3	4.17%
Others	2.9	2.9	2.9	3.1	1.22%
Total	3,956.9	4,321.1	4,716.3	7,522.3	9.79%

e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor.

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for AESS, by technology, is estimated to be USD 4,716.3 million in 2017 and is projected to reach USD 7,522.3 million by 2022, at a CAGR of 9.79%, during the forecast period. The market for electro chemical is projected to reach USD 6,820.8 million by 2022, at a CAGR of 10.50%, during the forecast period. The market for mechanical is projected to reach USD 713.3 million by 2022, at a CAGR of 4.17%, during the forecast period.

TABLE 31 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015-2022 (USD MILLION)

Technology	2015	2016	2017	2022	CAGR (2017-2022)
Lithium-ion battery	2,727.9	3,028.0	3,362.1	5,807.7	11.55%
Lead acid battery	2.8	3.0	3.0	3.3	1.65%
Sodium sulfur	47.9	50.8	52.4	61.1	3.11%
Flow battery	21.9	23.3	24.1	31.7	5.65%
NiMH and NiCd	612.1	647.0	687.3	899.0	5.52%
Total	3,412.7	3,752.0	4,128.9	6,802.8	10.50%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

The market for AESS, by electro chemical technology, is estimated to be USD 4,128.9 million in 2017 and is projected to reach USD 6,802.8 million by 2022, at a CAGR of 10.50%, during the forecast period. The market for Lithium-ion battery is projected to reach USD 5,807.7 million by 2022, at a CAGR of 11.55%, during the forecast period. The market for NiCd and NiMH batteries is projected to reach USD 899.0 million by 2022, at a CAGR of 5.52%, during the forecast period.

TABLE 32 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD THOUSAND)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Sensible thermal	739.7	752.7	745.9	792.1	1.21%
Latent thermal	2,077.7	2,124.9	2,116.7	2,306.1	1.73%
Total	2,817.5	2,877.5	2,862.7	3,098.2	1.59%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for latent thermal had the largest share of 73.8% in 2016 and is projected to grow at a CAGR of 1.73%, during the forecast period. The sensible thermal market is expected to grow at a CAGR of 1.21% during the forecast period.

TABLE 33 ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Pumped Hydro	532.7	557.1	575.5	706.0	4.17%
Flywheel	5.9	6.1	6.2	7.3	3.27%
Total	538.6	563.3	581.7	713.3	4.17%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for AESS, by mechanical technology, is estimated to be USD 581.7 million in 2017 and is projected to reach USD 713.3 million by 2022, at a CAGR of 4.17%, during the forecast period. The market for pumped hydro is projected to reach USD 706.0 million by 2022, at a CAGR of 4.17%, during the forecast period. The market for flywheel is projected to reach USD 7.3 million by 2022, at a CAGR of 3.27% during the forecast period.

8.2.1 CHINA

China has the second largest pumped hydro plant in the world, including 2400 MW in the Southern Grid near Hong Kong, and 1800 MW near Shanghai in the Eastern Grid. The country is still looking to invest in battery storage in large amount. The country has the world's longest high-speed rail operation and is trending towards EVs for private transport. By 2013, the country installed 57MW of demonstration projects, however larger projects are now underway. The country also has major share of the electric car market, as it has more sale for the electric car market, when compared to other countries. Thus, EV market will play a large role in the growing demand for advanced battery systems, supporting the development of new large manufacturing facilities.

TABLE 34 CHINA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–22)
Transportation	1,345.9	1,470.8	1,606.5	2,571.3	9.86%
Grid Storage	440.8	478.5	519.2	803.6	9.13%
Total	1,786.7	1,949.4	2,125.7	3,375.0	9.69%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The Chinese AESS market, by application, is projected to reach USD 3,375.0 million by 2022, from an estimated USD 2,125.7 million in 2017, at a CAGR of 9.69%, during the forecast period. The transportation segment is estimated to be the largest market at USD 1,606.5 million in 2017 and is projected to reach USD 2,571.3 million by 2022, at a CAGR of 9.86% from 2017 to 2022. The grid storage segment is expected to reach USD 803.6 million by 2022, growing at a CAGR of 9.13%, during the forecast period.

8.2.2 INDIA

The Indian AESS market is expected to grow in coming years as there is a tremendous growth potential for the wind and solar energy, with the growing interest in micro-grids. The market in the country is gearing up with large number of pilot projects and large number of sales for lead acid batteries. The government is integrating 160 MW of wind and solar by 2022 and plans to rollout EVs by 2020. The Indian government is also aiming to support EV initiatives in 2017 for the increase in electric cars. Thus, the growing factors is the key reason for the increase in AESS market.

TABLE 35 INDIA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–22)
Transportation	75.3	84.0	93.7	166.2	12.15%
Grid Storage	176.1	195.5	216.9	375.1	11.58%
Total	251.4	279.5	310.5	541.3	11.75%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The Indian AESS market, by application, is projected to reach USD 541.3 million by 2022, from an estimated USD 310.5 million in 2017, at a CAGR of 11.75%, during the forecast period. The grid storage segment is estimated to be the largest market at USD 216.9 million in 2017 and is projected to reach USD 375.1 million by 2022, at a CAGR of 11.58% from 2017 to 2022. The grid storage segment is expected to reach USD 166.2 million by 2022, growing at the highest CAGR of 12.15% during the forecast period.

8.2.3 SOUTH KOREA

The market in South Korea is the major manufacturer of batteries. It is also involved in the promotion of grid-scale and distributed energy storage for load leveling and frequency regulation. The country has a small solar and wind capacity at present. The energy storage in the country is driven by industrial and macro energy policy. South Korea is among the top energy importers as it has extreme shortage of domestic energy reserves. The South Korean government has also announced major plans for the new capacity of energy storage. Thus, new capacity addition can lead to increase in demand for the AESS market.

TABLE 36 SOUTH KOREA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–22)
Transportation	126.4	146.2	168.2	333.0	14.64%
Grid Storage	192.7	222.2	254.9	497.4	14.31%
Total	319.1	368.3	423.1	830.5	14.44%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market in South Korea for AESS, by application, is projected to reach USD 830.5 million by 2022, from an estimated USD 423.1 million in 2017, at a CAGR of 14.44%, during the forecast period. The grid storage segment is estimated to be the largest market at USD 254.9 million in 2017 and is projected to reach USD 497.4 million by 2022, at a CAGR of 14.31%, from 2017 to 2022. The transportation segment is expected to reach USD 333.0 million by 2022, growing at the highest CAGR of 14.64%, during the forecast period.

8.2.4 JAPAN

Japan is the potentially attractive market as the country is a leader in energy storage technology and electronics. The country is shifting to rapid expansion in the solar and wind power, which will increase the energy storage in renewable integration. The government and industrial base are investing in energy storage applications. The country also had considerable growth in the electric car stock between 2015 and 2016. Thus, the AESS market in Japan is expected to grow in the years to come.

TABLE 37 JAPAN: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–22)
Transportation	596.6	639.8	686.5	1,002.8	7.87%
Grid Storage	803.9	864.9	929.4	1,373.6	8.13%
Total	1,400.5	1,504.7	1,615.9	2,376.4	8.02%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The Japanese advanced storage systems market, by application, is projected to reach USD 2,376.4 million by 2022, from an estimated USD 1,615.9 million in 2017, at a CAGR of 8.02%, during the forecast period. The grid storage segment is estimated to be the largest market at USD 929.4 million in 2017 and is projected to reach USD 1,373.6 million by 2022, at a CAGR of 8.13%, from 2017 to 2022. The

transportation segment is expected to reach USD 1,002.8 million by 2022, growing at a CAGR of 7.87%, during the forecast period.

8.2.5 THE REST OF ASIA PACIFIC

The rest of Asia Pacific includes Malaysia, Singapore, the Philippines, Australia and Vietnam among others. In 2016, the rest of Asia Pacific accounted for 5.1% market share of the total AESS market in the region. It is expected to reach USD 399.2 million by 2022, at a CAGR of 10.6% during the forecast period.

TABLE 38 THE REST OF ASIA PACIFIC: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	30.0	33.4	37.1	65.0	11.86%
Grid Storage	169.2	185.8	203.9	334.2	10.39%
Total	199.2	219.2	241.0	399.2	10.62%

e- Estimated; p- Projected

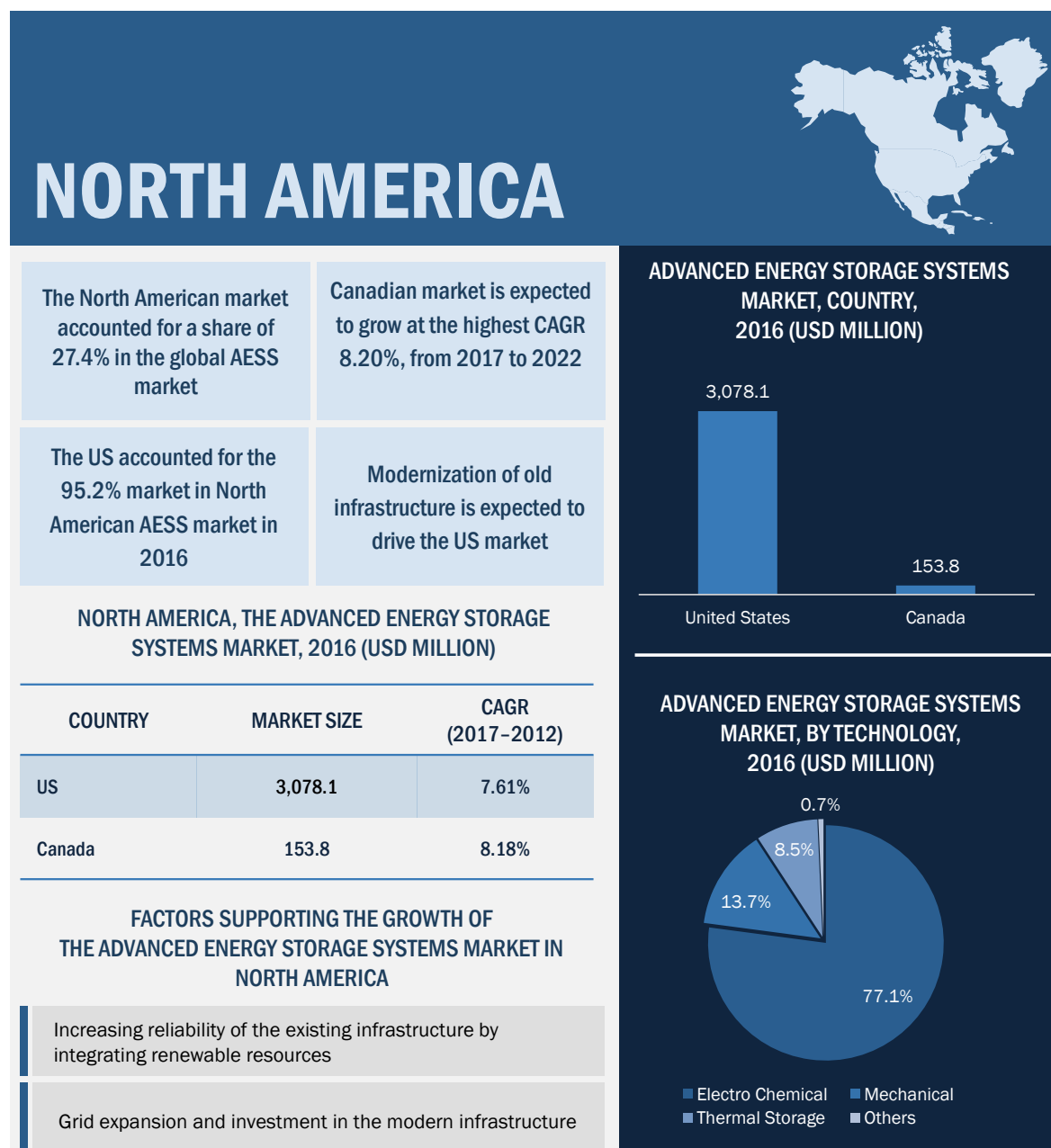
Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The transportation segment is estimated to be at USD 37.1 million in 2017 and is projected to reach USD 65.0 million by 2022, at a CAGR of 11.86 %, from 2017 to 2022. The grid storage segment market is expected to reach USD 334.2 million by 2022, growing at a CAGR of 10.39%, during the forecast period.

8.3 NORTH AMERICA

The market in North America is expected to account for the second largest share of 27.3% in 2017, projected to hold on its position during the forecast period. The New York Independent System Operator (NYISO) is encouraging the market to grow along with some other projects, such as Substation Energy Storage Project and Waiawa High PV Penetration Circuit, among others.

FIGURE 41 NORTH AMERICA, ADVANCED ENERGY STORAGE SYSTEMS MARKET SNAPSHOT



Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 39 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY COUNTRY, 2015–2022 (USD MILLION)

Country	2015	2016	2017-e	2022-p	CAGR (2017–2022)
United States	2,859.6	3,078.1	3,313.1	4,780.9	7.61%
Canada	142.1	153.8	166.5	246.6	8.18%
Total	3,001.7	3,231.9	3,479.6	5,027.6	7.64%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by country, is projected to reach USD 5,027.6 million by 2022, from an estimated USD 3,479.6 million in 2017, at a CAGR of 7.64%, during the forecast period. The U.S is estimated to be the largest market at USD 3,313.1 million in 2017 and is projected to reach USD 4,780.9 million by 2022, at a CAGR of 7.61%, from 2017 to 2022.

TABLE 40 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–22)
Transportation	1,956.8	2,112.8	2,281.1	3,463.1	8.71%
Grid Storage	1,044.9	1,119.1	1,198.5	1,564.5	5.47%
Total	3,001.7	3,231.9	3,479.6	5,027.6	7.64%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by application, is projected to reach USD 5,027.6 million by 2022, from an estimated USD 3,479.6 million in 2017, at a CAGR of 7.64%, during the forecast period. The transportation market is estimated to be the largest market at USD 2,281.1 million in 2017 and is projected to reach USD 3,463.1 million by 2022, at a CAGR of 8.71%, from 2017 to 2022. The grid storage segment is expected to reach USD 1,564.5 million by 2022, growing at a CAGR of 5.47%, during the forecast period.

TABLE 41 NORTH AMERICA: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion	1,561.5	1,701.9	1,854.9	2,952.1	9.74%
NiMH and NiCd	395.3	410.8	426.2	510.9	3.70%
Total	1,956.8	2,112.8	2,281.1	3,463.1	8.71%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American transportation AESS market is projected to reach USD 3,463.1 million by 2022, from an estimated USD 2,281.1 million in 2017, at a CAGR of 8.71%, during the forecast period. The lithium-ion battery technology market in the transportation application is estimated to be the largest market at USD 1,854.9 million in 2017 and is projected to reach USD 2,952.1 million by 2022, at a CAGR of 9.74%, from

2017 to 2022. The NiMH and NiCd market for the transportation application is expected to reach USD 510.9 million by 2022, growing at the CAGR of 3.70%, during the forecast period.

TABLE 42 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017	2022	CAGR (2017-22)
Electro chemical	2,295.0	2,486.9	2,695.8	4,102.9	8.76%
Thermal storage	261.5	274.4	288.0	339.8	3.36%
Mechanical	422.2	446.0	469.8	551.2	3.25%
Others	23.1	24.6	26.1	33.7	5.25%
Total	3,001.7	3,231.9	3,479.6	5,027.6	7.64%

e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by technology, is projected to reach USD 5,027.6 million by 2022, from an estimated USD 3,479.6 million in 2017, at a CAGR of 7.64%, during the forecast period. The electro chemical market is estimated to be the largest market at USD 2,695.8 million in 2017 and is projected to reach USD 4,102.9 million by 2022, at a CAGR of 8.76%, from 2017 to 2022.

TABLE 43 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion battery	1,807.5	1,978.4	2,166.0	3,467.8	9.87%
Lead acid battery	27.5	29.1	30.8	38.1	4.31%
Sodium sulfur	21.3	22.7	24.3	31.1	5.10%
Flow battery	43.4	45.9	48.6	55.0	2.50%
NiMH and NiCd	395.3	410.8	426.2	510.9	3.70%
Total	2,295.0	2,486.9	2,695.8	4,102.9	8.76%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by electro chemical technology, is projected to reach USD 4,102.9 million by 2022, from an estimated USD 2,695.8 million in 2017, at a CAGR of 8.76%, during the forecast period. The lithium-ion battery segment is estimated to be the largest market at USD 2,166.0 million in 2017 and is projected to reach USD 3,467.8 million by 2022. The lithium-ion battery segment is projected to grow at the highest CAGR of 9.87%, during the forecast period.

TABLE 44 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD THOUSAND)

Technology	2015	2016	2017	2022	CAGR (2017-22)
Thermo chemical	260.2	279.8	309.2	330.3	1.33%
Sensible thermal	36,480.7	38,956.4	41,596.5	53,503.7	5.16%
Latent thermal	2,24,732.2	2,35,160.5	2,46,050.5	2,85,932.3	3.05%
Total	2,61,473.1	2,74,396.8	2,87,956.2	3,39,766.3	3.36%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by thermal storage, is projected to grow at a CAGR of 3.36%, during the forecast period. The latent thermal segment is the largest segment with a market share of 85.7% in 2016 and is projected to grow at a CAGR of 3.05%, from 2017 to 2022. The sensible thermal segment is expected to grow at the highest CAGR of 5.16%, during the forecast period.

TABLE 45 NORTH AMERICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Pumped Hydro	206.9	217.2	227.9	268.8	3.36%
Flywheel	162.0	172.3	182.2	211.2	3.00%
CAES and LAES	53.3	56.5	59.7	71.2	3.58%
Total	422.2	446.0	469.8	551.2	3.25%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The North American AESS market, by mechanical technology, is projected to reach USD 551.2 million by 2022, from an estimated USD 469.8 million in 2017, growing at a CAGR of 3.25%, during the forecast period. The pumped hydro technology segment is estimated to be the largest market at USD 227.9 million in 2017 and is projected to reach USD 268.8 million by 2022, at a CAGR of 3.36%, from 2017 to 2022. The CAES and LAES segment is expected to reach USD 71.2 million by 2022, growing at the highest CAGR of 3.58%, during the forecast period.

8.3.1 US

The US is considered as a world leader for innovation in energy technology and energy policy. Overall, the country has advanced designs for generation markets, capacity markets, transmissions, ancillary services, and renewable energy credits. The US has supporting policies such as the federal policy for energy storage, which will help to meet the national energy goals with improved reliability, higher system efficiency, and with better integration of renewable energy. According to EV Outlook, 2017, country accounted for the largest portion of the global electric car stock.

TABLE 46 US: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	1,800.7	1,947.2	2,105.5	3,107.5	8.10%
Grid Storage	1,058.9	1,130.9	1,207.7	1,673.4	6.74%
Total	2,859.6	3,078.1	3,313.1	4,780.9	7.61%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The US AESS market, by application, is projected to reach USD 4,780.9 million by 2022, from an estimated USD 3,313.1 million in 2017, at a CAGR of 7.61%, during the forecast period. The transportation segment is estimated to be the largest market at USD 2,105.5 million in 2017 and is projected to reach USD 3,107.5 million by 2022, at a CAGR of 8.10%, from 2017 to 2022. The grid storage segment is expected to reach USD 1,673.4 million by 2022, growing at the CAGR of 6.74%, during the forecast period.

8.3.2 CANADA

The energy storage in Canada enables substantial carbon emission reductions, by making the electricity grid more efficient, integrating renewables, and allowing clean and renewable electricity to be injected into other parts of Canada's energy system. The AESS increases the reliability and resiliency of local grids. Energy Storage Ontario (ESO) is the organization that represents a broad range of companies engaged in the energy storage business in Ontario and across Canada.

TABLE 47 CANADA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	107.7	116.8	126.6	189.4	8.38%
Grid Storage	34.4	37.0	39.8	57.2	7.53%
Total	142.1	153.8	166.5	246.6	8.18%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The Canadian AESS market, by application, is projected to reach USD 246.6 million by 2022, from an estimated USD 166.5 million in 2017, at a CAGR of 8.18%, during the forecast period. The transportation segment is estimated to be the largest market at USD 126.6 million in 2017 and is projected to reach USD 189.4 million by 2022, at a CAGR of 8.38%, from 2017 to 2022. The grid storage segment is expected to reach USD 57.2 million by 2022, growing at the CAGR of 7.53%, during the forecast period.

8.4 EUROPE

Europe has various opportunities for energy storage such as policies on environmental and carbon emission which promotes renewable energy, increasing the need for storage. The European government is also taking steps for market structure and incentives for the promotion of energy storage.

TABLE 48 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY COUNTRY, 2015–2022 (USD MILLION)

Country	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Germany	800.0	867.8	941.3	1,426.9	8.68%
France	505.5	545.9	589.6	874.1	8.20%
United Kingdom	305.9	329.8	355.5	522.5	8.00%
Rest of Europe	1,019.8	1,106.7	1,200.9	1,823.2	8.71%
Total	2,631.2	2,850.2	3,087.3	4,646.7	8.52%

e- Estimated; p- Projected

Note: The rest of Europe includes Norway, Sweden, Spain and Italy.

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by region, is projected to reach USD 4,646.7 million by 2022, from an estimated USD 3,087.3 million in 2017, at a CAGR of 8.52%, during the forecast period. Germany is estimated to be the largest market at USD 941.3 million in 2017 and is projected to reach USD 1,426.9 million by 2022, at a CAGR of 8.68%, from 2017 to 2022. The rest of Europe is expected to reach USD 1,823.2 million by 2022, growing at a CAGR of 8.71%, during the forecast period.

TABLE 49 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	1,905.2	2,071.8	2,252.8	3,494.2	9.18%
Grid Storage	726.0	778.4	834.5	1,152.5	6.67%
Total	2,631.2	2,850.2	3,087.3	4,646.7	8.52%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by application, is projected to reach USD 4,646.7 million by 2022, from an estimated USD 3,087.3 million in 2017, at a CAGR of 8.52%, during the forecast period. The transportation segment is estimated to be the largest market at USD 2,252.8 million in 2017 and is projected to reach USD 3,494.2 million by 2022, at a CAGR of 9.18%, from 2017 to 2022. The grid storage segment is expected to reach USD 1,152.5 million by 2022, growing at the CAGR of 6.67%, during the forecast period.

TABLE 50 EUROPE: TRANSPORTATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion	1,520.3	1,668.9	1,832.0	2,978.7	10.21%
NiMH and NiCd	384.9	402.9	420.9	515.5	4.14%
Total	1,905.2	2,071.8	2,252.8	3,494.2	9.18%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European transportation AESS market, by technology, is projected to reach USD 3,494.2 million by 2022, from an estimated USD 2,252.8 million in 2017, at a CAGR of 9.18%, during the forecast period. The lithium-ion batteries segment is estimated to be the largest market at USD 1,832.0 million in 2017 and is projected to reach USD 2,978.7 million by 2022, at a CAGR of 10.21%, from 2017 to 2022. The NiMH and NiCd segment is expected to reach USD 515.5 million by 2022, growing at a CAGR of 4.14%, during the forecast period.

TABLE 51 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Electro chemical	2,153.5	2,339.2	2,549.2	3,949.7	9.15%
Thermal storage	15.3	16.3	17.4	23.1	5.84%
Mechanical	459.7	491.9	517.8	670.6	5.31%
Others	2.7	2.8	2.9	3.3	2.86%
Total	2,631.2	2,850.2	3,087.3	4,646.7	8.52%

e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by technology, is projected to reach USD 4,646.7 million by 2022, from an estimated USD 3,087.3 million in 2017, at a CAGR of 8.52%, during the forecast period. The electro chemical segment is estimated to be the largest market at USD 2,549.2 million in 2017 and is projected to reach USD 3,949.7 million by 2022, at a CAGR of 9.15%, from 2017 to 2022. The thermal storage segment is expected to reach USD 23.1 million by 2022, growing at a CAGR of 5.84%, during the forecast period.

TABLE 52 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTRO CHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion battery	1,715.1	1,879.3	2,067.5	3,352.1	10.15%
Lead acid battery	5.6	6.0	6.4	8.8	6.46%
Sodium sulfur	47.0	50.1	53.4	72.0	6.14%
Flow battery	0.9	1.0	1.0	1.3	4.94%
NiMH and NiCd	384.9	402.9	420.9	515.5	4.14%
Total	2,153.5	2,339.2	2,549.2	3,949.7	9.15%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by electro chemical technology, is projected to reach USD 3,949.7 million by 2022, from an estimated USD 2,549.2 million in 2017, at a CAGR of 9.2%, during the forecast period. The lithium-ion battery segment is estimated to be the largest market at USD 2,067.5 million in 2017 and is projected to reach USD 3,352.1 million by 2022, at a CAGR of 10.15%, from 2017 to 2022. The lead acid battery segment is expected to reach USD 8.8 million by 2022, growing at a CAGR of 6.46%, during the forecast period.

TABLE 53 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Sensible thermal	4.0	4.3	4.6	6.0	5.76%
Latent thermal	11.3	12.0	12.8	17.0	5.87%
Total	15.3	16.3	17.4	23.1	5.84%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by thermal storage technology, is projected to grow at a CAGR of 5.84%, during the forecast period. The latent thermal battery segment had the largest market share of 73.8% in 2016 and is projected to grow at a CAGR of 5.87%, from 2017 to 2022. The sensible thermal segment is expected to grow at a CAGR of 5.76%, during the forecast period.

TABLE 54 EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Pumped Hydro	435.4	466.4	491.3	638.6	5.39%
Flywheel	13.5	14.1	14.8	18.4	4.40%
CAES and LAES	10.9	11.3	11.7	13.6	3.09%
Total	459.7	491.9	517.8	670.6	5.31%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The European AESS market, by mechanical technology, is projected to reach USD 670.6 million by 2022, from an estimated USD 517.8 million in 2017, at a CAGR of 5.31%, during the forecast period. The pumped hydro segment is estimated to be the largest market at USD 491.3 million in 2017 and is projected to reach USD 638.6 million by 2022, at a CAGR of 5.39%, from 2017 to 2022. The flywheel segment is expected to reach USD 18.4 million by 2022, growing at a CAGR of 4.40%, during the forecast period.

8.4.1 THE UK

The UK experience pumped hydro and battery-based storage with advanced deregulated electricity markets. The country has major plans for retirements of the conventional thermal generation with new approaches for integrating renewable energy. The country has decent experiences with energy storage and well-established R&D for development of battery technology. The country is also promoting for the deployment of new energy technologies, which will help to increase the AESS market. The country registered maximum electric cars in 2016.

TABLE 55 THE UK: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	245.9	265.9	287.4	428.5	8.31%
Grid Storage	60.0	64.0	68.1	94.0	6.66%
Total	305.9	329.8	355.5	522.5	8.00%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The UK AESS market, by application, is projected to reach USD 522.5 million by 2022, from an estimated USD 355.5 million in 2017, at a CAGR of 8.00%, during the forecast period. The transportation segment is estimated to be the largest market at USD 287.4 million in 2017 and is projected to reach USD 428.5 million by 2022, at a CAGR of 8.31%, from 2017 to 2022. The grid storage segment is expected to reach USD 94.0 million by 2022, growing at a CAGR of 6.66%, during the forecast period.

8.4.2 GERMANY

Germany has interconnected grids in the European region. The country also exchanges electricity between the neighboring countries. In Germany, the government financially supports the R&D in energy storage and also facilitates the development of energy storage projects given the growing need for energy storage in the country. This will increase the AESS market in the country.

TABLE 56 GERMANY: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	594.4	645.6	701.0	1,073.0	8.89%
Grid Storage	205.5	222.2	240.3	353.9	8.05%
Total	800.0	867.8	941.3	1,426.9	8.68%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The AESS market in Germany, by application, is projected to reach USD 1,426.9 million by 2022, from an estimated USD 941.3 million in 2017, at a CAGR of 8.68%, during the forecast period. The transportation segment is estimated to be the largest market at USD 701.0 million in 2017 and is projected to reach USD 1,073.0 million by 2022, at a CAGR of 8.89%, from 2017 to 2022. The grid storage segment is expected to reach USD 353.9 million by 2022, growing at a CAGR of 8.05%, during the forecast period.

8.4.3 FRANCE

France is a leading country in energy storage technologies. The country also has number of small projects, undergoing outside of the metropolitan. The country also leads in battery manufacturing, including lithium-ion batteries. The French government plans for the reduction of nuclear targeting, increasing the renewable energy, which would drive the energy storage systems market in the country.

TABLE 57 FRANCE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	426.1	461.1	499.0	747.4	8.42%
Grid Storage	79.4	84.8	90.6	126.8	6.95%
Total	505.5	545.9	589.6	874.1	8.20%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The France AESS market, by application, is projected to reach USD 874.1 million by 2022, from an estimated USD 589.6 million in 2017, at a CAGR of 8.20%, during the forecast period. The transportation segment is estimated to be the largest market at USD 499.0 million in 2017 and is projected to reach USD 747.4 million by 2022, at a CAGR of 8.42%, from 2017 to 2022. The grid storage segment is expected to reach USD 126.8 million by 2022, growing at a CAGR of 6.95%, during the forecast period.

8.4.4 THE REST OF EUROPE

The rest of Europe consists of countries, such as Norway, Sweden, Spain and Italy. It is expected to account for 38.9% of the European AESS market in 2017. Its market size is expected to reach USD 1,823.2 million by 2022, at a CAGR of 8.71% during the forecast period. The transportation sector is the major factor behind the growth of the AESS market in the region.

TABLE 58 THE REST OF EUROPE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	572.2	621.9	675.8	1,033.8	8.87%
Grid Storage	447.6	484.8	525.1	789.4	8.50%
Total	1,019.8	1,106.7	1,200.9	1,823.2	8.71%

e- Estimated; p- Projected

Source: Company Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The transportation segment is estimated to be the largest market at USD 675.8 million in 2017 and is projected to reach USD 1,033.8 million by 2022, at a CAGR of 8.87%, from 2017 to 2022. The grid storage segment is expected to reach USD 789.4 million by 2022, growing at a CAGR of 8.50%, during the forecast period.

8.5 REST OF THE WORLD

The rest of the world includes Middle Eastern, African, and South American countries. The AESS market is segmented based on application, technology, and country. The market, by technology, has been segmented into electro chemical, thermal storage, mechanical, and others. The market based on application has been segmented into transportation and grid storage segments.

TABLE 59 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET SIZE, BY COUNTRY, 2015–2022 (USD MILLION)

Country	2015	2016	2017-e	2022-p	CAGR (2017–2022)
South Africa	400.4	423.4	447.5	600.2	6.05%
Chile	325.2	341.3	358.2	464.3	5.32%
Others	602.5	624.1	646.4	782.8	3.90%
Total	1,328.1	1,388.8	1,452.1	1,847.3	4.93%

e- Estimated; p- Projected

Note: The rest of the world includes other Middle Eastern, African, and South American countries

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market in RoW is estimated to be at USD 1,452.1 million in 2017 and is projected to reach USD 1,847.3 million by 2022, at a CAGR of 4.93%, from 2017 to 2022. The market in South Africa is expected to reach USD 600.2 million by 2022, growing at the highest CAGR of 6.05%, during the forecast period.

TABLE 60 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	149.0	156.7	164.7	215.3	5.49%
Grid Storage	1,179.1	1,232.1	1,287.4	1,632.0	4.86%
Total	1,328.1	1,388.8	1,452.1	1,847.3	4.93%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for RoW AESS, by application, is estimated to be USD 1,452.1 million in 2017 and is projected to reach USD 1,847.3 million by 2022, at a CAGR of 4.93%, during the forecast period. The market for transportation is projected to reach USD 215.3 million by 2022, at a CAGR of 5.49%, during the forecast period.

TABLE 61 REST OF THE WORLD: TRANSPORTATION APPLICATION ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion	118.8	125.9	133.4	181.3	6.33%
NiMH and NiCd	30.3	30.8	31.4	34.0	1.60%
Total	149.0	156.7	164.7	215.3	5.49%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The lithium-ion batteries market in RoW for the transportation application is projected to reach USD 181.3 million by 2022, at a CAGR of 6.33%, during the forecast period. The market for NiMH and NiCd batteries for the transportation application in RoW is projected to reach USD 34.0 million by 2022, at a CAGR of 1.60%, during the forecast period.

TABLE 62 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Electro chemical	331.5	346.7	362.7	462.3	4.97%
Thermal storage	290.2	304.1	318.7	410.5	5.19%
Mechanical	705.5	737.0	769.7	973.4	4.81%
Others	0.9	1.0	1.0	1.1	2.09%
Total	1,328.1	1,388.8	1,452.1	1,847.3	4.93%

e- Estimated; p- Projected

Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitor.

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for mechanical technology is projected to reach USD 973.4 million by 2022, from USD 769.7 million growing at a CAGR of 4.81%, during the forecast period. The thermal storage technology is expected to grow at the highest CAGR of 5.19%, during the forecast period.

TABLE 63 REST OF WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY ELECTROCHEMICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Lithium-ion battery	280.3	294.3	309.0	401.7	5.38%
Lead acid battery	3.1	3.2	3.3	3.8	2.72%
Sodium sulfur	9.5	9.8	10.1	12.1	3.58%
Flow battery	8.2	8.6	8.9	10.8	4.04%
NiMH and NiCd	30.3	30.8	31.4	34.0	1.60%
Total	331.5	346.7	362.7	462.3	4.97%

e- Estimated; p- Projected

Note: NiMH and NiCd refers to Nickel metal hydride and Nickel cadmium

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for lithium-ion battery is projected to reach USD 401.7million by 2022, at a CAGR of 5.38%, during the forecast period. The market for flow batteries is projected to reach USD 10.8 million by 2022, at a CAGR of 4.04%, during the forecast period.

TABLE 64 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY THERMAL STORAGE TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Sensible thermal	106.7	110.7	114.8	140.4	4.10%
Latent thermal	183.5	193.4	203.9	270.1	5.79%
Total	290.2	304.1	318.7	410.5	5.19%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for sensible thermal is projected to reach USD 140.4 million by 2022, at a CAGR of 4.10%, during the forecast period. The market for latent thermal is projected to reach USD 270.1 million by 2022, at a CAGR of 5.79%, during the forecast period.

TABLE 65 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY MECHANICAL TECHNOLOGY, 2015–2022 (USD MILLION)

Technology	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Pumped Hydro	662.6	692.4	723.6	917.3	4.86%
Flywheel	43.0	44.5	46.1	56.0	3.96%
Total	705.5	737.0	769.7	973.4	4.81%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The market for pumped hydro is projected to reach USD 917.3 million by 2022, at a CAGR of 4.86%, during the forecast period. The market for flywheel is projected to reach USD 56.0 million by 2022, at a CAGR of 3.96%, during the forecast period.

8.5.1 SOUTH AFRICA

South Africa is working for new renewable and gas fired energy for improving the electricity and distribution. The players in the country are looking for partners that can integrate and store the renewable energy adding to the electricity grid. The stored energy with improved flexibility grid and increased diversification can be used when needed.

TABLE 66 SOUTH AFRICA: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	43.4	44.8	46.2	54.0	3.17%
Grid Storage	357.0	378.6	401.4	546.3	6.36%
Total	400.4	423.4	447.5	600.2	6.05%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The transportation market is estimated to be the largest market at USD 46.2 million in 2017 and is projected to reach USD 54.0 million by 2022, at a CAGR of 3.17%, from 2017 to 2022. The grid storage market is expected to reach USD 546.3 million by 2022, growing at a CAGR of 6.36%, during the forecast period.

8.5.2 CHILE

In Chile, the second largest advanced energy project was undertaken by the subsidiary of AES Energy Storage, 'Angamos', which was commissioned in 2012. The project is integrated with a 544 MW coal power plant near the town of Mejillones. The project gives advantage to the energy market in Chile and separates entities for generation, transmission, and retail sales of electricity in the wholesale markets. The government in Chile is deploying a large amount of solar PV. The advanced energy project will help to facilitate the integration of these new resources and help the country meet a goal of generating 20 percent of its energy from renewables by 2025. The government in Chile, allows open competition among energy generation and ancillary service providers and this has been key factor for several developments in Chile.

TABLE 67 CHILE: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	28.0	28.8	29.6	34.4	3.00%
Grid Storage	297.1	312.5	328.6	429.9	5.52%
Total	325.2	341.3	358.2	464.3	5.32%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The transportation market is estimated to be USD 29.6 million in 2017 and is projected to reach USD 34.4 million by 2022, at a CAGR of 3.00%, from 2017 to 2022. The grid storage market is expected to reach USD 429.9 million by 2022, growing at the highest CAGR of 5.52%, during the forecast period.

8.5.3 REST OF ROW

The rest of RoW includes Middle Eastern, other African and other South American countries. The AESs market has been segmented based on application, technology, and country. The market, by technology, has been categorized into electro chemical, thermal storage, mechanical, and others. The market based on application has been segmented into transportation and grid.

TABLE 68 REST OF THE WORLD: ADVANCED ENERGY STORAGE SYSTEMS MARKET, BY APPLICATION, 2015–2022 (USD MILLION)

Application	2015	2016	2017-e	2022-p	CAGR (2017–2022)
Transportation	51.9	52.7	53.5	57.9	1.62%
Grid Storage	550.6	571.4	592.9	724.8	4.10%
Total	602.5	624.1	646.4	782.8	3.90%

e- Estimated; p- Projected

Source: Company Press Releases, Investor Presentations, Journals, Expert Interviews, and MarketsandMarkets Analysis

The transportation segment is estimated to be USD 53.5 million in 2017 and is projected to reach USD 57.9 million by 2022, at a CAGR of 1.62%, from 2017 to 2022. The grid storage segment is expected to reach USD 724.8 million by 2022, growing at the highest CAGR of 4.10%, during the forecast period.

9 COMPETITIVE LANDSCAPE

9.1 OVERVIEW

This section of the report provides an overview of the prevailing competitive scenario within the advanced energy storage system market.

The popular growth strategies adopted by a few of the leading companies, globally, include mergers & acquisitions, contracts & agreements, new product launches, investments and expansions, and joint ventures and partnerships. These strategies show the current market and competitive scenario of the AEES market.

A benchmark of players within the market had also been carried out using proprietary competitive leadership mapping, which analyzes market players on various parameters within broad categories of business and product strategy.

9.2 KEY PLAYERS OF THE ADVANCED ENERGY STORAGE SYSTEMS MARKET

This report on the AEES market considers top 25 industry players that are involved in providing solutions for energy storage. The companies, ABB Ltd. (Switzerland), LG Chem, Ltd. (South Korea), General Electric Company (US), Samsung SDI Co., Ltd. (South Korea), GS Yuasa Corporation (Japan), Siemens AG (Germany), Panasonic Corporation (Japan), CALMAC (US), Maxwell Technologies Inc. (US), Exide Technologies (US), Baltimore Aircoil Company, Inc (US) AES Corporation (US), Nippon Chemi-Con Corporation (Japan), Tesla, Inc. (US), Toshiba Corporation (Japan), NGK Insulators, Ltd. (Japan), Evapco, Inc. (US), Saft Groupe S.A (France) Beacon Power, LLC. (US), A123 Systems, LLC. (US), The Furukawa Battery Co., Ltd. (Japan), Hitachi, Ltd. (Japan), among others are considered as a sample size to study the trends of the AEES market key company focus area

9.2.1 GROWTH STRATEGIES IN THE ADVANCED ENERGY STORAGE SYSTEM MARKET

From 2013 to July 2017, AEES companies focused on new product launches and contracts and agreements as a key strategy to increase their market share and business presence, globally. Investments and expansions were the second most followed strategy, by leading industry players to cater the increasing demand for energy storage.

FIGURE 42 COMPANIES ADOPTED NEW PRODUCT LAUNCHES AND CONTRACTS AND AGREEMENTS AS KEY STRATEGIES TO CAPTURE THE MARKET, 2013– JULY 2017

COMPANY NAME	ORGANIC GROWTH STRATEGIES	INORGANIC GROWTH STRATEGIES
	New Product Launches	Partnerships/Agreements/ Collaborations
ABB LTD. (SWITZERLAND)		ABB Ltd. (Switzerland) provided an innovative microgrid, combining battery and flywheel based storage technologies, designed to test the scalability and improve power stability .
LG CHEM, LTD.(SOUTH KOREA)		Corvus Energy (Canada) announced the agreement with LG Chem Ltd.to supply lithium-ion pouch cells for Corvus' next generation maritime Energy Storage Systems (ESS).
GENERAL ELECTRIC COMPANY (US)		GE signed a contract with Con Edison to supply an 8 MWh energy storage system for a site in Central Valley, California. The system uses lithium-ion batteries with a GE Mark Vie control system and Brilliance inverters
SAMSUNG SDI CO., LTD.(SOUTH KOREA)	Samsung SDI unveiled two residential energy storage systems (ESS) at Intersolar Europe 2015 in Munich, Germany. The new 5.5kWh and 8.0kWh units join the existing 3.6kWh system.	
TESLA, INC. (US)		Tesla, Inc, (US) and AMS (US) won another major energy storage contract, 34 MWh of battery capacity for water treatment facilities.

Note:  - Indicates events that will have a significant impact on the market

Source: Company Websites, Press Releases, and Company Annual Reports

9.3 MARKET RANKING, ADVANCED ENERGY STORAGE SYSTEMS, 2016

Most of the advanced energy storage system market is dominated by a few leading industry players due to their high-end technological advantages. These top players have a wide network of distribution and sales, globally. The rest of the markets have many small regional players. ABB Ltd. (Switzerland), General Electric Company (US), LG Chem, Ltd. (South Korea), Siemens AG (Germany), Tesla, Inc. (US), BYD Company Ltd (China), GS Yuasa Corporation (Japan) and Panasonic Corporation (Japan) are a few of the leading players in the advanced energy storage system market. A market ranking analysis had been carried out based on the revenues generated by key players in various business segments, and key developments undertaken by them.

TABLE 69 MARKET RANKING, 2016

Rank	Company Name
1	ABB Ltd. (Switzerland)
2	LG Chem, Ltd. (South Korea)
3	General Electric Company (US)
4	Samsung SDI Co.,Ltd (South Korea)
5	Tesla, Inc. (US)

Source: Expert Interviews, Company Annual Reports, and MarketsandMarkets Analysis

9.4 COMPETITIVE SITUATIONS & TRENDS

New product launches is one of the key strategies adopted by market players to strengthen their product portfolio and to expand their market size. It is also a key revenue generating tool for players operating in the market. From 2013 to July 2017, this strategy was adopted to achieve growth in the advanced energy storage system market. Industry players who adopted this strategy included GS Yuasa Corporation (Japan), Toshiba Corporation (Japan), Beacon Power, LLC. (US), and Panasonic Corporation (Japan), among others.

Contracts and agreements also served as a successful approach for companies. In terms of developments, new product launches held 31%, whereas, contracts & agreements accounted and investments & expansions accounted for 29% and 15%, respectively, of the total market share. The companies who adopted contracts & agreements as their key strategy includes, LG Chem, Ltd. (South Korea), Tesla, Inc. (US) and EnerSys (US), among others.

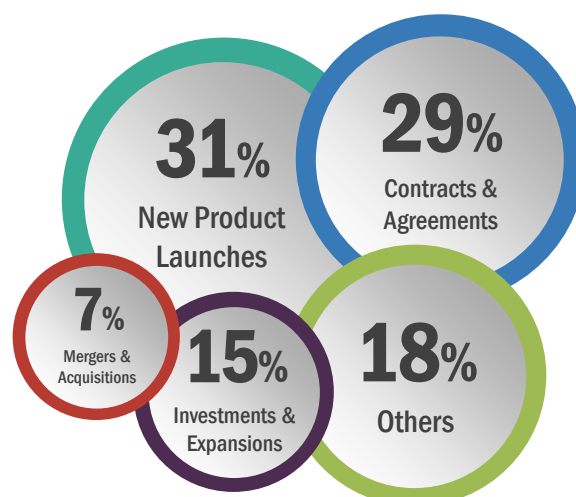
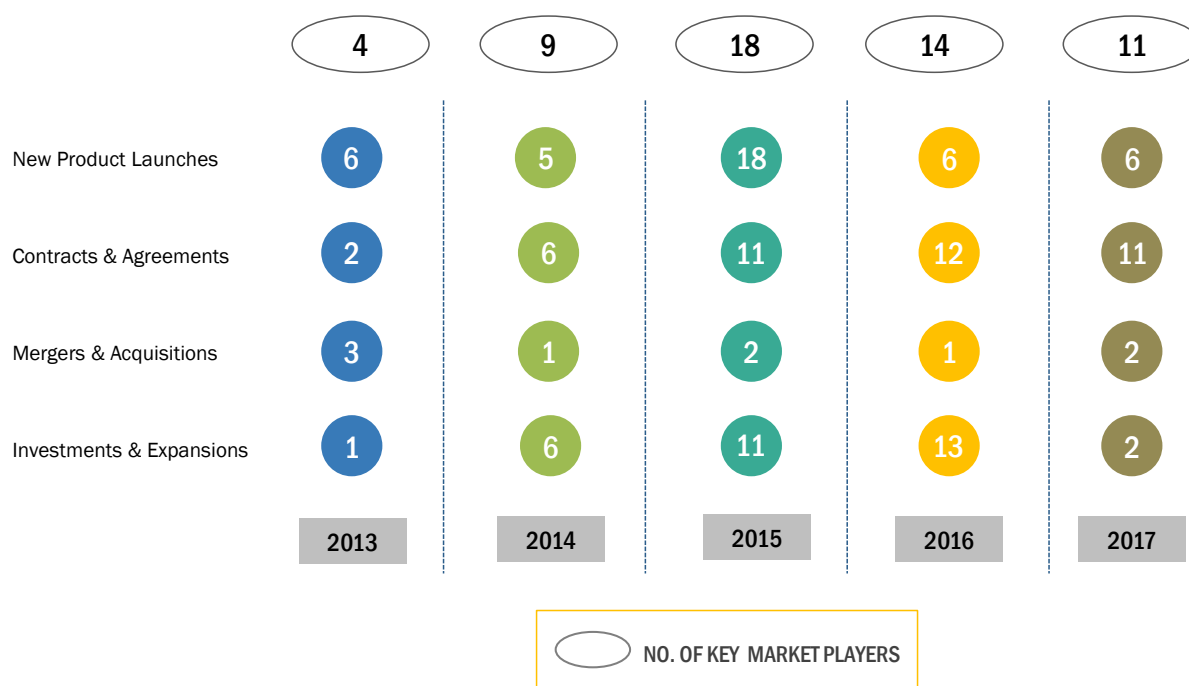


FIGURE 43 MARKET EVALUATION FRAMEWORK: NEW PRODUCT LAUNCHES, MERGERS & ACQUISITIONS, AND CONTRACTS & AGREEMENTS HAVE FUELED THE GROWTH OF COMPANIES, 2013– JULY 2017



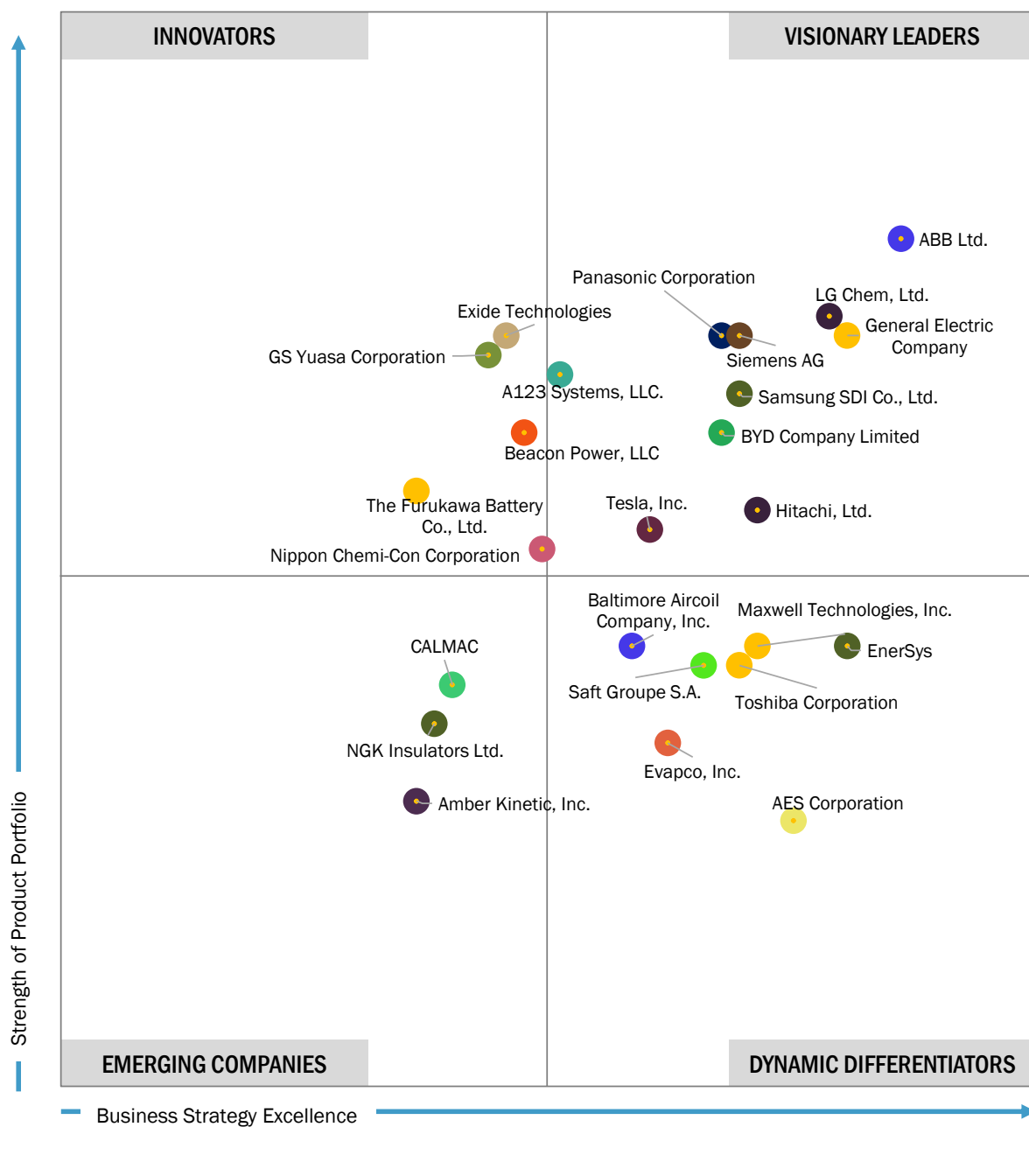
Source: Company Website, Press Releases, and Company Annual Report

The competitive mapping provides information about major players that offer advanced energy storage system and outlines the findings and analysis on how well each advanced energy storage system market companies, perform within our criteria. We have placed vendors into four categories based on their performance in each criterion, namely, “Visionary Leaders,” “Innovators,” “Emerging Companies,” and “Dynamic Differentiators”.

9.5 COMPETITIVE LEADERSHIP MAPPING

The competitive mapping provides information about major players that offer advanced energy storage system and outlines the findings and analysis on how well each advanced energy storage system market companies, perform within our criteria. We have placed vendors into four categories based on their performance in each criterion, namely, “Visionary Leaders,” “Innovators,” “Emerging Companies,” and “Dynamic Differentiators”

FIGURE 44 ADVANCED ENERGY STORAGE SYSTEM MARKET (GLOBAL): COMPETITIVE LEADERSHIP MAPPING, 2017



Source: Company Website, Press Releases, Company Annual Report, Expert Interviews, and MarketsandMarkets Analysis

9.5.1 VISIONARY LEADERS

Market leaders who fall into this category generally receive high scores for most evaluation criteria. They have strong and established service portfolio and have a very strong market presence. They should provide mature and reputable advanced energy storage system services. They also have strong business strategies. The visionary leaders in advanced energy storage system market includes ABB Ltd. (Switzerland), General Electric Company (US), LG Chem, Ltd. (South Korea), Panasonic Corporation (Japan), Siemens AG (Germany), and BYD Company Limited (China). A123 Systems, LLC. (US), Hitachi, Ltd. (Japan), Tesla, Inc. (US),

9.5.2 DYNAMIC DIFFERENTIATORS

They are established vendors with very strong business strategies. However, they are low in service portfolio. They generally focus on a specific type of technology related to the product. AES Corporation. (U.S), EnerSys (US), Maxwell Technologies Inc. (US), Saft Groupe S.A. (France), Evapco, Inc. (US) and Baltimore Aircoil Company, Inc. (US) are key dynamic players in AESS market.

9.5.3 INNOVATORS

Innovators in the competitive leadership mapping are vendors that have demonstrated substantial service innovations as compared to their competitors. They should have very focused service portfolio. However, they do not have a very strong growth strategy for their overall business. Key innovators in AESS market includes GS Yuasa Corporation (Japan), Nippon Chemi-Con Corporation (Japan), Beacon Power, LLC. (US), The Furukawa Battery Co., Ltd. (Japan) and Exide Technologies (US).

9.5.4 EMERGING COMPANIES

They are vendors with niche product offering and are starting to gain their position in the market. They do not have much strong business strategies as compared to other established vendors. They might be new entrants in the market and require some more time before getting significant traction in the market. NGK Insulators, Ltd. (US), Amber Kinetic, Inc, (US), CALMAC (US), are emerging players in the advanced energy storage system market

9.6 STRENGTH OF PRODUCT PORTFOLIO (25 COMPANIES)

Vendor	Breadth and Depth of Product Offering	Product Features and Functionality	Focus on Product Innovation	Product Branding
A123 Systems, LLC. (US)	● ● ●	● ● ●	● ● ●	● ● ●
ABB Ltd. (Switzerland)	● ● ●	● ● ●	● ● ●	● ● ●
AES Corporation (US)	● ● ●	● ● ●	● ● ●	● ● ●
Amber Kinetic, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Baltimore Aircoil Company, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Beacon Power, LLC. (US)	● ● ●	● ● ●	● ● ●	● ● ●
BYD Company Limited (China)	● ● ●	● ● ●	● ● ●	● ● ●
CALMAC (US)	● ● ●	● ● ●	● ● ●	● ● ●
EnerSys (US)	● ● ●	● ● ●	● ● ●	● ● ●
Evapco, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Exide Technologies (US)	● ● ●	● ● ●	● ● ●	● ● ●
General Electric Company (US)	● ● ●	● ● ●	● ● ●	● ● ●
GS Yuasa Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Hitachi, Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
LG Chem, Ltd. (South Korea)	● ● ●	● ● ●	● ● ●	● ● ●
Maxwell Technologies Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
NGK Insulators, Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Nippon Chemi-Con Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Panasonic Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Saft Groupe S.A. (France)	● ● ●	● ● ●	● ● ●	● ● ●
Samsung SDI Co., Ltd.(South Korea)	● ● ●	● ● ●	● ● ●	● ● ●
Siemens AG (Germany)	● ● ●	● ● ●	● ● ●	● ● ●
The Furukawa Battery Co., Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Toshiba Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Tesla, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Penetration Rating : ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH				

Note: The analysis has been carried out on the basis of the information available in the public domain for each parameter

Breadth and Depth of Product Offering: This parameter reflects the breadth and depth in the company's product line in terms of key components provided for power monitoring. The rating for breadth of offering is based on the company's product/service offerings catering to power monitoring market (power meters, smart meters, communication networks, software among others)

Product Features and Functionality: This parameter considers all available product features in the product line and its varied functionality.

Focus on Product Innovation: This parameter considers investments toward new product developments and actual product launches.

Product Branding: This parameter considers how the product branding has helped the company either with revenue growth or with enhancing its competitive position in the industry

Source: Company Website, Press Releases, Company Annual Report, Expert Interviews, and MarketsandMarkets Analysis

9.7 BUSINESS STRATEGY EXCELLENCE (25 COMPANIES)

Vendor	Regional Footprint	Effectiveness of Organic Growth Strategy	Mergers & Acquisitions Strategy	Breadth of Applications Served
A123 Systems, LLC. (US)	● ● ●	● ● ●	● ● ●	● ● ●
ABB Ltd. (Switzerland)	● ● ●	● ● ●	● ● ●	● ● ●
AES Corporation (US)	● ● ●	● ● ●	● ● ●	● ● ●
Amber Kinetic, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Baltimore Aircoil Company, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Beacon Power, LLC. (US)	● ● ●	● ● ●	● ● ●	● ● ●
BYD Company Limited (China)	● ● ●	● ● ●	● ● ●	● ● ●
CALMAC (US)	● ● ●	● ● ●	● ● ●	● ● ●
EnerSys (US)	● ● ●	● ● ●	● ● ●	● ● ●
Evapco, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Exide Technologies (US)	● ● ●	● ● ●	● ● ●	● ● ●
General Electric Company (US)	● ● ●	● ● ●	● ● ●	● ● ●
GS Yuasa Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Hitachi, Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
LG Chem, Ltd. (South Korea)	● ● ●	● ● ●	● ● ●	● ● ●
Maxwell Technologies Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
NGK Insulators, Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Nippon Chemi-Con Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Panasonic Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Saft Groupe S.A. (France)	● ● ●	● ● ●	● ● ●	● ● ●
Samsung SDI Co., Ltd.(South Korea)	● ● ●	● ● ●	● ● ●	● ● ●
Siemens AG (Germany)	● ● ●	● ● ●	● ● ●	● ● ●
The Furukawa Battery Co., Ltd. (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Toshiba Corporation (Japan)	● ● ●	● ● ●	● ● ●	● ● ●
Tesla, Inc. (US)	● ● ●	● ● ●	● ● ●	● ● ●
Penetration Rating : ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH				

Note: The analysis has been carried out on the basis on the information available in the public domain for each parameter

Regional Footprint: The parameter considers the direct and indirect presence of the company across all regions

Effectiveness of growth strategy: The parameters considers strategies, such as investments and expansions, collaborations, partnerships, mergers & acquisitions, and contracts & agreements. The ratings of the companies are based on number of developments covering all the mentioned parameters

Mergers & Acquisition Strategy: The parameters considers level of growth has the company experienced as a result of those deals

Breadth of Applications Served: This parameter reflects how well the company serves the wide range of applications, and how well is company able to address the requirements of diverse applications

10 COMPANY PROFILES

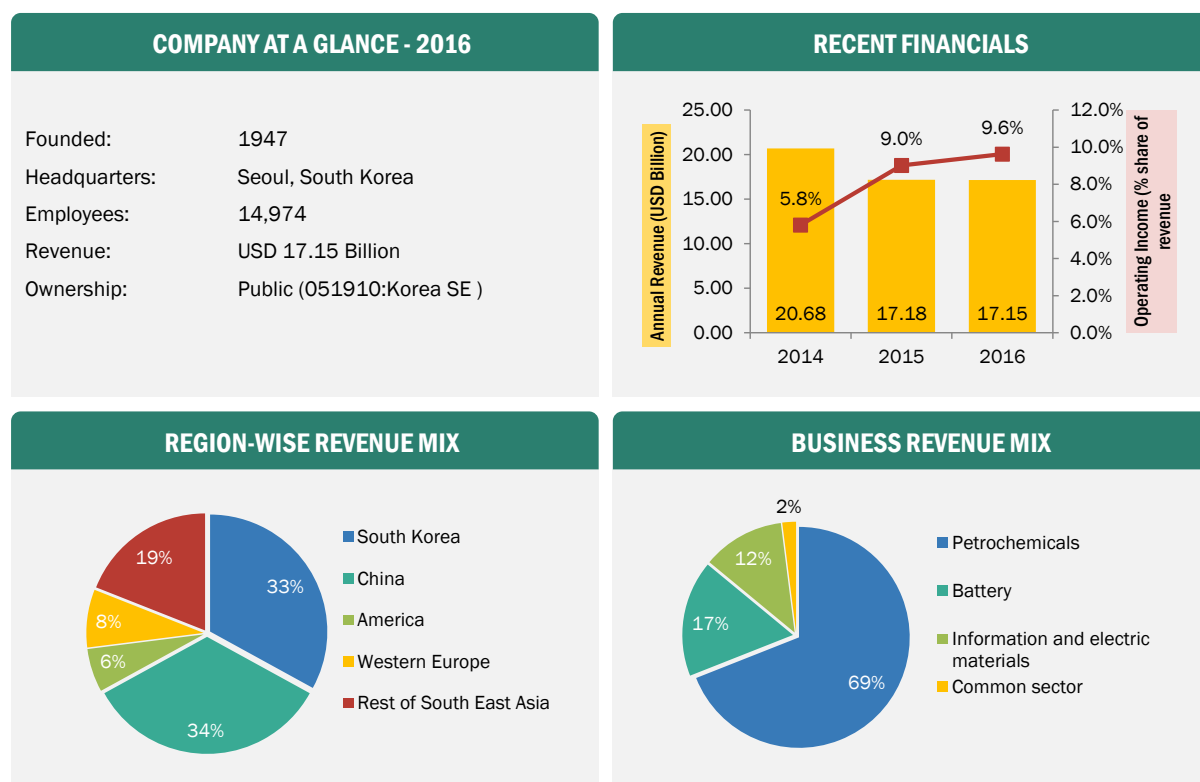
10.1 LG CHEM, LTD.

10.1.1 OVERVIEW

LG Chem, Ltd., is a chemical company. The company operates through four business segments namely, basic materials and chemicals, IT and electronic material, energy solutions, and advanced materials. The company offers advanced energy storage systems through its energy solutions segment which consist of mobile batteries and automotive batteries. LG Chem, Ltd. is gaining market in various energy storage system sectors which includes smart grids, homes, and UPS. The company has its existence in countries, such as, South Korea, India, China, Hong Kong, Taiwan, Germany, Vietnam, Poland, Brazil, and the US.

The company manufactures a wide range of products, ranging from petrochemical goods to high-value added plastics and extends its chemical expertise in high-tech areas, such as electronic materials and rechargeable batteries.

FIGURE 45 LG CHEM, LTD.: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.1.2 PRODUCTS OFFERED

Category	Product
Energy Solutions	• Mobile batteries • Automobile batteries • Energy storage systems

Source: Company Website, Press Releases, and Company Annual Reports

10.1.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The company has strong product offering, catering to the energy storage system. It is the leading manufacturer of lithium-ion batteries and provide energy storage system solutions globally. The company provides optimal energy solutions to the users, using energy storage system with a long lifespan and excellent quality.

10.1.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

LG Chem, Ltd. has adopted regional expansion as its key strategy to strengthen its business for advanced energy storage systems. The company also signed contracts and entered into agreements and partnerships as a part of the inorganic growth strategy to provide AESS

10.1.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
April 2017	New Product Launches	LG Chem, Ltd. (South Korea) formally launched its range of residential battery systems, which offers AC- and DC-coupled solutions with capacities up to 9.8 kWh in the North American market.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size If Available)
January 2017	Expansions & Investments	LG Chem, Ltd. (South Korea)	Eguana Technologies Inc. (Canada)	Eguana Technologies Inc. (Canada) expanded its energy storage partnership with LG Chem, Ltd. (South Korea) to develop, optimize, integrate, and deliver LG Chem, Ltd.'s new industry- JH3 battery cell technology for stationary storage systems, as a part of Eguana's AC battery portfolio.	
October 2016	Partnerships	LG Chem, Ltd. (South Korea)	Sunrun Inc. (US)	Sunrun Inc. (US) and LG Chem, Ltd. (South Korea) announced a partnership to launch an energy storage technology in the US residential solar market, allowing more families, the access to clean and affordable energy.	
March 2016	Contracts & Agreements	LG Chem, Ltd. (South Korea)	Corvus Energy (Canada)	Corvus Energy (Canada) announced a multi-year agreement with LG Chem, Ltd. (South Korea). Under this agreement, LG Chem, the world's leading manufacturer of advanced lithium-ion batteries will be a key supplier of lithium-ion pouch cells for Corvus' next generation maritime Energy Storage Systems (ESS).	

December 2015	Contracts & Agreements	LG Chem, Ltd. (South Korea)	AES Energy Storage(US)	LG Chem, Ltd. (South Korea) and AES Energy Storage (US) announced a multi-year agreement that provides access to gigawatt-hour (GWh) of lithium-ion battery capacity with the option to procure additional capacity for the AES Advancion energy storage solution.
May 2015	Contracts & Agreements	LG Chem, Ltd. (South Korea)	ONEnergy Inc. (Canada)	ONEnergy Inc. announced an important sales and distribution agreement with LG Chem, Ltd. (South Korea) whereby ONEnergy's Sunwave Energy Efficiency division ("Sunwave") will distribute and market LG Chem's state of the art lithium-ion-based energy storage systems.
March 2015	Expansions & Investments	LG Chem, Ltd. (South Korea)	Ideal Power Inc. (US)	Ideal Power Inc. (US) a developer of power conversion technology, announced a multi-year alliance with LG Chem, Ltd., to supply Ideal Power's patented Power Conversion Systems (PCS) alongside LG Chem, Ltd.'s battery solutions
November 2014	Contracts & Agreements	LG Chem, Ltd. (South Korea)	Siemens AG (Germany)	LG Chem, Ltd. (South Korea) and Siemens AG (Germany) will collaboratively work on developing and marketing industrial battery storage systems. The companies will accelerate the dominance of the ESS market by collaborating on several major ESS projects. This agreement will help in responding to the growing demand for energy storage systems that are required to stabilize distribution grids.

Source: Company Website, Press Releases, and Company Annual Reports

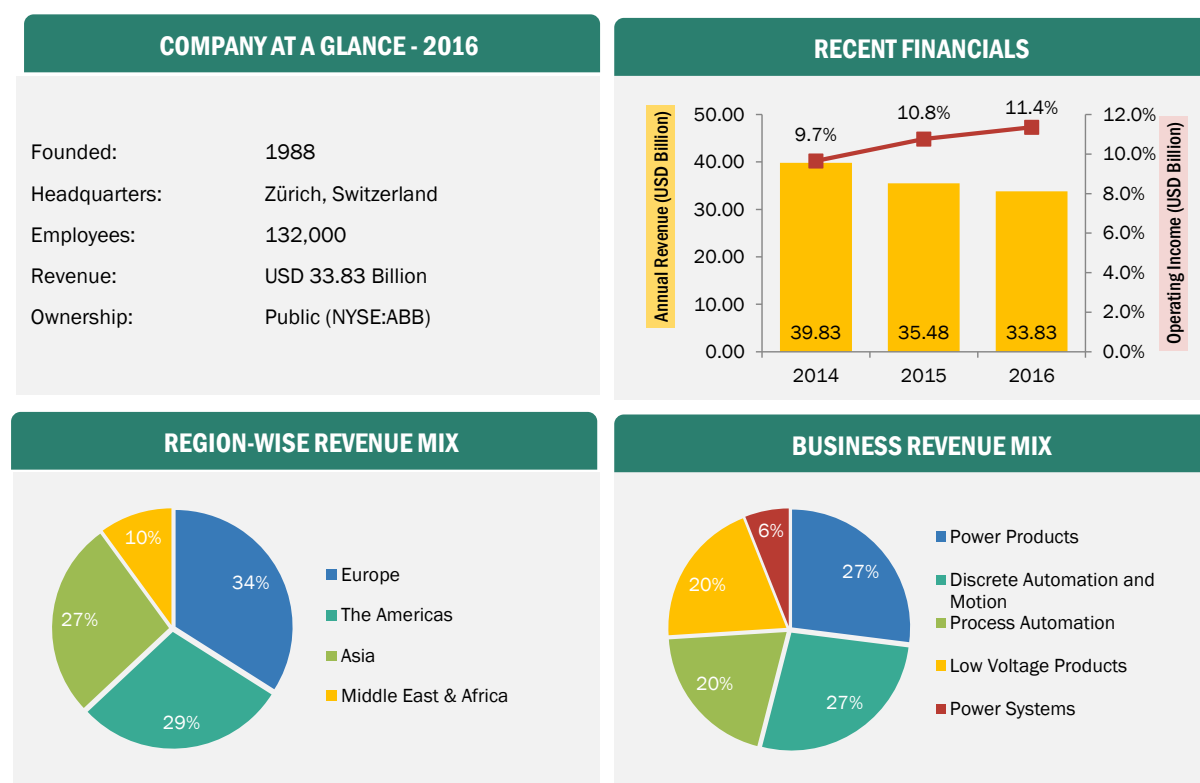
10.2 ABB LTD.

10.2.1 OVERVIEW

ABB Ltd. is one of the leading players in the electrical product industry. It is a multinational company, which operates in the fields of power technology and industrial automation. The company operates through four business segments, namely, electrification product, robotics and motion, industrial automation, and power grid.

ABB Ltd. has its presence in more than 100 countries along with its subsidiaries. The company has its manufacturing unit in Europe, the Middle East, and Africa (EMEA) and North America. The company has more than 125 years of experience in power engineering. It has proficiency in energy storage implementation and majorly deal in community energy storage system, distributed energy storage system, battery energy storage system, high voltage capacitors and filters, medium voltage capacitors and filters, and low voltage capacitors and filters.

FIGURE 46 ABB LTD.: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.2.2 PRODUCTS OFFERED

Category	Product
Energy Storage Solutions	- Battery energy storage systems - Pumped hydro storage systems

Source: Company Website, Press Releases, and Company Annual Reports

10.2.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

ABB Ltd. is one of the leading companies in energy storage solutions. It has a strong product offering catering to the community energy storage systems, distributed energy storage systems, battery energy storage systems, and pumped hydro energy storage systems. It offers one of the top notch features and functionality along with technology innovation. Due to its product quality and features, it has become a strong brand in the energy storage systems market.

10.2.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

ABB Ltd. is spread across the globe and is present in Europe, the Americas, the Middle East and Africa, and Asia and Oceania. Hence, regional expansion is one of the key strategy of the company, for the strong foothold in the energy and power industry. It offers an array of applications, such as, peak shaving, load shifting, voltage regulation, renewable integration, and backup power. The company also adopted mergers & acquisitions and contracts & agreements to grow inorganically.

10.2.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
February 2017	New Product Launches	ABB Ltd. launched an innovative microgrid, combining battery and flywheel-based storage technologies, designed to test the scalability and improve the power stability for around 300,000 people in Anchorage, Alaska, USA.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
March 2015	Contracts & Agreements	ABB Ltd. (Switzerland)	Samsung SDI Co., Ltd. (South Korea)	ABB Ltd. (Switzerland) and Samsung SDI Co., Ltd. (South Korea) signed an agreement to develop micro-grid solutions and lithium-ion batteries for the purpose of energy storage. This agreement will not only help ABB to easily access the market already captured by Samsung but also improve its energy storage portfolio.	
September 2014	Joint Ventures	ABB Ltd. (Switzerland)	BYD Company Limited (China)	ABB Ltd. (Switzerland) and BYD Company Limited (China) entered into a joint venture to introduce upgraded energy storage technologies. This will inspire innovations in the field of EV charging and storage of renewable energy sources in off-grid and on-grid solutions as well as battery and energy storage solutions. This joint venture will help ABB to gain importance in the energy storage market worldwide.	

Source: Company Website, Press Releases, and Company Annual Reports

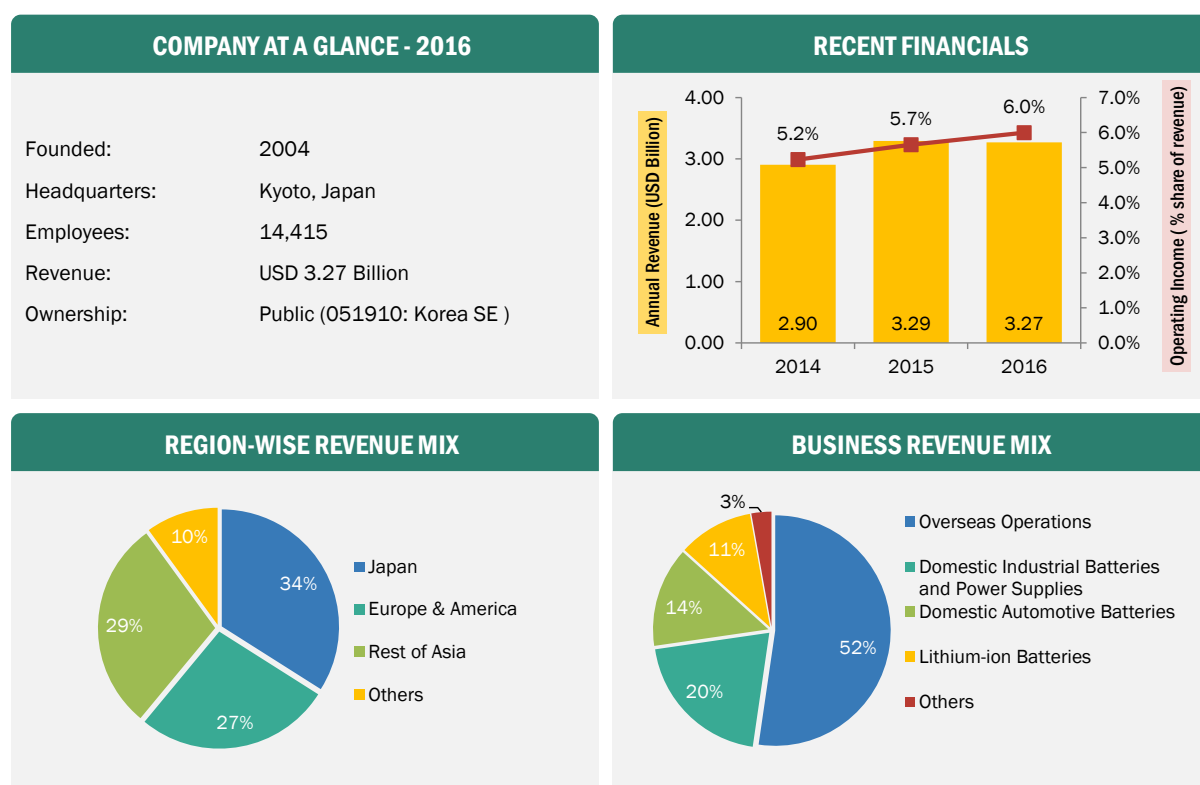
10.3 GS YUASA CORPORATION

10.3.1 OVERVIEW

GS Yuasa Corporation is a group company of GS Yuasa International Ltd. The company manufactures and markets automotive batteries, industrial batteries, power supply systems, switch gears, lighting equipment, ultraviolet systems, specialty equipment, and other electrical equipment. It operates through five business segments which are domestic automotive batteries, domestic industrial batteries and power supplies, overseas applications, lithium-ion batteries, and others.

With the increase in demand for energy storage technologies, GS Yuasa Corporation focuses on photovoltaic power and lithium-ion batteries as its new business areas. The expansion in the product line is expected to increase its global market share in the lead acid batteries by capturing the untapped markets of Asia, Africa, the Middle East, and Central and Latin America. The company has 38 overseas bases in 17 countries which are used to manufacture and market automotive, motorcycle, and industrial lead acid batteries. GS Yuasa Corporation has a large presence in China and Southeast Asia.

FIGURE 47 GS YUASA CORPORATION: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.3.2 PRODUCTS OFFERED

Category	Product
Energy Storage Solutions	- Automotive batteries - Industrial batteries - Power supply systems - Lithium-ion batteries - Special batteries and nickel-metal hydride batteries

Source: Company Website, Press Releases, and Company Annual Reports

10.3.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

GS Yuasa Corporation has a strong engineering team and modern production lines with a high level of automation. It adopted new product launches, technologies, and innovation to cater to the advanced energy storage systems market along with its product offering, for e.g. the introduction of new advanced Nano-Carbon Lead Acid battery. It provides an unprecedented 5000 cycles at 70% depth of discharge. Its product quality is reflected in every aspect of competitiveness and is a part of its brand identity.

10.3.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

GS Yuasa Corporation adopted joint ventures to strengthen its solutions for AESS. The Joint Venture between Bosch, GS Yuasa Corporation, and Mitsubishi Corporation is expected to make the EV a successful mass product in the next decade. The company also adopted partnerships as a part of its inorganic strategies to provide solutions to the advanced energy storage systems.

10.3.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
March 2016	New Product Launches	GS Yuasa Corporation (Japan) announced that the company has introduced a new advanced Nano-Carbon Lead Acid battery. The new SLR-1000 is a two volt advanced lead battery that provides an unprecedented 5000 cycles at 70% depth of discharge.
March 2015	New Product Launches	GS Yuasa Corporation (Japan) introduced a new lithium-ion battery module, LIM25H Series, for industrial applications with high input and output performance. The Li-ion battery utilizes regenerative energy for stabilizing electric power. The product, which has enhanced the product portfolio of the company, will help the company by increasing its product line and reducing the cost of the products as compared to the other ones.
January 2015	New Product Launches	GS Yuasa Corporation (Japan) launched a power conditioner, equipped with lithium-ion batteries to generate solar power optimal for use in creating disaster-resistant areas with low environmental impact and reducing energy costs.
November 2014	New Product Launches	GS Yuasa International Ltd., developed a new advanced lithium secondary battery which has a "Sulfur-Porous Carbon Composite" positive electrode with eight-time higher capacity than that of the conventional lithium-ion battery. This will help the company in achieving its goal of reducing the carbon footprint.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
November 2016	Contracts & Agreements	GS Yuasa Corporation (Japan)	Orbital ATK, Inc (US)	GS Yuasa Lithium Power (GYLP), a US subsidiary of GS Yuasa Corporation (Japan), announced that Orbital ATK has selected lithium-ion cells manufactured by GS Yuasa Technology to power its Mission Extension Vehicle (MEV-1).	
September 2014	Collaborations	GS Yuasa Corporation (Japan)	Nippon Sharyo (Nagoya)	GS Yuasa Corporation (Japan) and Nippon Sharyo (Nagoya) jointly developed NeGEM, a lithium-ion battery module to be used in the electric powered carrier. This reduced the dependence on large diesel engines to be used as the power source for heavy load vehicles that carry heavy objects (carriers) at	

locations, such as steelworks or shipyards.				
February 2014	Joint Ventures	GS Yuasa Corporation (Japan)	Robert Bosch GmbH (Germany) Mitsubishi Corporation (Japan)	GS Yuasa Corporation (Japan), Robert Bosch GmbH (Germany), and Mitsubishi Corporation (Japan) entered into a joint venture and formed a new company, Lithium Energy and Power GmbH & Co.

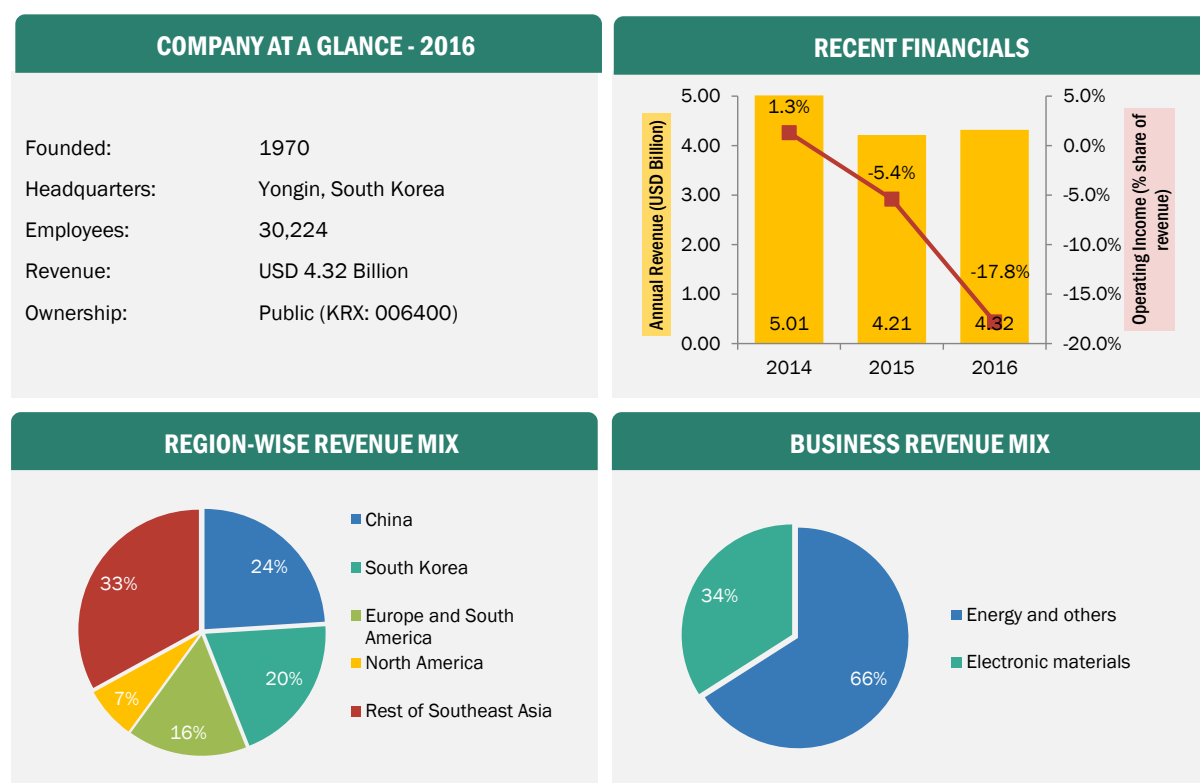
10.4 SAMSUNG SDI CO., LTD.

10.4.1 OVERVIEW

Samsung SDI Co., Ltd. is a manufacturer of rechargeable batteries for the IT industry, automobiles, and energy storage systems. The company operates through three segments namely, chemicals, electronic materials, and energy and others. Small lithium ion batteries are used in laptops, tablets, mobile phones, wearable devices, power banks, power tools, garden tools, vacuum cleaners, e-bikes, e-scooters, and ignition products, energy storage systems used in base transceiver stations, as well as in residential, utility, commercial, and UPS applications, and in automotive batteries.

The secondary lithium-ion batteries manufactured by Samsung SDI Co., Ltd. are rapidly being used in applications ranging from digital mobile devices, such as cell phones and laptops to EVs, such as Hybrid and EV and ESS. Currently, the company is trying to change its business portfolio from electronic display products to eco-friendly energy solutions.

FIGURE 48 SAMSUNG SDI CO., LTD.: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.4.2 PRODUCTS OFFERED

Category	Product
Energy Storage Solutions	- Lithium-ion battery - Electric car battery - E-bike battery - Rechargeable battery

Source: Company Website, Press Releases, and Company Annual Reports

10.4.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The company offers a strong product portfolio ranging from small lithium-ion batteries, automotive batteries, and ESSs. There are various forms of a cell, such as prismatic, cylindrical, and polymer. Samsung SDI Co., Ltd. is known for its robust technology and innovation. Recently, the E2 model, a high-capacity ESS battery is launched that has substantially increased the number of cells integrated in the container by adopting the innovative design technology.

10.4.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Samsung SDI Co., Ltd. is expanding its efforts to develop a global network, hence expansions is one of the key strategies to strengthen its business. It has wide ESS applications, such as, UPS, telecom tower, residential, commercial, and utility which helps it in penetrating the energy and power market.

10.4.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
June 2015	New Product Launches	Samsung SDI Co., Ltd. unveiled two residential energy storage systems at Intersolar Europe 2015 in Munich, Germany. The new 5.5 and 8.0 kWh units joined the existing 3.6 kWh system.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
December 2016	Partnerships	Samsung SDI Co., Ltd. (South Korea)	Dynapower Company, LLC (US)	Leading battery storage producer Samsung SDI Co., Ltd. (South Korea) is teaming up with leading storage inverter maker Dynapower to develop an all-in-one storage solution for commercial and industrial customers. It is a move that will remove intermediaries, lower the installed costs, and encourage a broader use of behind-the-meter batteries.	
March 2015	Strategic Alliances	Samsung SDI Co., Ltd. (South Korea)	ABB Ltd. (Switzerland)	Samsung SDI Co., Ltd. (South Korea) and ABB Ltd. (Switzerland) formed a strategic commercial alliance to develop and market microgrid systems that comprise energy storage solutions. The alliance will help Samsung to expand its leading position in the microgrid market.	
February 2015	Mergers & Acquisitions	Samsung SDI Co., Ltd. (South Korea)	Magna International Inc (Ontario)	Samsung SDI Co., Ltd. (South Korea) agreed to acquire the battery pack business of Magna International (Ontario).	

				The acquisition helped Samsung SDI Co., Ltd. to enhance its capabilities in batteries for EVs. The acquisition is a key strategic step for Samsung SDI Co., Ltd. to strengthen the competitiveness of its automotive battery business.
October 2014	Contracts & Agreements	Samsung SDI Co., Ltd. (South Korea)	Green Charge Networks (New York)	Samsung SDI Co., Ltd. (South Korea) and Green Charge Networks (New York) have agreed to provide intelligent energy storage worldwide to deliver 25 megawatt-hours (MWh) of lithium-ion batteries over the next two years. The agreement will benefit Samsung SDI Co., Ltd. to emerge as the leading solution provider in the distributed energy storage segment.
July 2014	Contracts & Agreements	Samsung SDI Co., Ltd. (South Korea)	BMW Group (Germany)	Samsung SDI Co., Ltd. (South Korea) will continue supplying the BMW Group (Germany) with battery cells for the BMW i3, BMW i8, and additional hybrid models over the coming years. The agreement will help the companies to cater to the growing demand for electro-mobility and secure their leadership in future technologies.

Source: Company Website, Press Releases, and Company Annual Reports

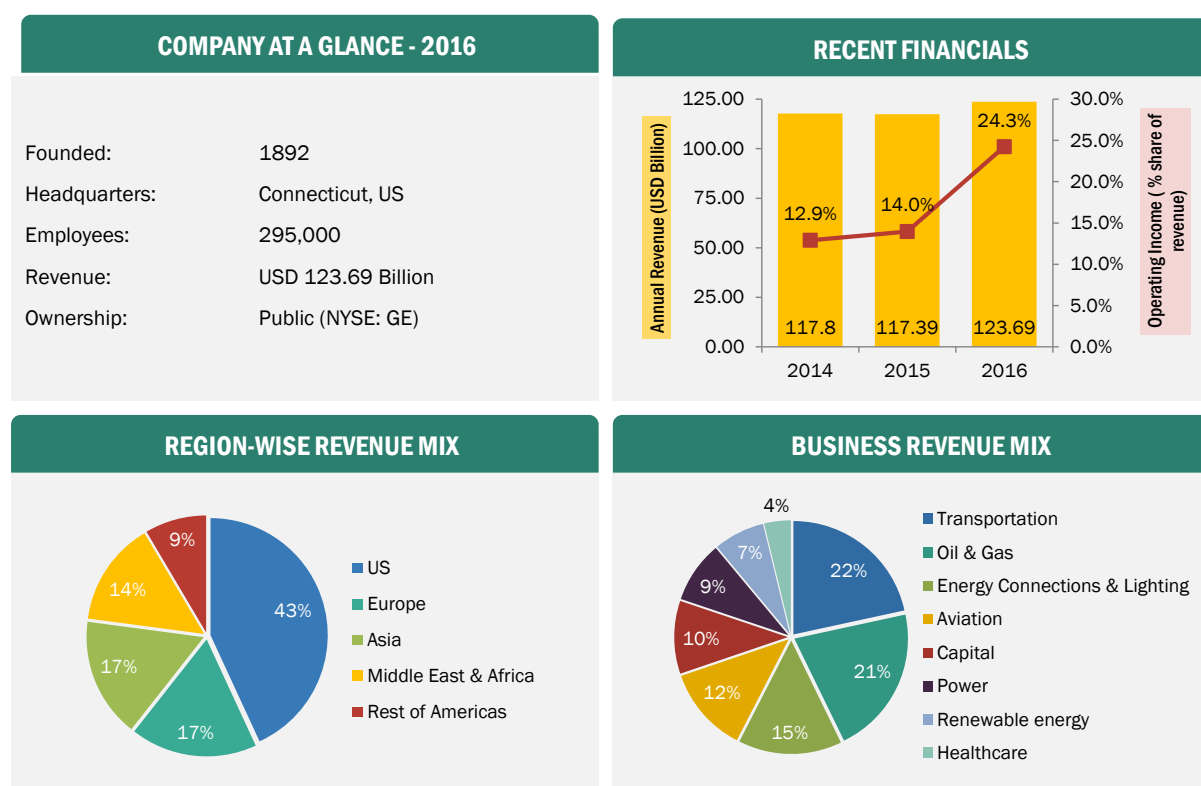
10.5 GENERAL ELECTRIC COMPANY

10.5.1 OVERVIEW

The General Electric Company manufactures products and services for generation, transmission and distribution, control, and utilization of electricity. It operates through nine business segments, namely, aviation, power, healthcare, oil & gas, appliances and lighting, energy management, transportation, renewable energy, and capital. The energy storage systems of the General Electric Company are a part of its energy management segment. The major products under the energy management are grid management solutions, power conversion technologies, and lighting and energy efficiency solutions.

Manufacturing operations are carried out at 184 manufacturing plants located in 38 states in the United States and Puerto Rico and at 325 manufacturing plants located in 40 other countries.

FIGURE 49 GENERAL ELECTRIC COMPANY: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.5.2 PRODUCTS OFFERED

Category	Product
Innovation Solutions	• Energy storage • Pump hydro turbine • Grid flexibility battery energy storage

Source: Company Website, Press Releases, and Company Annual Reports

10.5.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

General Electric Company's energy storage solutions are present in 25 countries with a multiplicity of applications. The company offers a portfolio of products, which includes pumped hydro storage, grid flexibility battery energy storage, and solar power solutions. GE's energy storage team concentrates in bringing the right technologies together to organize custom solutions to meet a diversity of needs. The company works with its customers to match the correct combination of technology with its applications, so the customers can get supreme value out of their system. This strategy acts as a very important element to strengthen the business.

10.5.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The General Electric Company has a wide regional reach as it spreads across the six continents, namely, Africa, Asia, Americas, Australia, Europe, and the Middle East. This is one of its key strategies, which has strengthened its business. It has also entered into agreements and contracts which has further helped it to grow inorganically. GE has expanded the energy storage business by becoming the reseller of other batteries and storage options. GE is moving towards a model that the company has adopted across several businesses, such as turbines, where production and facility are key parts of the business model. The company offers a wide array of applications which includes energy management, transmission, distribution, and microgrid, and thermal and renewable power.

10.5.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
March 2015	Expansions & Investments	The General Electric Company (US) opened a USD 170 million Durathon battery manufacturing plant in Schenectady, New York. The company is creating a consulting and services business to help the customers determine which batteries or technology they should use to solve their energy demands.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
September 2016	Strategic Alliances	General Electric Company (US)	Corvus Energy (US)	GE's Power Conversion business selected Corvus' new Orca ESS to install in its marine power test facility and customer demonstration center. The center integrates various GE and supplier technologies, such as the Orca ESS from Corvus Energy, for merchant, naval, and offshore applications.	
August 2015	Contracts & Agreements	General Electric Company (US)	Coachella Energy Storage Partners, LLC (US)	General Electric Company (US) signed a contract with Coachella Energy Storage Partner, LLC (US) to supply a 30 MW battery energy storage system as a part of the supply contract with the Imperial Irrigation District (IID). The plant would be located in California's Imperial Valley and the facility is expected to feature grid flexibility and increased reliability.	
April 2015	Contracts & Agreements	General Electric Company (US)	Con Edison (US)	General Electric Company (US) signed a contract with Con Edison to supply an 8 MWh energy storage system for a site in Central Valley, California. The system uses lithium-ion batteries with a GE Mark VIe control system and Brilliance MW inverters.	

Source: Company Website, Press Releases, and Company Annual Reports

10.6 SAFT GROUPE S.A.

10.6.1 OVERVIEW

Saft Groupe S.A., a wholly owned subsidiary of Total, is a manufacturer of batteries. Saft's nickel and lithium-ion battery serves as the backup power during power shutdowns. The lithium-ion energy solutions are used for ancillary services, large renewable integrations, microgrids, smart distribution grids, and end-user energy optimization.

Saft Groupe S.A. caters to a wide range of end-users, such as, aerospace and defense, buildings and industries, medical, networks, oil & gas, transportation, and utilities. The company has presence in 19 countries with 14 manufacturing sites and 31 sales offices worldwide.

COMPANY AT A GLANCE

Founded:	1913
Headquarter:	Paris, France
Ownership:	Private

Note: This is a privately held company. Hence no financial information is available in the public domain
Source: Company Website

10.6.2 PRODUCTS OFFERED

Category	Product
Energy Storage Solutions	- Lithium-ion batteries - Lithium-thionyl chloride batteries - Lithium-Sulphur dioxide batteries - Lithium manganese dioxide

Source: Company Website, Press Releases, and Company Annual Reports

10.6.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Saft Groupe S.A. is a company involved in the design, development, and manufacturing of batteries. The multitechnology battery systems include nickel-based and primary lithium batteries in industrial infrastructure, transportation, civil, and military electronics and lithium-ion solutions in energy storage, telecoms, industrial vehicles, space, and defense. Saft Groupe S.A. also offers system integration and turnkey solutions that include services, such as, installation, commissioning, operation, training, maintenance, and supply of spare parts for battery systems. Hence a combination of products and services helps the company to penetrate the market and strengthen its business.

10.6.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Saft Groupe S.A. has a wide regional presence in 19 countries with 14 manufacturing sites which makes the market penetration easy and strengthen its business. It serves a vast range of market segments from rail to telecoms and health and aerospace to marine, offering battery solutions for many different types of applications.

10.6.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
April 2015	New Product Launches	Saft Groupe S.A. (France) introduced the specialized Sunica plus nickel-based batteries that offer off-grid energy storage. Sunica plus battery delivers off-grid back-up power and is able to perform in extremely low temperatures, with erratic charging patterns. The battery system stores energy to support all the lodge services, including lighting and an 18 kW water pump, for 24 hours. This development will help the company to come closer towards its objective of becoming a provider of high-tech batteries.
December 2014	Expansions & Investments	AMCO Saft (New Delhi), the joint venture of SAFT in India, presented a wide portfolio of the industrial battery technology that is suitable for India's fast growing energy storage sector. The product application includes grid integration of renewable energy, micro-grids, smart grids, and off-grid installations. This will help the country to get reliable and high-quality power and will also help the company to expand its regional reach.
December 2014	Expansions & Investments	Saft Groupe S.A. (France) successfully produced the 100 millionth primary lithium batteries for its highly demanding market at Zhuhai, China. This shows the strong presence of SAFT in China.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
April 2016	Contracts & Agreements	Saft Groupe S.A. (France)	Fortum (Finland)	Saft Groupe S.A. (France) had been awarded a megawatt-scale Li-Ion (lithium-ion) battery energy storage contract by Fortum, the Finnish energy company.	
July 2015	Contracts & Agreements	Saft Groupe S.A. (France)	Schneider Electric SA (France)	Schneider Electric SA (France) and Saft Groupe S.A. (France) won two projects from Langa Group for the installation of energy storage systems for two solar power plants in Corsica (France), awarded under the French public tender schemes – Commission de Régulation de l'Energie (CRE).	

Source: Company Website, Press Releases, and Company Annual Reports

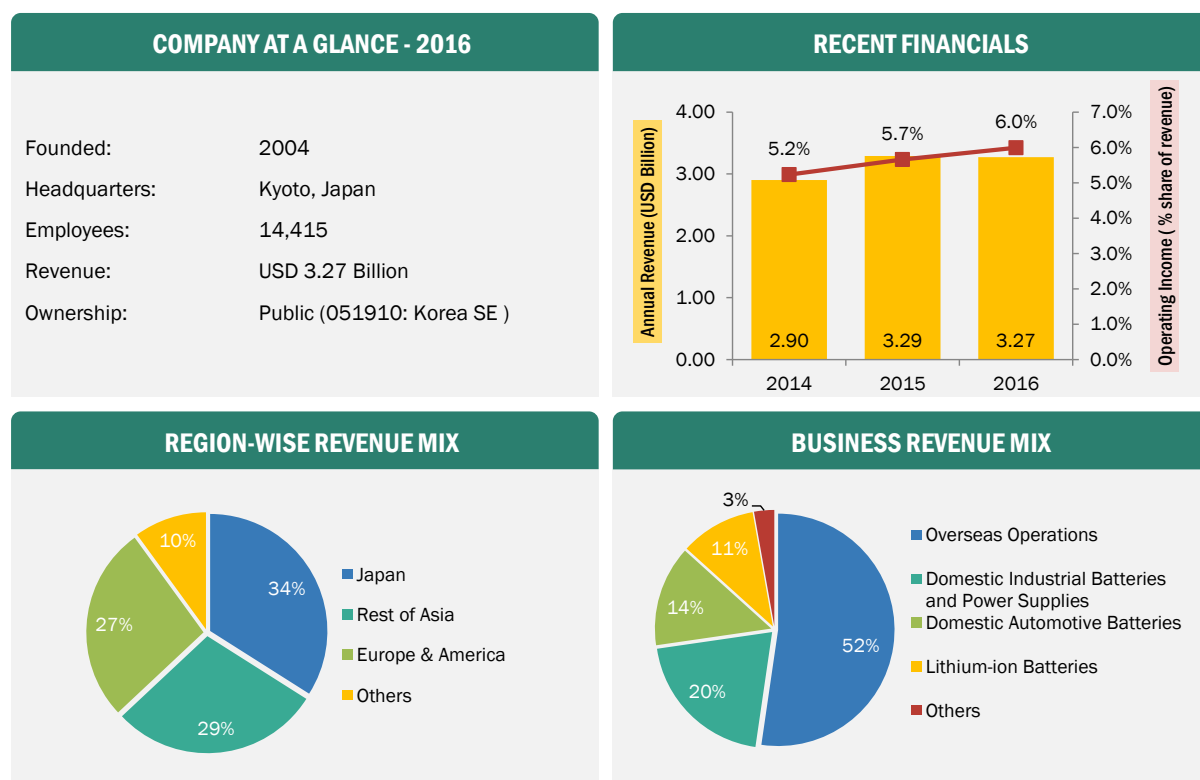
10.7 TESLA, INC.

10.7.1 OVERVIEW

Tesla, Inc. is a developer and manufacturer of fully EVs and energy storage systems, as well as it install, operate, and maintain solar and energy storage products. The company's products are categorized as Model S, Model X, Model 3, stationary energy storage applications, and powertrain. Tesla's stationary energy storage products are used in homes, commercial sites, and utilities. These battery systems are used in backup power, peak demand reduction, and electric market services. The company offers energy innovation solutions for a sustainable energy future with batteries, enabling homes, business, and utilities to store sustainable and renewable energy to manage power demand, provide backup power, and increase grid resilience.

The company operates through two business segments, namely, automotive and energy generation and storage. The energy generation and storage segment includes the designing, manufacturing, installation, and sale or lease of products. The electricity generated by solar energy systems are sold to the customers. The company has its presence in North America, Europe, Asia Pacific, and the Middle East.

FIGURE 50 TESLA, INC.: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.7.2 PRODUCTS OFFERED

Category	Product
Energy Storage Solutions	- Powerwall - Powerpack

Source: Company Website, Press Releases, and Company Annual Reports

10.7.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Tesla, Inc. has been building integrated battery systems in cars for over 10 years. The identical degree of proficiency, quality mechanism, and technical innovation has an informed process of developing high-performance batteries for the grid. Powerpack, a product of Tesla, Inc. contains 16 individual battery pods, each with an isolated DC-DC converter. Its pod architecture and onboard power electronics, optimize the performance across the assortment and enable easy swapping at any time. Tesla, Inc. is entering into the next step in energy storage known as aggregation in which Powerwall and Powerpack batteries are put into a single portfolio to make the grid cleaner and more efficient.

10.7.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Tesla, Inc. operates in many parts of the world, which includes the Americas, Germany, the UK, Australia, and China, among others. The company is working with energy retailers, grid operators, utilities, and aggregators across the globe to expose the ability for Tesla batteries to deliver grid services while providing dependable power during the whole day. Hence, penetration across the globe is one of the key strategy to strengthen its business.

10.7.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
September 2015	New Product Launches	Tesla, Inc. (US) expanded its market presence by delivering Powerwall products first in Australia. North America is also expected to receive deliveries of Powerwall residential battery storage systems.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
January 2017	Joint Ventures	Tesla, Inc. (US)	Southern California Edison (US)	Tesla, Inc. (US) announced that it was working with Southern California Edison (SCE) (US) to create the largest lithium-ion battery in the world.	
September 2016	Contracts & Agreements	Tesla, Inc. (US)	Advanced Microgrid Solutions (US)	Tesla, Inc. (US) and Advanced Microgrid Solutions won another major energy storage contract, a 34 MWh of battery capacity for water treatment facilities.	
September 2016	Contracts & Agreements	Tesla, Inc. (US)	Southern California Edison (US)	Tesla, Inc. (US) won a bid to supply grid-scale power in Southern California to help prevent electricity shortages following the biggest natural gas leak in the US history.	
June 2015	Contracts & Agreements	Tesla, Inc. (US)	Advanced Microgrid Solutions (US)	Advanced Microgrid Solutions (AMS) selected Tesla, Inc. (US) as the primary technology provider for its groundbreaking utility-scale energy storage projects.	
April 2015	Contracts & Agreements	Tesla, Inc. (US)	SolarCity (US)	Tesla, Inc. (US) signed a contract with SolarCity (US) to supply battery energy storage system for businesses and government organizations. SolarCity would incorporate the new Tesla battery into its DemandLogic energy storage system to increase the utility cost savings.	

Source: Company Website, Press Releases, and Company Annual Reports

10.8 EVAPCO, INC.

10.8.1 OVERVIEW

Evapco, Inc. is a manufacturing company that manufactures products for evaporative cooling and industrial refrigeration. The end-use markets for the company are commercial HVAC, industrial process, power, and industrial refrigeration.

Evapco, Inc. focuses primarily on its research and development and manufacturing plans to adopt its product with the latest technologies. The company has its manufacturing plants in 22 locations in 10 countries across the globe. The products manufactured by Evapco, Inc. are distributed through their 170 sales offices.

COMPANY AT A GLANCE

Founded:	1947
Headquarter:	Maryland (US)
Ownership:	Private

Note: This is a privately held company. Hence no financial information is available in the public domain
Source: Company Website

10.8.2 PRODUCTS OFFERED

Product	Description
Extra-Pak Ice Coil	- Capacity: up to 700 tons - Temperature: 34 to 42° F - Applications: Commercial HVAC, district energy, industrial processes, power, and industrial refrigeration

Source: Company Website, Press Releases, and Company Annual Reports

10.8.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating:	● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH

Evapco Inc. deals in closed circuit coolers, cooling towers, thermal ice storage, worldwide air-conditioning, industrial process cooling, and industrial refrigeration market to provide solutions to the energy storage system. It is also dedicated to design and manufacture the highest quality products around the globe. The company also supports its universal customers with industrialized facilities and sales offices strategically located around the world, thereby strengthening its business.

10.8.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Building on its core values, Evapco Inc. continued to grow through the 1990s and strengthened international operations by opening facilities in the Americas, Africa, Asia Pacific, and Europe. Hence expansions is one of the key strategies for the strong foothold in the energy storage business.

10.9 CALMAC

10.9.1 OVERVIEW

CALMAC is a manufacturer of cool energy related technologies. The IceBanks energy storage system is an air conditioning solution that makes ice at night to cool buildings during the day. These energy storage systems are installed at over 4,000 locations in 60 countries.

IceMat and Alumazorb are the two products of CALMAC which are well recognized worldwide. CALMAC provides one of the best ice skating surfaces and is a member of the Ice Skating Institute, US Green Building Council, and the New Buildings Institute.

COMPANY AT A GLANCE

Founded:	1851
Headquarter:	New Jersey (U.S)
Ownership:	Private

Note: This is a privately held company. Hence no financial information is available in the public domain
 Source: Company Website

10.9.2 PRODUCTS OFFERED

Product	Application
Energy storage systems	- Air conditioning - Smart grid - Hybrid cooling
1045C	- Capacity: 41 ton hrs, 144 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa) - Applications: Air conditioning, district cooling, indoor, smart grid
1082C	- Capacity: 82 ton hrs, 288 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa) - Applications: Leed certification, hybrid cooling, retrofit
1109C	- Capacity: 162 ton hrs, 570 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa)
1105C	- Capacity: 105 ton hrs, 369 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa)
1098C	- Capacity: 98 ton hrs, 345 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa)
1220C	- Capacity: 214 ton hrs, 753 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa)
1320C	- Capacity: 324 ton hrs, 1140 kWh - Temperature: 100°F (38°C) - Operating Pressure: 90 psi (620 kpa)

1500C

- Capacity: 486 ton hrs, 1710 kWh
- Temperature: 100 °F (38 °C)
- Operating Pressure: 90 psi (620 kpa)

Source: Company Website, Press Releases, and Company Annual Reports

10.9.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

CALMAC has over two hundred years of combined thermal energy storage engineering and application knowledge. Hence, it capitalizes on its product features and functionality to strengthen its business. Over 60 countries rely on CALMAC's thermal energy storage to cool their buildings.

10.9.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

CALMAC adopted contracts & agreements as its key strategy to strengthen its business. The recent contract with the Public Service Enterprise Group Incorporated to install a thermal ice storage system at John T. Mather Memorial Hospital, received USD 1 million rebate for system installation, approved by the Public Service Enterprise Group Incorporated. This is expected to reduce the peak energy demand usage by 20%.

10.9.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
March 2015	New Product Launches	CALMAC (US) introduced a technology according to which the commercial building owners and facility or energy managers can analyze the electricity rates charged and can help in saving electricity by shifting a building's cooling load to night time off-peak hours.
September 2014	Investments & Expansions	CALMAC (US) provides energy storage systems in South Florida that is able to store 20 MW of energy and has saved USD 2 million over 20 years.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal size
December 2016	Contracts & Agreements	CALMAC (US)	Public Service Enterprise Group Incorporated (US)	CALMAC (US) signed a contract with Public Service Enterprise Group Incorporated (US) to install a thermal ice storage system at John T. Mather Memorial Hospital. It is the first hospital in the Long Island to install a thermal (ice) storage system. The Public Service Enterprise Group Incorporated (US) approved a USD 1 million rebate for this system installation.	USD 1 million
November 2016	Contracts & Agreements	CALMAC (US)	Integrated Multi- Physics Renewable Energy Laboratory (US).	CALMAC (US) signed a contract with the Naval Post Graduate School to integrate the ice-based energy storage technology into a microgrid project at the Integrated Multi- Physics Renewable Energy Laboratory (US).	
July 2013	Contracts & Agreements	CALMAC (US)	Dundalk Institute of Technology (Ireland).	CALMAC (US) signed a contract to install its IceBank energy storage tanks at the Dundalk Institute of Technology (Ireland).	

10.10 BALTIMORE AIRCOIL COMPANY, INC.

10.10.1 OVERVIEW

Baltimore Aircoil Company, Inc. is the world's largest manufacturer of heat transfer and thermal energy storage equipment. The product portfolio comprises of cooling towers, evaporative condensers, ice thermal storage equipment, and controls. These products caters to various end-users, namely, the commercial building market for air conditioning systems, food industry for air conditioning and refrigeration applications, and to a broad range of industries for process and power installation equipment cooling.

COMPANY AT A GLANCE

Founded:	1938
Headquarter:	Maryland (US)
Ownership:	Private

Note: This is a privately held company. Hence no financial information is available in the public domain
Source: Company Website

The ice thermal storage system can be used to cool offices, schools, hospitals, malls, and other buildings at a convenient cost. These systems help in lower energy consumption and reduce greenhouse gas emissions.

10.10.2 PRODUCTS OFFERED

Product	Application
Energy storage systems	- Air conditioning - Hybrid cooling - Food processing - Industrial and process cooling

Source: Company Website, Press Releases, and Company Annual Reports

10.10.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The Baltimore Aircoil Company, Inc. deals in cooling towers, closed circuit cooling towers, evaporative condensers, ice thermal storage, and hybrid technology. Cooling with ice thermal storage can be a lucrative, dependable system method for cooling offices, schools, hospitals, malls, and other buildings and provides a steady source of low temperature fluids for process cooling applications, thus adding to the product portfolio.

10.10.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	
Effectiveness of Organic Growth Strategy	
Mergers & Acquisitions	
Breadth of Applications Served	
Penetration Rating:  LOW  MEDIUM  HIGH	

The ice thermal storage technology has reduced the electricity costs for cooling systems, commercial and institutional buildings, hospitals, and manufacturing facilities. The company has its own research and development center in Baltimore (US) called the Baltimore Aircoil's Research and Development Centre. The research team tests various combinations of components and develops new products and materials with higher performance, easy installation, and easy maintenance, thus, strengthening the business.

10.10.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
February 2015	New Product Launches	Baltimore Aircoil Company, Inc. (US) announced the release of the industry's only direct-drive variable speed solution in a cooling tower, the Series 3000 with the ENDURADRIIVE Fan System. The ENDURADRIIVE Fan System provides unmatched reliability and superior energy savings with the lowest maintenance costs in the industry.

Source: Company Website, Press Releases, and Company Annual Reports

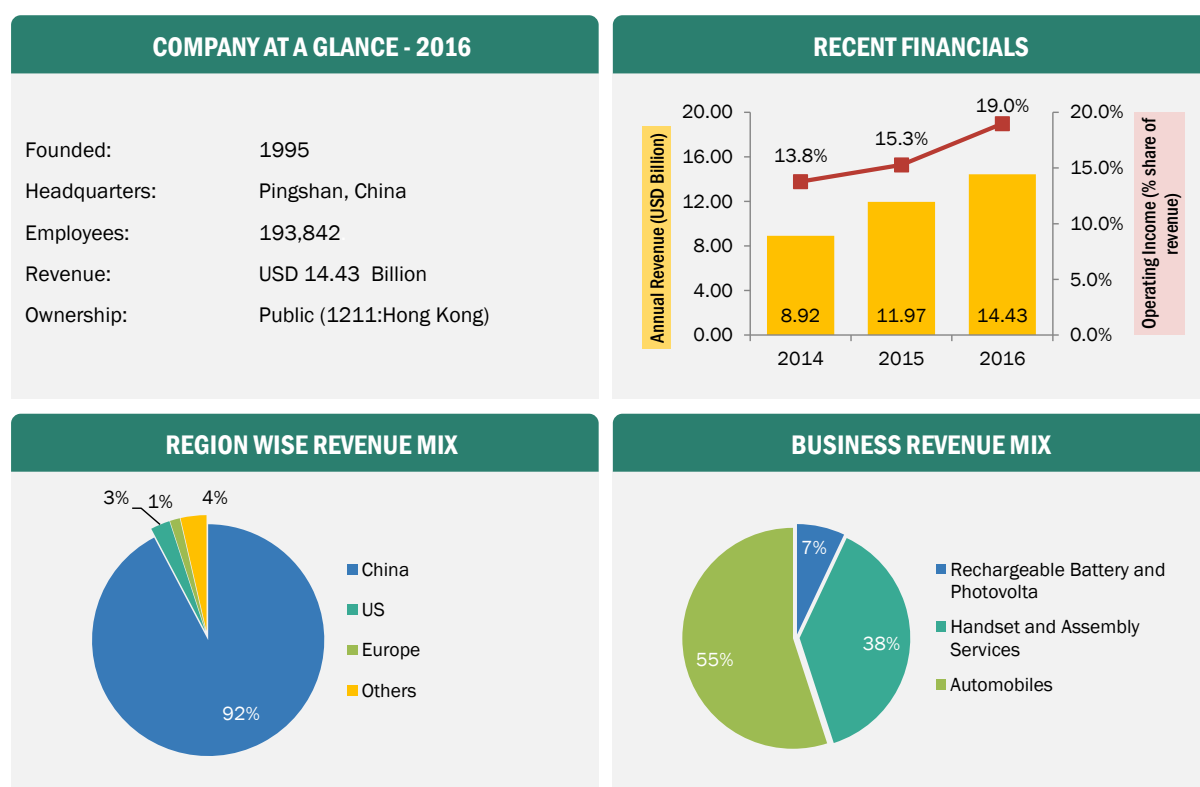
10.11 BYD COMPANY LIMITED

10.11.1 OVERVIEW

BYD Company Limited specializes in IT, automobile, and new energy related products. The company operates through three business segments, namely, automobiles, handset and assembly services, and rechargeable battery and photovoltaic. The rechargeable battery and photovoltaic business segment manufactures lithium-ion batteries and nickel batteries, photovoltaic products and iron battery products (including energy storage stations and iron battery pack), mainly for mobile phones, electric tools, and other portable electronic instruments, photovoltaic products, energy storage products, and EVs. BYD Company Limited holds the largest market share for the nickel-cadmium batteries, handset Li-ion batteries, cell phone chargers, and keypads globally. The end-users for the company are Samsung, Nokia, Huawei, Bosch, and TTI among others.

The business strategy of BYD Company Limited for rechargeable batteries is to expand its applications of lithium-ion batteries and nickel batteries. For photovoltaic business the company plans to expand its domestic and overseas market. The photovoltaic business is concerned with improving capacity utilization and enhanced sales volume to increase its revenue.

FIGURE 51 BYD COMPANY LIMITED: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.11.2 PRODUCTS OFFERED

Category	Product Type
Energy storage systems	• Distributed energy storage systems • Energy storage solutions • Emergency power systems • Home energy systems • Moveable emergency power systems

Source: Company Website, Press Releases, and Company Annual Reports

10.11.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The BYD Company Limited is the largest provider of rechargeable batteries across the globe and has the largest market share for the nickel-cadmium batteries, handset lithium-ion batteries, cell-phone chargers, and keypads worldwide. BYD Company Limited provides large-scale energy storage, distributed energy storage, and micro-grid, which forms a complete set and series of energy storage system solutions. The product features and functionalities are of the excellent quality. According to a study conducted by J.D. Power Quality, the BYD Company Limited ranks at the top, when quality is concerned

10.11.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The BYD Company Limited ESS technology offers a sectional and flexible scheme and can be easily personalized to meet a diverse set of customer needs. The ESS application fields are wind power plant, photovoltaic power station, Large load center (factory, commercial center etc.), emergency power supply occasion, areas of limited situations of environment and field (airport, isolated island etc.), "Black start" of grid, and grid frequency modulation. The BYD Company Limited has a strong foothold in North America, Asia Pacific, and Europe.

10.11.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments/Awards & Recognitions

Date	Approach	Description
June 2017	Expansions & Investments	BYD Company Limited a global leader in clean energy and the world's largest manufacturer of batteries and EVs, signed a letter of cooperation with Universidade Estadual de Campinas (Unicamp), in which the company commits to transfer over USD 5 million (approximately 1.5 million USD) by 2020 to set up BYD's first overseas Photovoltaic Research Center at the university.
February 2017	New product Launches	BYD Company Limited launched its newly expanded line-up of B-Box Battery Energy Storage Systems in Australia, one of the world's fastest growing residential energy storage markets.
September 2016	New Product Launches	BYD Company Limited has now officially launched its "B-BOX" in the UK, an energy storage system for household and commercial applications featuring the same battery technology used in buses running throughout central London.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Date	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
December 2014	Partnerships	BYD Company Limited(China)	Buyers World BV (Netherlands)	The BYD Company Limited and Buyers World BV, a Dutch solar wholesale company, expanded the distributorship to supply solar instruments. This partnership will help the company to grow strongly in the Netherlands and Europe	
August 2014	Contracts & Agreements	BYD Company Limited (China)	Fotowatio Renewable Ventures (US)	Fotowatio Renewable Ventures (FRV) awarded BYD Company Limited. one of the Latin America's biggest solar project in Uruguay for production of 65 MW PV system. This will help the company in regional expansion.	
May 2014	Joint Ventures	BYD Company Limited (China)	Hindustan Clean Energy Limited, (India)	Hindustan Clean Energy Limited (New Delhi, India) and BYD Company Limited jointly formed Kindle Engineering &	

Construction Pvt. The company will provide 50 MW capacity power. This will be BYD Company Limited's largest PV project in India which will promote large-scale installation of photovoltaic systems and reduce the cost of solar power in India. This strategy was adopted by the company for its regional expansion

Source: Company Website, Press Releases, and Company Annual Reports

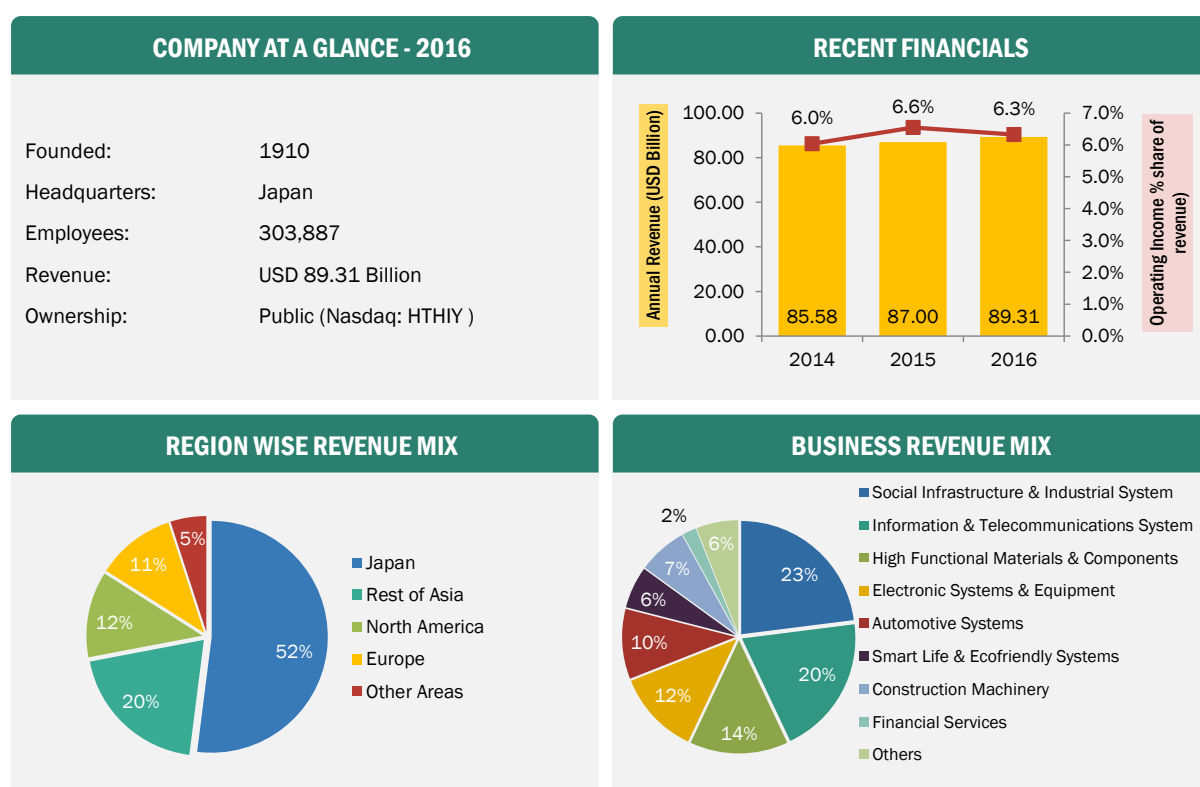
10.12 HITACHI, LTD.

10.12.1 OVERVIEW

Hitachi, Ltd. is a manufacturing company which operates through eight business segments, namely, information and telecommunication systems, social infrastructure and industrial systems, electronic systems and equipment, construction machinery, high functional materials and components, automotive systems, smart life and ecofriendly systems, and others. The company offers ESS through its high functional materials and components segment. Through its products and services, Hitachi, Ltd. caters to a wide range of sectors, including power, energy and utilities, transportation, finance, government and public, and healthcare.

The company has a wide regional reach with its presence in the Americas, Asia, the Middle East and Africa, Europe, and Oceania.

FIGURE 52 HITACHI, LTD.: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.12.2 PRODUCTS OFFERED

Category	Product Type
Energy storage systems	- Industrial batteries - Industrial lithium-ion batteries - Electrical equipment

Source: Company Website, Press Releases, and Company Annual Reports

10.12.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The entire Hitachi group contributes end to end quality assurance as a part of their practice. Hitachi, Ltd. is working towards the commercialization of safe and economical battery energy storage systems to provide a solution. Hitachi, Ltd. also contributed in the project to improve the knowledge for safe and economical large-scale energy storage systems supported by the New Energy and Industrial Technology Development Organization (NEDO), hence an important aspect leading to product branding.

10.12.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Hitachi, Ltd. develops and supplies an extensive range of highly efficient and reliable power generation systems throughout the world. The company has expanded widely across the globe in countries, such as Americas, Asia, and Europe where it has 24 offices, Middle East and Africa, and Oceania. Recently Hitachi, Ltd. has announced the integration and expansion of its businesses serving the market in America. This strategy is expected to strengthen its business.

10.12.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments

Date	Approach	Description
June 2016	Expansions & Investments	Hitachi, Ltd. opened a new division to expand its energy solution business in the North American Market.

Source: Company Website, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
February 2015	Joint Ventures	Hitachi, Ltd. (Japan)	Demansys Energy, Inc. (US)	Hitachi America, Ltd., a wholly owned subsidiary of Hitachi, Ltd. (Japan), in association with Demansys Energy, Inc. (US) launched 1 MW lithium-ion energy storage facility utilizing Hitachi's CrystEna*1 energy storage system and have started a demonstration project at Somerdale, New Jersey.	

Source: Company Websites, Press Releases, and Company Annual Reports

10.13 SIEMENS AG

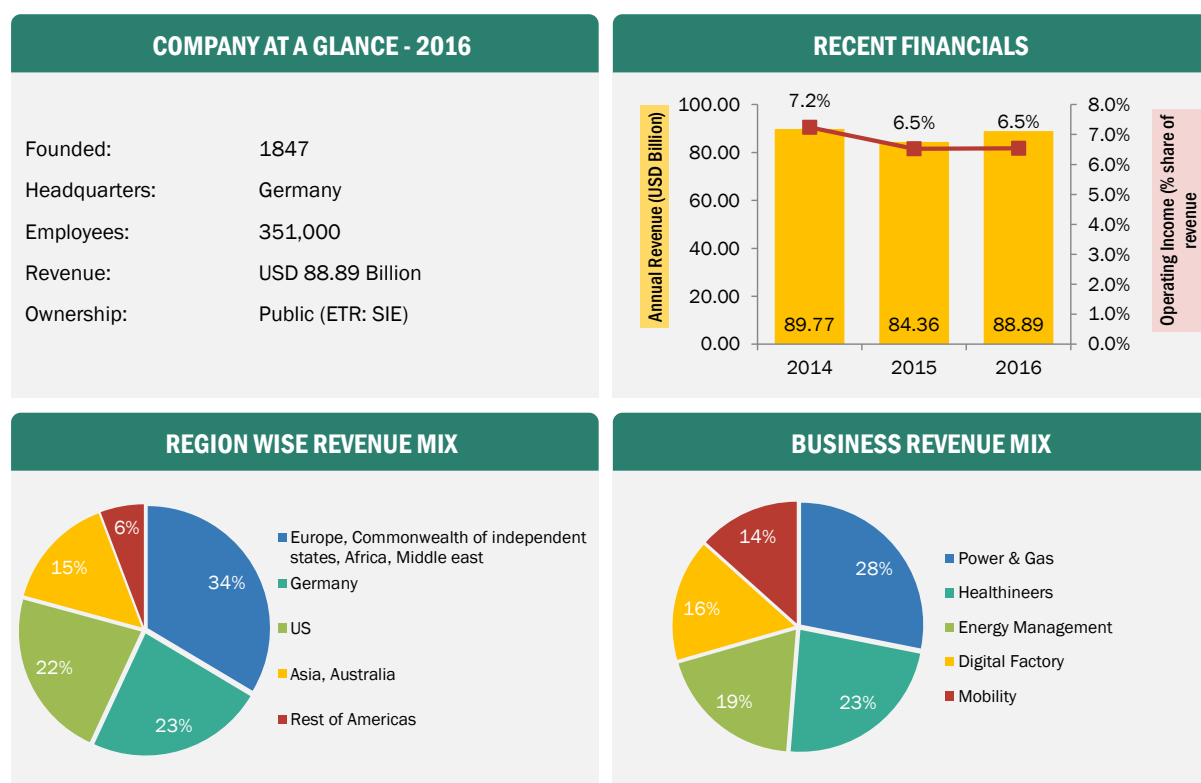
10.13.1 OVERVIEW

Siemens AG is a technology company with core activities in the fields of electrification, automation, and digitalization. It operates through nine business segments, namely, power and gas, healthcare, energy management, digital factory, process industries and drives, mobility, building technologies, wind power and renewables, and financial services. The company offers energy storage systems through its energy management division. This division enables the solution and energy storage system for integrating renewable energy into power grids.

In the battery energy storage system market, the company offers SIESTORAGE, a system that secures a stable and reliable power supply and integrates renewable energy sources by optimizing the use of fossil generation to a modern eco-friendly grid. This system combines power electronics for grid applications and high-performance lithium-ion batteries. The SIESTORAGE system also has applications in utilities, industries, cities, and infrastructure.

The company is present in approximately 190 countries and operates 289 production and manufacturing plants across the globe. It has a global presence through a strong network of regional offices, warehouses, R&D development facilities, and sales offices.

FIGURE 53 SIEMENS AG: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.13.2 PRODUCTS OFFERED

Category	Product
Energy storage systems	• SIESTORAGE • Hybrid Power and Storage • Hydro Power

Source: Company Website, Press Releases, and Company Annual Reports

10.13.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Siemens AG is a universal center focusing on the areas of electrification, mechanization, and digitalization. The products, systems, and solutions for low- and medium-voltage, make power distribution efficient, reliable, and safe. Siemens AG portfolio ranges from state-of-the-art compressors, turbines, and generators to virtual power plants, intelligent grid management, and innovative storage solutions, thus strengthening the business portfolio.

10.13.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

The company has expanded widely across the globe in regions, such as, the Americas, Asia Pacific, Middle East, and Europe. The company also entered into joint ventures and partnerships as a part of the inorganic growth strategy. From positioning along the electrification value chain, it has the knowledge that encompasses from power generation to power transmission and power distribution and smart grid to the efficient application of electrical energy.

10.13.5 RECENT DEVELOPMENTS

New Product Launches/ Expansions & Investments

Date	Approach	Description
October 2015	New Product launches	Siemens AG (Germany) extended the power range of its Sinamics DCP DC/DC power converter from 30 to 120 kilowatt. This converter is used in energy storage systems in photovoltaic installations or wind power farms, alongside press applications, diesel power harbor cranes, rack handling systems, or quick-charging stations for eCars.

Source: Company Websites, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
July 2017	Joint Ventures	Siemens AG (Germany)	AES Corporation (US)	The American energy firm AES Corporation and German engineering conglomerate Siemens AG formed a joint venture to serve the rapidly growing energy storage market.	
November 2014	Partnerships	Siemens AG (Germany)	LG Chem, Ltd., Ltd. (South Korea)	Siemens AG and LG Chem, Ltd. signed a memorandum of understanding and collaborated on project development and marketing of industrial battery storage systems. The companies are responding to the growing demand for energy storage systems which are needed more to stabilize distribution grids.	

Source: Company Websites, Press Releases, and Company Annual Reports

10.14 PANASONIC CORPORATION

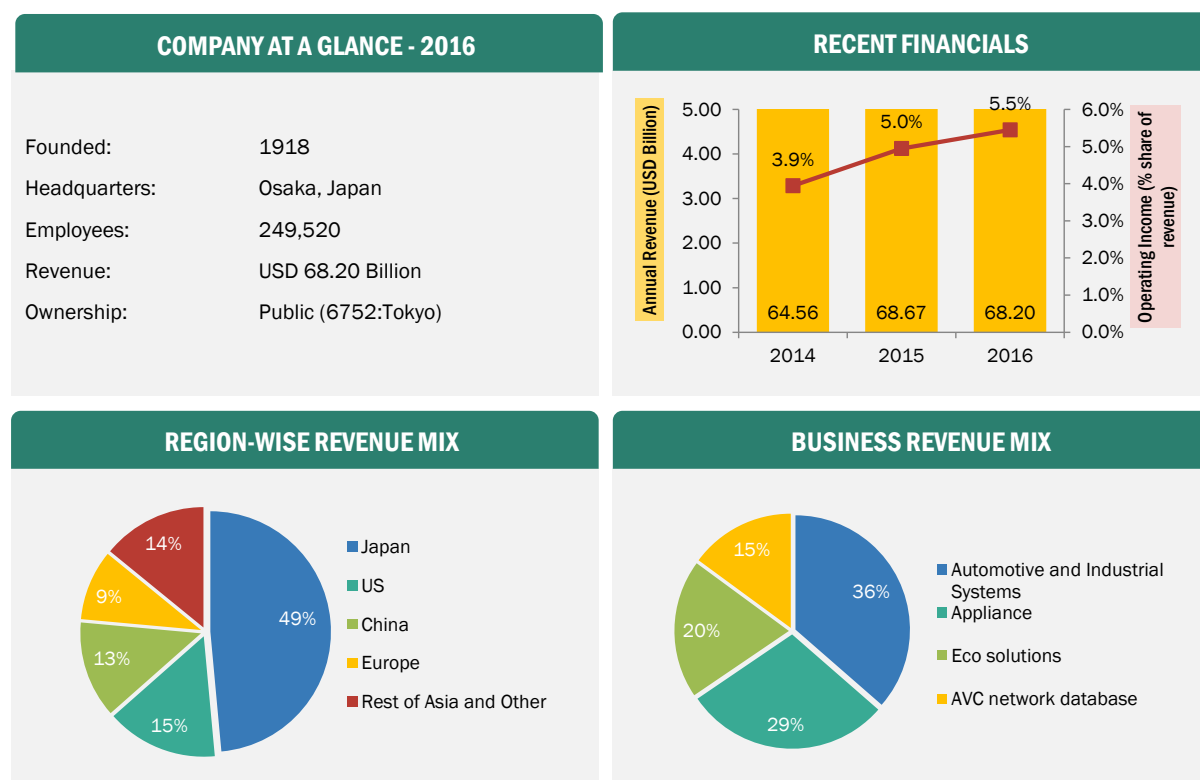
10.14.1 BUSINESS OVERVIEW

Panasonic Corporation was founded in 1918 and is headquartered at Osaka, Japan. The company manufactures and sells electrical products. It operates through the appliances, eco solutions, Audio Visual Computer (AVC) networks, automotive and industrial systems, and other segments. The company's batteries and energy storage systems are based on modular lithium-ion battery systems, consisting of battery modules and battery management systems. These systems can be integrated to external power conditioning and control systems.

The company focuses on storage applications, such as home energy storage for self-consumption of PV energy, commercial-scale energy storage for peak-shaving, UPS systems, and large-scale energy storage for grid stabilization, among others.

The company has expanded its battery energy storage systems business in various countries, including Australia, Germany, Japan, New Zealand, and the US.

FIGURE 54 PANASONIC CORPORATION: COMPANY SNAPSHOT



Source: Company Website, Press Releases, and Company Annual Reports

10.14.2 PRODUCTS OFFERED

Category	Product
Batteries and energy products	- Storage battery systems - Automotive batteries - Industrial batteries

Source: Company Website, Press Releases, and Company Annual Reports

10.14.3 STRENGTH OF PRODUCT PORTFOLIO

Particulars	Rating
Breadth and Depth of Product Offering	● ● ●
Product Features and Functionality	● ● ●
Product/Technology Innovation	● ● ●
Product Branding	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Panasonic Corporation (Japan) deals in four business areas, namely, consumer electronics, housing, automotive, and B2B solutions. Panasonic supports Energy Storage Systems with Battery Management ICs, Communication ICs between equipment in accord with changing condition and Display ICs to help reduce energy consumption for commercial or residential use. It has a huge product portfolio and it contributes to the society by providing the cutting edge technology.

10.14.4 BUSINESS STRATEGY EXCELLENCE

Particulars	Rating
Regional Footprint	● ● ●
Effectiveness of Organic Growth Strategy	● ● ●
Mergers & Acquisitions	● ● ●
Breadth of Applications Served	● ● ●
Penetration Rating: ● ● ● LOW ● ● ● MEDIUM ● ● ● HIGH	

Panasonic offers home storage, industrial, and automotive battery products in the battery energy storage systems market. The company focuses on expanding its battery business in various countries, including the US, Germany, the UK, and Japan. Its expansion strategy has supported its growth in the advanced energy storage systems market.

New Product Launches/ Expansions & Investments

Date	Approach	Description
October 2015	Expansions	Panasonic Eco Solutions Canada Inc. installed and started its first pilot residential solar photovoltaic and energy storage system in North America. The battery storage system used in this installation is powered by Panasonic's lithium-ion batteries. When combined with solar PV panels in an installation, the panel-generated electricity can work in synergy with the battery via a hybrid inverter to provide energy management, security, and cost performance benefits.
September 2015	Expansions	Panasonic Corporation announced its plans to expand its battery business by selling its home batteries across Europe, starting with Germany. A home battery is a lithium-ion-based system with a storage capacity of 8 kWh, with a 2 kW output and lightweight, sleek design that is easy to install and typically designed for home energy management systems.
September 2015	Expansions	Panasonic Corporation, through Panasonic Eco Solutions (its North American division) expanded its business by selling the grid energy storage technology in the US, working with energy management partner, PowerOasis, on the wireless industry system called Green Tower
June 2015	New product developments	Panasonic developed a battery storage technology in Australia, expected to be available in New Zealand as well.

Source: Company Websites, Press Releases, and Company Annual Reports

Partnerships/Collaborations/Joint Ventures/Contracts & Agreements/Mergers & Acquisitions/Strategic Alliances

Year	Nature of Agreement	Company 1	Company 2	Synergy	Deal Size (If Available)
October 2017	Partnerships	Panasonic Corporation (Japan)	Tesla, Inc. (US)	Tesla, Inc. (US) announced its partnership with Panasonic Corporation (Japan) to make photovoltaic cells and modules in Buffalo, New York, to open the largest solar panel factory in North America.	
April 2016	Contracts & Agreements	Panasonic Corporation (Japan)	AES Corporation (US)	Panasonic India Pvt. Ltd (a subsidiary of Panasonic Corporation) and AES India Private Limited (a subsidiary of AES Corporation) announced an agreement to construct a 10 MW energy storage array at Panasonic's Technopark manufacturing facility in Jhajjar, Haryana.	
January 2016	Mergers & Acquisitions	Panasonic Corporation (Japan)	GS Yuasa Corporation (China)	Panasonic Corporation (Japan) agreed to transfer its lead acid battery business, which is mainly operated by its wholly-owned subsidiary, Panasonic Storage Battery Co Ltd (PSB), to GS Yuasa Corporation	

Source: Company Websites, Press Releases, and Company Annual Reports

11 APPENDIX

11.1 INSIGHTS OF INDUSTRY EXPERTS



MARKETSANDMARKETS REPORT: ADVANCED ENERGY STORAGE SYSTEMS MARKET - GLOBAL FORECAST TO 2022

"There is a need for flexibility in grid storage due to fluctuation in the nature of renewable energy, such as solar and wind. This has increased the installation of AESS, where the storage technologies helps to increase self-consumption and perform peak-shaving and peak-shifting offering and Uninterruptible Power Supply (UPS)."



MARKETSANDMARKETS REPORT: ADVANCED ENERGY STORAGE SYSTEMS MARKET - GLOBAL FORECAST TO 2022

"High demand for battery energy storage systems is expected from North America; however, the market in Asia-Pacific is expected to grow rapidly owing to the growing installations in China, India, and Japan."



MARKETSANDMARKETS REPORT: ADVANCED ENERGY STORAGE SYSTEMS MARKET - GLOBAL FORECAST TO 2022

"The growing implementation of batteries in Electric Vehicles (EV) and Plug-in Hybrid Electric Vehicles (PHEV) supports the growth of the AESS market."



MARKETSANDMARKETS REPORT: ADVANCED ENERGY STORAGE SYSTEMS MARKET - GLOBAL FORECAST TO 2022

"The advanced energy storage system is expected to add to the efficiency of residential and non-residential applications in the future."

11.2 DISCUSSION GUIDE

- Q. 1.** The advanced energy storage systems market evolved at a faster rate in the past few years. This has been evident with the fast evolution of battery technology and emergence of new storage technologies, such as molten salt. What are your views on the prospects of the advanced energy storage systems market? Could you please describe the current scenario and how is the market expected to grow in the future?

Primary's viewpoint: _____

- Q. 2.** We estimated the current market size for the advanced energy storage systems market in 2016, based on the ongoing and completed projects, encompassing all the storage technologies included in the scope of the report. What are your views on the same?

Region	Market size (USD Million) 2016	Your Views
North America		
Europe		
Asia Pacific		
Rest of the World (RoW)		

Primary's viewpoint: _____

- Q. 3.** Based on the installed capacities, secondary literature surveys and expert interviews, we have derived the percentage contribution of the following technologies in the advanced energy storage systems market? Request your inputs on our findings.

Technology	Market share (%), 2016	Your views
Batteries (Lithium-ion, lead acid, sodium sulfur, flow batteries)		
Thermal Storage (sensible and latent)		
Pumped Hydro		
Flywheel		
Compressed Air Energy Storage		
Liquid Air Energy Storage		

Primary's viewpoint: _____

- Q. 4.** The battery technology is the fastest evolving technology in the advanced energy storage systems market. Most of the battery types have become more reliable and less costly year on year. What is your opinion on the derived percentage contribution of the battery types in the battery energy storage systems market?

Battery type	Market share (%), 2016	Your Opinion
Lead acid batteries		
Flow batteries		
Lithium-ion batteries		
Sodium Sulfur (NaS) batteries		
Others		

Note: Others includes nickel-metal hydride and nickel-cadmium cell

Primary's viewpoint: _____

- Q. 5.** As per our understanding there are two major applications for advanced energy storage systems that is grid storage and transportation. We observed that the transportation application is the largest and fastest growing market for advanced energy storage systems, especially the battery storage systems. What is your opinion on the application of advanced energy storage systems market?

Battery type	Market share (%), 2016	Your Opinion
Grid		
Transportation (Electric vehicles)		

Primary's viewpoint: _____

- Q. 6.** Through literature survey and discussions with subject matter experts we compiled following key drivers, restraints, opportunities, and challenges for the advanced energy storage systems market? We would like to know your inputs for the same.

Factor	Description
Drivers	
Restraints	
Opportunities	
Challenges	

Primary's viewpoint: _____

- Q. 7.** What are the upcoming technologies that are expected to have a significant impact on the advanced energy storage systems market in the near future?

Primary's viewpoint: _____

Q. 8. What would be the key revenue pockets for the advanced energy storage systems market in the next five years?

Primary's viewpoint: _____

Q. 9. What are the key factors that players should consider to succeed in the advanced energy storage systems market?

Primary's viewpoint: _____

Q. 10. What are the latest trends in the advanced energy storage systems market?

Primary's viewpoint: _____

Q. 11. Which are the top companies in the advanced energy storage systems market in order of their ranking?

Primary's viewpoint: _____

11.3 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

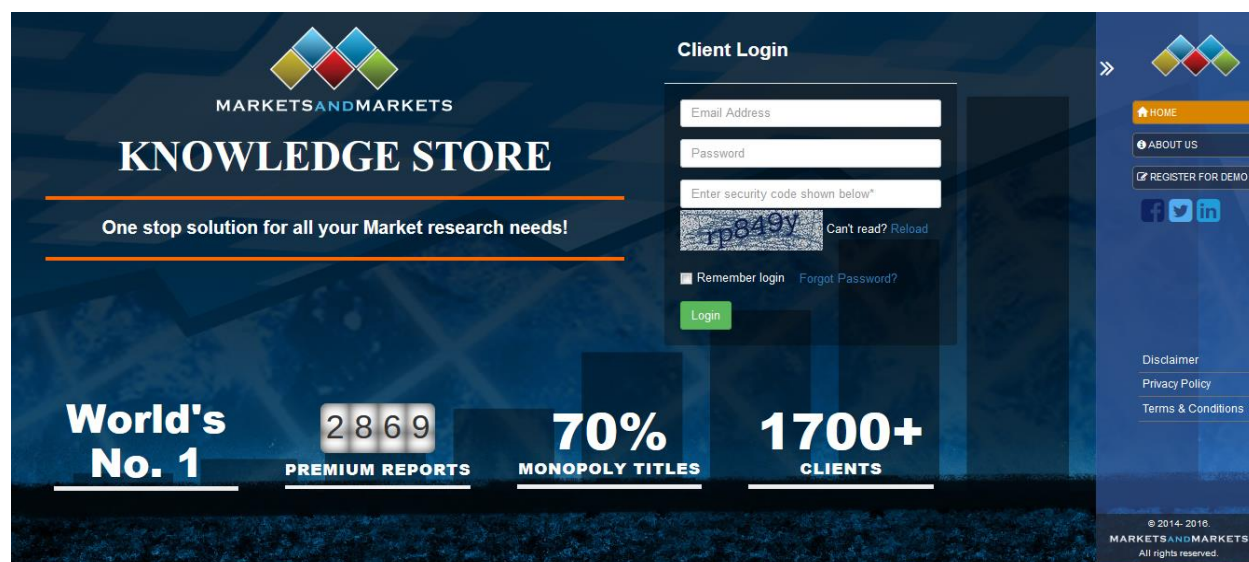
Knowledge Store contains MarketsandMarkets' unique market insights. As a subscriber you have access to all of the published titles and any new ones added over the course of the subscription period. You can subscribe to a specific number of reports or domains of your interest.

Knowledge Store enables you to:

- View and download all subscribed reports from a single, online platform
- Contact our analysts to answer any questions related to our reports or request a custom research
- Share comments on specific reports with other users of your organization
- Suggest titles or topics to our research teams that you would like us to cover in our future reports
- Identify reports published on high growth markets within your industry

Get started now by requesting a demo and learning more about Knowledge Store - <https://www.mnmks.com/>

MarketsandMarkets Knowledge Store Snapshot



MarketsandMarkets Knowledge Store: Energy & Power


MARKETSANDMARKETS KNOWLEDGE STORE

[Home](#)
[Access Reports](#)
[High Growth Markets](#)
[Admin](#)

[Advanced Search](#)
[Mm Admin](#)



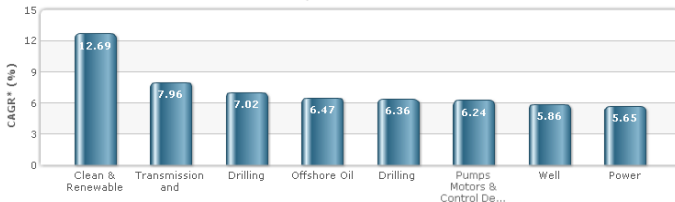
[My Industries](#)
[All Industries](#)

SORT BY : CAGR

[Reset Selection](#)

- Chemicals & Material (466)
- Healthcare (439)
- Semiconductor & Electronics (419)
- Information & Communications Technology (396)
- Energy & Power (158)
 - Transmission and Distribution (40)
 - Clean & Renewable Energy (25)
 - Pumps Motors & Control Devices (24)
 - Drilling Services (20)
 - Power Generation (19)
 - Drilling Equipment (12)
 - Offshore Oil & Gas (10)
 - Well Intervention (7)
- Food & Beverage (143)
- Aerospace & Defence (104)
- Automotive & Transportation (94)
- Packaging, Construction, Mining & Gases (85)
- Agriculture (83)

High Growth Markets



*CAGR (Compound Annual Growth Rate) represents Market Value (USD) growth forecast for a period of 5 years. It is an average CAGR based on the high growth niche markets covered in MarketsandMarkets syndicated reports and should not be interpreted as Industry CAGR.
Source: MarketsandMarkets Reports published since Jan 2015

[Ask Analyst](#)
[Request for Consulting](#)
[Suggest Title](#)

Energy & Power

CAGR %

Transmission and Distribution

Advanced Phase Change Material (PCM) Market: Global Forecast (2010 – 2015)	31.7	Jul, 2010		
Smart Transformers Market by Type (Power, Distribution, Specialty, and Instrument), Application (Smart Grid, Traction Locomotive, Electric Vehicle Charging, and Others) & by Region - Global Trends & Forecast to 2020	19.3	Jun, 2015		
Converter Transformer Market by Configuration (Back-to-Back, Monopolar, Bipolar, and Multi-terminal), Voltage level, Application (Windfarms, Oil & Gas, and Grid interconnections), Power rating, and by Region - Global trends and forecast to 2020	18.2	Aug, 2015		

[View All Reports](#)

Clean & Renewable Energy

North America Fuel Cell Technology Market: By Applications (Portable, Stationary, Transport), Types (PEMFC, DMFC, PAFC, SOFC, MCFC), Fuel (Hydrogen, Natural Gas, Methanol, Anaerobic Digester Gas) & Geography- Trends And Forecast To 2018	33.6	Jun, 2013		
Thin Film Photovoltaics (PV)- Advanced Technologies and Global Market (2008-2015)	32.2	Nov, 2010		
Carbon Capture & Sequestration Market by Service (Capture, Transport & Sequestration) Technology (Post-Combustion, Pre-Combustion, Oxy-Fuel Combustion & Industrial Separation) and Geography - Global Trends & Forecast to 2019	27.18	May, 2014		

[View All Reports](#)

11.4 INTRODUCING RT: REAL-TIME MARKET INTELLIGENCE

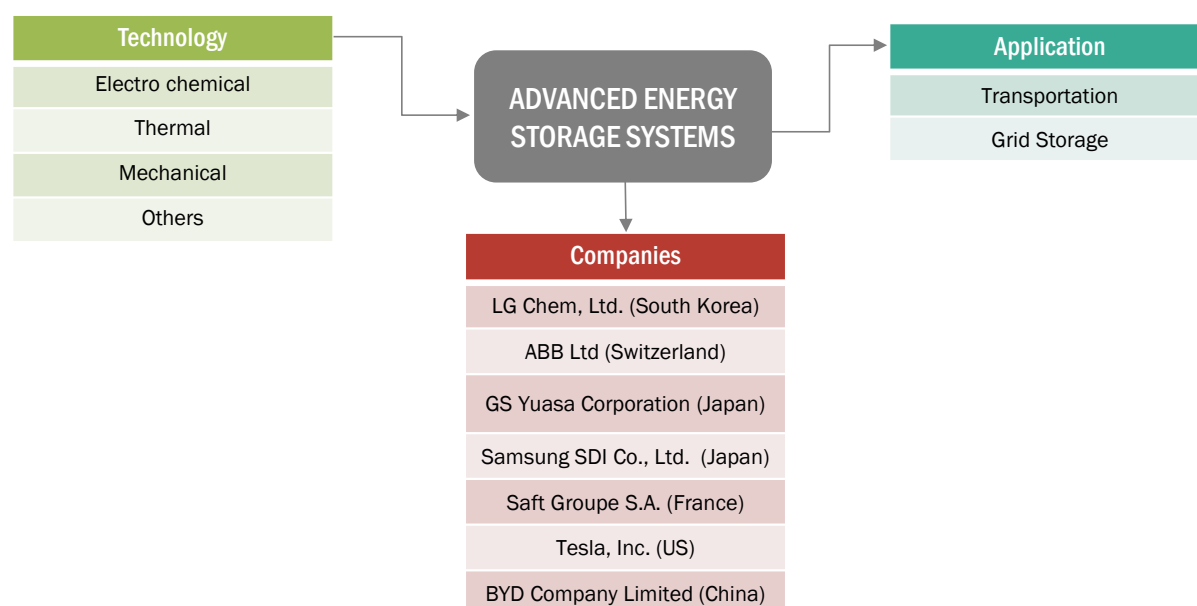
Revenue Tree (RT) is an interactive platform that systematically helps the user enhance revenue by identifying high-potential growth markets and track the same on a real-time basis on a competitive, customer-centric, and ecosystem level.

The in-built intelligence of the tool auto-identifies competitors, allied markets, market value, and customers, and helps the user identify tangible hotspots of revenue growth, both in existing and new markets. The unique feature of RT is the Ecosystem, which is linked across a wide-range of industries and markets.

Features and benefits of RT:

- Visual presentation of market data
- Real-time information on market trends and developments
- Information specific to an application, technology, generation and delivery mode, or region
- Competitive intelligence (by tracking developments of major players)
- Forward integration in the value chain to find potential clients
- Comparison of two or more markets

Market Dashboard



Note: Others includes hydrogen storage, synthetic natural gas, and supercapacitors

11.5 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations based on company-specific needs. The following customization options are available for the report:

Product Analysis

- Competitive intelligence
 - Product benchmarking of key players
 - Market analysis in detail by product type on the country as well as global level

Regional Analysis

- Further analysis of other countries of the region, such as Europe and Asia Pacific
- Further country-level analysis of regions on the basis of generation and delivery mode

Company Information

- Detailed analysis and profiling of additional market players (up to five)

11.6 RELATED REPORTS

Sr. No.	Report Title	Published Date
1	BATTERY ENERGY STORAGE SYSTEM MARKET Battery Type (Lithium-Ion, Advanced Lead Acid, Flow Batteries, & Sodium Sulfur), Connection Type (On-Grid and Off-Grid), Ownership, Revenue Source, Application, and Region - Forecast to 2022 http://www.marketsandmarkets.com/Market-Reports/battery-energy-storage-system-market-61500371.html	March 2016
2	THERMAL ENERGY STORAGE MARKET Technology (Sensible, Latent, and TCS), Storage Material (Water, Molten Salt, and Phase Change Material), Application (Power Generation, District Heating & Cooling, and Process Heating & Cooling), End-User, and Region - Global Forecast to 2022 http://www.marketsandmarkets.com/Market-Reports/thermal-energy-storage-market-61500371.html	March 2017

11.7 AUTHOR DETAILS

Rajat Kumar

*Vice President
Energy & Power*

Rajat manages the India-based syndicated research team of MarketsandMarkets. He brings over 10 years of experience in establishing, managing, and growing the research & analysis teams. An astute strategic planner and operations leader, he has been recognized by the top leadership for valuable contribution in engaging large accounts on many occasions throughout his career.

Prior to joining MarketsandMarkets, Rajat worked with GlobalData Research Center for nearly 8 years. He led a team of 100+ analysts and was instrumental in developing and successfully commercializing their oil and gas subscription product. He worked with all major companies across the oil & gas value chain including oil & gas supermajors, national oil companies, independents, oil & gas service companies and professional services companies. Rajat has also worked with the European Energy team of Frost & Sullivan, specializing in the distributed generation technologies such as renewables, fuels cells and power rentals.

Rajiv Roychaudhuri

*Associate Director
Energy & Power*

Rajiv carries 15 years of overall experience, most of which are in the consulting, strategy, marketing and market research industry pivoting around power and oil & gas domain, with due focus on allied industry verticals like industrial equipment including automation & electronics, environment & building technologies and general engineering. He started his career with due focus on industrial prime movers of the nation working in the marketing and strategy domain of a professionally managed business house like the Kirloskar Group & Thapar Group and gradually moving on to take larger mantles in MNC companies like Accenture & Frost & Sullivan in the Management Consulting & Market Research space. He has directed several client engaging consulting assignments and has also been an instrumental brand ambassador for marketing industrial products in competitive and challenging Global environments, thereby embellishing brands.

Prashanth K

*Associate Manager
Energy & Power*

Prashanth carries nearly ten years of market intelligence and research experience across multiple industry verticals including, Oil and Gas Upstream, Power Generation and T&D, Alternative Energy, Industrial and Heavy Engineering, Automotive, Consumer Goods and others. His past associations include GlobalData, HSBC, and First American Corporation. He has delivered multiple market assessment and forecast studies, Product Benchmarking, Customer Surveys, Buyer Behavior, and also developed industry statistics databases

Contributors

Karishma Batra - *Senior Research Analyst*

Akanksha Sinha - *Research Analyst*

Disclaimer: MarketsandMarkets' strategic analysis services are limited publications containing valuable market information that is provided to a select group of customers in response to orders. Our customers acknowledge, when ordering, that MarketsandMarkets' strategic analysis services are for their internal use and not for general publication or disclosure to any third party. Quantitative market information is primarily based on interviews, and is therefore subject to fluctuations.

MarketsandMarkets does not endorse any vendor, product, or service depicted in its research publications. MarketsandMarkets' strategic analysis publications are opinions based on its research and should not be construed as statements of fact. MarketsandMarkets disclaims all warranties, expressed or implied, with respect to this research, including any warranties of merchantability or fitness, for any particular purpose.

MarketsandMarkets takes no responsibility for any incorrect information supplied to it by manufacturers or users.

Trademarks, copyrights, and other forms of intellectual property belong to MarketsandMarkets or their respective owners and are protected by law. Under no circumstance may any of these be reproduced, copied, circulated in any form, without the prior written approval of MarketsandMarkets or its owner—as the case may be.

No part of this strategic analysis service may be given, lent, resold or disclosed to any third party, without written permission from MarketsandMarkets.

Reproduction and/or transmission in any form and by any means, including photocopying, mechanical, electronic, recording or otherwise, without the permission of MarketsandMarkets, is prohibited.

For information regarding permission, contact:
Tel: 1-888-600-6441